

Predicting Patent Validity Outcomes in the United Kingdom

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Abstract

Predicting litigation outcomes often requires combining information from documents that serve different roles and cannot always be interpreted independently. I examine three questions: which documents provide useful predictive information, whether documents should be encoded separately or jointly, and how to represent text whose meaning depends on other text. To investigate these questions, I construct a new dataset of U.K. patent litigation outcomes from 1995 to 2026 and compare predictive models at the patent, claim, and ground-of-invalidity levels. Patent litigation provides a setting in which these choices matter: assessing validity may require comparing a claim with the patent description or earlier technical disclosures, while dependent claims incorporate limitations from other claims. I define maximum dependency depth to measure the longest chain of these dependencies and construct expanded claim representations that explicitly incorporate inherited limitations. The results suggest that encoding documents separately and then combining their representations can outperform jointly encoding the concatenated texts in this setting.

1 Introduction

Patents can constitute economically important intangible assets. Hall et al. (2005), using Tobin’s q regressions for publicly traded U.S. firms, find that an increase of one citation in the average number of citations received per patent is associated with an approximately 3% increase in firm market value. Naturally, litigation outcomes concerning the invalidity of a patent can be extremely important events for a company. Panattoni (2011) shows that patent owners that won the litigation experienced an average increase of more than \$1.9 billion in market capitalization, while those that lost experienced an average decrease of approximately \$1 billion over the two trading days covering the court decision and the following day.

The value of patent protection depends in part on whether a patent can withstand a legal challenge. Predicting validity outcomes before a court issues its decision could therefore help firms assess litigation risk. Much of the literature on predicting patentability, however, concerns patent-office examination rather than litigation. Studies of patent litigation include Hicks (2021), Janicke and Ren (2006), and Cowart et al. (2014), which use conventional statistical or machine-learning methods.

This paper examines how textual information should be selected and represented to predict patent-validity outcomes in litigation. I compare predictive models using representations ranging from Word2Vec embeddings to pretrained transformer embeddings from Qwen. The analysis addresses three research questions.

First, how should a model represent text whose meaning depends on other text? A dependent patent claim incorporates limitations from the claims to which it refers. For example, claim 2 may add a restriction to claim 1, while claim 3 may refer to either claim 1 or claim 2 and add a further restriction. The literal text of a dependent claim therefore does not state all the limitations that define its scope.

Second, which texts provide useful predictive information? A patent contains a description of the invention and a set of claims defining the scope of protection. A validity challenge may also rely on earlier technical disclosures, known as prior art. The challenger may argue that the claimed invention is obvious because a person skilled in the relevant technical field would have found it straightforward to arrive at the invention in light of that prior art. These documents serve different roles, and their predictive value may depend on the ground of challenge. I therefore compare models using different combinations of the challenged claim, the complete claim set, the patent description, and the corresponding texts of the cited prior-art patents.

Third, how should these texts be combined? One approach concatenates the texts and encodes them jointly; another encodes each text separately and then combines the resulting representations. Whether joint encoding yields better predictive representations than separate encoding is an empirical question.

To answer these questions, I construct a new dataset covering U.K. patent litigation from 1995 to 2026. I use an LLM with extraction prompts to obtain structured information from published judgments. The judgments supply outcome labels and identify the patents and claims under attack, the grounds invoked, and the documents relied upon in those attacks. The court’s reasonings are excluded from the predictive inputs.

I provide predictions at three levels, corresponding to ex-ante, semi-ex-ante, and ex-post information sets defined relative to the hearing. Here, the ex-post information set refers to the period after the hearing but before the judgment is issued.

At the ex-ante-hearing level, I predict patent-level outcomes. I restrict the analyst’s information set to the patent under attack, without assuming knowledge of the specific claims, grounds, or prior-art documents that will ultimately be pursued at the hearing.

At the semi-ex-ante-hearing level, I predict claim-level outcomes. A patent consists of multiple claims, each defining a scope of protection sought by the owner and potentially subject to a separate challenge. Although the challenged claims may be identified from pleadings before the hearing, some may subsequently be dropped or rewritten. The claims ultimately pursued at the hearing may therefore differ from those initially identified. I describe this information set as semi-ex-ante because the prediction task conditions on the claims ultimately pursued.

At the ex-post-hearing level, I predict outcomes for individual grounds of attack against a claim,

conditional on the claims, grounds, and documents pursued at the hearing. A ground of attack is a legal basis for challenging the validity of a patent or claim. I focus on insufficiency, novelty, and obviousness, the three most common grounds of attack in the dataset. A claim lacks *novelty* when a prior-art disclosure—previously available technical information—discloses every feature of the claimed invention in enough detail for a person skilled in the relevant field to carry it out. A claim is *obvious* if such a person would have found it straightforward to arrive at the claimed invention in light of the prior art. *Insufficiency* arises when the patent does not disclose the claimed invention clearly and completely enough for such a person to carry it out.

This is where my approach departs most sharply from the existing literature. For example, PANORAMA (Lim et al., 2025) and FiNE-Patents (Knappich et al., 2026) draw on examiners’ written opinions to identify relevant prior-art documents or passages for prediction. Although this information is useful in itself, it is not available until the decision-maker issues the written opinion, at which point the outcome reported in that opinion is already known.

Being able to predict outcomes before the hearing is important for both patent owners and challengers. Patent owners could anticipate which claims are vulnerable, improve the wording of their claims, and better prepare their defence. Challengers could assess which grounds of attack to use against which claims.

Using these prediction tasks, I compare the predictive value of alternative textual inputs, encoding strategies, and treatments of claim dependencies. The results provide three main answers to the research questions set out above.

First, I address the problem of text whose meaning depends on other text by introducing the concept of an *“expanded claim”*. With the exception of Gao (2024), the existing literature has largely handled claim dependency either by restricting attention to independent claims or by treating dependent claims in the same way as independent claims. I instead represent the dependency structure as a tree, in which each path from a dependent claim eventually terminates at an independent claim. A dependent claim is then expanded by tracing each of these paths and incorporating the text of the claims on which it depends. This produces a self-contained representation that explicitly includes the limitations inherited through the dependency structure. Intuitively, one might expect prediction accuracy to deteriorate as dependency depth increases, since deeper claims require information from a longer chain of claims. In my test set, however, I do not observe such a decline.

Second, I examine which texts should be used as inputs for prediction. The relevant inputs differ across prediction tasks. For patent-level prediction, both the description and the claim set should intuitively contain useful information. For insufficiency, plausible inputs include the claim itself, the entire claim set together with the description, or the expanded claim together with the description, since sufficiency is assessed in the context provided by the patent specification. For obviousness, by contrast, the prior art should be particularly important, while the description of the patent itself may provide less additional information. The empirical results are consistent with these expectations: the best-performing models use the combinations of textual inputs predicted by

the legal structure of each task.

Third, I ask whether different texts should be encoded separately or concatenated before being passed through a single encoder. In the insufficiency analysis, separate encoding achieves higher court-test MCC and macro- F_1 than combined encoding at both depth regimes, although combined encoding retains predictive information. One possible explanation is the large imbalance in text length. For obviousness, for example, the claim and the prior art are both substantively important, but the claim is typically much shorter than the prior-art material. Concatenating them into a single input may therefore cause the representation of the claim to be overwhelmed by the much longer prior-art text. Separate encoding preserves a distinct representation of each source before their information is combined for prediction.

The remainder of the paper is organized as follows. The remainder of Section 1 reviews the related literature. Section 2 describes the U.K. patent system and the institutional and legal setting for validity disputes. Section 3 explains the construction of the dataset, the outcome definitions, and the training and evaluation samples. Section 4 describes the prediction models and evaluation metrics. Section 5 defines the ex-ante, semi-ex-ante, and post-hearing information sets used in the analysis. Section 6 presents the empirical results at the patent, claim, and ground-of-attack levels. Section 7 concludes.

1.1 Related Literature

1.1.1 Machine-learning literature

Predicting patent-validity outcomes requires identifying informative texts, relating these texts to one another, and interpreting claims whose meaning depends on other claims. I discuss the machine-learning literature along three dimensions: the information used to predict litigation outcomes, the selection and combination of patent texts, and the treatment of claim dependencies.

Information used to predict litigation outcomes. Studies of patent litigation use both structured information about disputes and the texts associated with them. Using case and party characteristics, Janicke and Ren (2006) estimate logistic regressions, while Cowart et al. (2014) compare logistic regression with a classification tree. Text-based studies instead examine information contained in patent claims or submissions made during a dispute. Hicks (2021) uses claim language to predict validity outcomes in U.S. litigation. Sokhansanj and Rosen (2022) use Patent Owner Preliminary Responses to predict whether the U.S. Patent Trial and Appeal Board will institute inter partes review. These responses allow patent owners to oppose a request to review their claims before the Board decides whether the administrative trial should proceed. The authors compare methods including TF-IDF with XGBoost and a convolutional neural network with attention. Their prediction target is therefore the decision to institute review, rather than the final determination of validity.

Other studies use the decision-maker’s written opinion itself. For example, Raghupathi et al. (2018) and Raghupathi (2026) apply text-mining and classification methods to published judicial

opinions in pharmaceutical patent cases. These opinions contain the court’s reasoning and generally state the outcome. Although this information is useful for classifying decided cases, it is not available until the court issues its written opinion, at which point the outcome is already known. Models requiring this information therefore cannot provide the same assistance to parties preparing for an unresolved dispute. My study uses judgments to obtain outcome labels and identify the patents, claims, grounds, and documents involved in an attack, while excluding the court’s reasoning and statements of the outcome from the predictive inputs.

Selecting and combining patent texts. Among text-based studies, an important distinction is whether prediction relies on claim language alone or also uses other documents. Hicks (2021) and Lo and Chu (2021) illustrate the first approach in different institutional settings.

Hicks studies U.S. district-court cases filed in 2008 and 2009, incorporating subsequent appellate decisions. He represents claim text using TF-IDF and applies random forests. His patent-level model using all claims achieves an accuracy of 66.0%, compared with a majority-class baseline of 54.7%. When asserted claims are grouped by their parent independent claim, the text-only model achieves 66.4%, rising to 68.5% with patent and litigation metadata, against a baseline of 56.5%. For novelty, obviousness, and indefiniteness, the text-only models achieve accuracies of 67.7%, 67.4%, and 75.6%, compared with respective baselines of 64.1%, 64.6%, and 71.7%. Thus, although the models pooling grounds improve appreciably on their baselines, the gains within individual grounds are smaller, ranging from 2.8 to 3.9 percentage points. The models do not use the patent description or prior-art text.

Lo and Chu instead study patent-office examination at the United States Patent and Trademark Office (USPTO). Using the first claim of an application as the principal input, they fine-tune BERT, RoBERTa, and XLNet to predict five binary rejection labels corresponding to different statutory grounds. Their reported micro-averaged metrics aggregate performance across these grounds, so they do not show how predictive performance differs by ground. Together, these studies provide evidence on prediction from claim wording alone, leaving open the question of what additional information other patent texts contribute.

Work on indefiniteness considers both the wording of claims and their relationship to the accompanying description (Ashtor, 2022; Knappich et al., 2025). Under U.S. law, the definiteness requirement asks whether claims define the scope of the invention with sufficient clarity. Although this requirement differs from insufficiency under U.K. law, both tasks can make the context supplied by the patent specification relevant to prediction. Ashtor (2022) predicts whether an application receives an indefiniteness rejection affecting at least one claim, using linguistic and structural features. PEDANTIC (Knappich et al., 2025) instead assigns labels to individual claims based on first Office Actions. Its logistic regression models use TF-IDF, linguistic features, or both, including claim length, readability, and lexical overlap with the description. It also evaluates Qwen 2.5 agents that can retrieve other claims and relevant description passages. The linguistic-feature baselines remain competitive with these agents. These approaches use relationships between texts, including

claim–description overlap, to supplement information contained in claim wording.

Prior-art comparison introduces a further relationship between documents. Under the U.S. novelty requirement in Section 102, a claim can be rejected when a single prior-art disclosure contains all its features. The non-obviousness requirement in Section 103 asks whether the claimed invention would have been obvious to a person skilled in the relevant field in light of the prior art. These grounds broadly correspond to novelty and inventive step under U.K. law.

Related work on prior-art retrieval addresses a different prediction question. PatentMatch (Risch et al., 2021), for example, pairs claims with paragraphs from examiner-cited prior-art documents and predicts their relevance to the claim. Its target is the relationship between a claim and a prior-art passage, rather than whether the claim survives a particular validity challenge.

PANORAMA (Lim et al., 2025) and FiNE-Patents (Knappich et al., 2026) construct examination benchmarks that pair claims with prior-art material. PANORAMA distinguishes novelty rejection, obviousness rejection, and allowance. Its prediction inputs include application text, cited prior-art text, and passages selected by the examiner. FiNE-Patents focuses on novelty at the European Patent Office (EPO), providing 3,658 first claims from eventually granted applications in selected computing and communications fields. Its main prediction task uses a claim and the closest prior-art document; examiner-provided passage references also support retrieval evaluation.

The construction of comparison examples matters in these benchmarks. For an allowed claim, PANORAMA supplies a passage cited against another claim in the same application. FiNE-Patents treats an initially filed claim rejected for lack of novelty as not novel and its amended, granted version as novel over the selected reference. A model could therefore learn differences between initial and amended wording without learning the relevant relationship to the prior art. The authors address this concern through controls for superficial differences and an adversarial test set.

These studies also illustrate different ways to process multiple texts. PANORAMA combines application and prior-art material within an LLM prompt and evaluates prompting and supervised fine-tuning. FiNE-Patents compares a single-step workflow, which evaluates the claim and prior-art document together, with a hierarchical workflow that assesses individual claim features before combining the assessments. PEDANTIC uses tools to obtain relevant claims and description passages during inference.

Nevertheless, these studies primarily focus on developing and evaluating prediction methods using specified sets of input texts. Although some, including FiNE-Patents, compare input configurations through ablation experiments, systematically determining which combinations of texts should be used is not their main research question. My paper makes this question a central focus. I compare predictive performance across alternative combinations of texts and examine whether including or excluding a particular text improves or worsens prediction. I also compare joint encoding with separate encoding followed by combination of the resulting representations. To my knowledge, this is the first study to systematically compare combinations of the challenged claim, the complete claim set, the patent description, and cited prior-art texts for predicting patent-validity outcomes in litigation.

Representing dependent claims. Claim dependencies create an additional challenge because the literal text of a dependent claim does not state all the limitations defining its scope. Gao et al. (2024) explicitly model these dependencies when predicting novelty rejections from USPTO prosecution data. They decompose claims into textual nodes and connect them through a graph representing relationships within claims and across parent and ancestor claims. Fixed text embeddings initialize the nodes, and graph neural networks combine information across connected nodes to produce a prediction. The model does not directly receive examiner-selected prior-art text, although the best specification includes a similarity measure based on retrieved prior patents. PEDANTIC (Knappich et al., 2025) provides another way to access dependency information: its agent can request parent claims when interpreting the target claim.

I incorporate claim dependencies into the text supplied to the encoder. For each dependent claim, I trace the dependency paths to independent claims and construct an expanded representation containing the inherited limitations. This differs from learning how to combine node representations through a graph neural network or allowing an agent to retrieve parent claims during inference. Expanding the text makes these limitations available before encoding and allows the same representation to be used with different text models. I compare literal and expanded claims and examine predictive performance across maximum dependency depths, testing whether this treatment remains useful as claims depend on longer chains of earlier claims.

1.1.2 Economics literature

The economics literature reviewed here provides two motivations for predicting patent validity outcomes. First, patents can generate substantial value for their owners. Second, the loss of patent protection can affect not only the owner’s value but also competition and subsequent innovation. These findings suggest that uncertainty over whether a patent will survive a validity challenge has economically meaningful consequences. They therefore motivate this paper’s investigation of whether patent texts and related information can predict claim-level validity outcomes. Whereas the studies reviewed below examine the value of patent protection and the consequences of its loss, this paper’s main focus is on predicting whether challenged claims survive.

The first branch of literature studies the value of patents to their owners. Pakes (1986) studies the distribution of the value of patent protection. He shows that the annual returns from patent protection amount to approximately 11–16% of annual R&D expenditure and that this value is highly concentrated, with about half accruing to only 5–10% of patents. Hall et al. (2005) ask whether patent counts and citations contain information about firms’ market values. They find that firms with more highly cited patent portfolios tend to have higher market valuations, conditional on their R&D and patent stocks. At the sample mean, the paper reports that one additional citation per patent is associated with approximately 3% higher firm value. Kogan et al. (2017) use stock-price reactions following patent grants to estimate the value of individual patents. They further show that this measure predicts firms’ future growth and productivity. They also find evidence of creative destruction: high-value patents benefit their owners, while capital and labor

move away from non-innovating firms and toward innovating firms. Farre-Mensa et al. (2020) show that the grant of a patent can accelerate startup growth and improve access to external financing.

The second branch studies what happens when patent protection is lost. Marco (2005) estimates that an invalidity decision reduces the patent owner’s stock-market value by approximately 1.4%. Panattoni (2011) also studies stock-market reactions and finds that branded pharmaceutical firms suffer cumulative abnormal market-value losses of slightly more than \$1 billion when they lose Paragraph IV cases. Suh et al. (2000) show that, following patent expiration and generic entry, the originator’s dollar sales fall by approximately 12% by the fourth year, while its market share measured by revenue falls to 78% of its original level. Galasso and Schankerman (2015) show that patent invalidation increases follow-on innovation, measured by subsequent citations. However, this effect is concentrated in computers, electronics, and medical instruments, with no significant effect in drugs, chemicals, or mechanical technologies. Branstetter et al. (2016) study a setting in which successful Paragraph IV challenges allow generic firms to enter the market before the relevant patents expire. They show that such entry reduces prices and incumbents’ market shares while increasing competition and consumer surplus. Gaessler et al. (2025) find a nonmonotonic effect of patent invalidation on follow-on innovation. The effect is strongest for low- and high-value patents and close to zero for patents of intermediate value.

2 Overview of the UK Patent System

2.1 Institutions and procedural settings

A patent that has effect in the United Kingdom may originate as a national patent granted by the UKIPO or as a European patent designating the United Kingdom and granted by the European Patent Office (EPO). The UKIPO is therefore the United Kingdom’s national patent office, but it is not the only institution that grants patents effective in the United Kingdom. This paper studies domestic UK decisions and does not include EPO examination, opposition, or appeal proceedings.

Within the domestic system studied here, questions concerning patentability or validity principally arise in three settings. First, before grant, a UKIPO examiner assesses whether a national patent application satisfies the statutory requirements. Strictly speaking, this stage concerns the *patentability of an application*, not the validity of an existing patent. The examiner normally communicates objections in writing, after which the applicant may submit observations or amend the application. If the objections are not resolved, the application may be refused. A contested matter may be determined by a UKIPO hearing officer on behalf of the Comptroller-General, either after an oral hearing or on the papers; the resulting document is an administrative *decision*, rather than a court judgment.

Second, after grant, any person may apply to the court or the Comptroller to revoke a patent on one or more of the grounds exhaustively specified in the Patents Act 1977. Third, when a patentee brings infringement proceedings, the defendant commonly denies infringement and challenges validity, usually by counterclaiming for revocation. The grounds on which validity may be put in issue in

such proceedings are the same as the grounds of revocation.

In England and Wales, first-instance patent litigation is heard principally in two specialist venues within the Business and Property Courts. The Patents Court generally handles larger and more complex patent disputes. IPEC provides a more streamlined and cost-controlled procedure for shorter, less complex, and lower-value intellectual-property disputes. The UKIPO also determines certain patent disputes through the Comptroller, but it is an administrative tribunal rather than a court. Appeals from most decisions of the Comptroller lie to the Patents Court. Judgments of the Patents Court and IPEC may be appealed, with permission, to the Court of Appeal, and a further appeal may reach the UK Supreme Court if permission is granted. Court of Appeal cases are generally heard by three judges, although a two-judge constitution is possible; Supreme Court appeals are normally heard by five Justices, with larger panels used in some cases.¹

2.2 The specification and the claim-level nature of validity

A patent specification ordinarily contains a description of the invention, one or more claims, any relevant drawings, and an abstract. The description explains the invention and how it may be performed. The claims perform a different legal function: they define the matter for which protection is sought and, after grant, delimit the scope of the patentee's exclusive right. Claims do not merely list the invention's industrial applications. They may instead be directed, for example, to a product, chemical compound, apparatus, process, use, or pharmaceutical composition.

Claims may be independent or dependent. An independent claim stands on its own. A dependent claim incorporates all the limitations of an earlier claim and adds one or more further limitations.

Validity is generally assessed claim by claim. Accordingly, one claim may be invalid while another claim in the same patent remains valid. Hence, the judgment may decide the patent to be partially valid, rather than outright revoking it or declaring that it is valid as a whole.

Throughout this paper, I use *Leo Pharma v Sandoz* as a running example.² The dispute concerned European Patent (UK) No. 0679 154 (EP0679154B1), covering calcipotriol monohydrate and pharmaceutical compositions containing it. LEO Pharma A/S, the patent owner, and LEO Laboratories Limited, its exclusive licensee, brought proceedings against Sandoz Limited.

I begin with the patent's claims to illustrate the distinction between independent and dependent claims. Example 2.1 presents the five claims and explains how they relate to one another.

Example 2.1 (Independent and dependent claims).

1. Calcipotriol monohydrate:

(1 α , 3 β , 5Z, 7E, 22E, 24S)-24-Cyclopropyl-9,10-secochola- 5,7,10(19),22-tetraene-1,3,24-triol, monohydrate.

¹The institutional descriptions in this subsection draw on UK Intellectual Property Office, nd, secs. 18, 72, 74, and 97, Courts and Tribunals Judiciary, ndb, Courts and Tribunals Judiciary, nda, and UK Supreme Court, nd.

²*Leo Pharma A/S and another v Sandoz Ltd* [2009] EWHC 996 (Pat), 15 May 2009. <https://www.bailii.org/ew/cases/EWHC/Patents/2009/996.html>.

2. A pharmaceutical composition containing the compound of claim 1.
3. A pharmaceutical composition according to claim 2 that is a cream.
4. A pharmaceutical composition according to claim 2 that is a gel.
5. A pharmaceutical composition according to any one of claims 2–4, with an active-component content of 1–100 $\mu\text{g/g}$ of the composition.

Claim 1 is an independent product claim directed to a particular chemical compound in monohydrate form. It is not a claim to every pharmaceutical that uses the compound. Claim 2 is a dependent composition claim that incorporates claim 1 and adds the requirement of a pharmaceutical composition containing the compound. Claims 3 and 4 narrow claim 2 to particular dosage forms, while claim 5 adds a concentration limitation and depends on one of claims 2–4.

If claim 1 is held invalid, the broad monopoly over the claimed compound is lost, but claims 2–5 do not automatically become invalid. A dependent claim may remain novel and inventive because of the additional limitations it contains, even when the broader claim from which it depends is invalid. For example, the invalidity of claim 2 neither invalidates claim 1 nor necessarily invalidates claims 3–5.

Each challenged claim must be tested on its own limitations and on the particular ground advanced against it. Only if no valid claim remains, and no allowable amendment cures the defect, will the patent necessarily be revoked in its entirety.

2.3 Grounds of invalidity

Section 72(1) of the Patents Act 1977 states the exhaustive statutory grounds on which a granted patent may be revoked. These grounds are divided into the following seven substantive categories plus entitlement, which I will not use in my analysis. The reason is that entitlement is an ownership dispute rather than an assessment of the claimed invention’s technical validity.

2.4 Key concepts

Before introducing the seven grounds of invalidity, it is useful to define three terms: the priority date, prior art, and the person skilled in the art (UK Intellectual Property Office, nd, secs. 2, 3, 5, and 14).

The *priority date* is the cut-off date used to decide what earlier information can be used against an invention. It is usually the filing date of the patent application, but the applicant may be entitled to an earlier date if the same invention was disclosed in an earlier application.

Prior art is information that was already available to the public before the priority date. It does not have to be an earlier patent or even something described as an invention. It may be a journal article, a product sold to the public, a conference presentation, a public demonstration, a website, or any other public disclosure anywhere in the world. By contrast, an earlier discovery that was

kept secret is not normally prior art because the public could not access it. Certain earlier-filed but later-published patent applications are treated differently for novelty.

The *person skilled in the art* is a hypothetical person who is competent in the relevant technical field but is not unusually inventive. The judge uses this benchmark to answer questions such as whether a claimed invention would have been obvious and whether the patent explains it well enough to be made or used.

In practice, the parties often call expert witnesses to help the judge understand the technology and reconstruct the perspective of this hypothetical skilled person. An expert witness is an actual person and is different from the legal concept of the skilled person. Experts are often instructed and paid by the parties, but their primary duty is to assist the court rather than to advocate for either side. The court may also appoint a scientific adviser where needed. (Civil Procedure Rules 35.1–35.4; Courts and Tribunals Judiciary, 2026, paras. 7.5 and 14).

Experts can nevertheless disagree about what was commonly known, what steps would have been routine, or what the patent teaches. The judge must decide which evidence is more convincing and how much weight to give it. Thus, the legal tests are designed to be objective, but their application still leaves room for differences across judges.

Below I outline the seven grounds of invalidity (UK Intellectual Property Office, nd, secs. 1–4, 14, and 76).

1. **Lack of novelty (anticipation).** A claim lacks novelty if a single prior-art disclosure satisfies two requirements: disclosure and enablement. First, it must disclose every feature of the claim, either expressly or by necessary implication. Second, it must enable a person skilled in the art, using their common general knowledge, to perform the claimed invention. Features from separate prior-art disclosures cannot be combined to establish anticipation.
2. **Lack of inventive step (obviousness).** A claim may be novel yet still be invalid because it is obvious. Under section 3, an invention involves an inventive step if it would not have been obvious to the notional person skilled in the art, having regard to the relevant state of the art. The inquiry is objective: it asks what the skilled person, equipped with the common general knowledge in the field at the priority date, would have found obvious, rather than whether the actual inventor subjectively found the invention difficult.
3. **Insufficiency.** Under sections 14(3) and 72(1)(c), the specification must disclose the invention clearly and completely enough for it to be performed by a person skilled in the art. The skilled person may use common general knowledge, but the patent must enable the invention across the scope of the claim without undue burden.
4. **Lack of industrial applicability.** Under section 4, an invention is capable of industrial application if it can be made or used in any kind of industry, including agriculture. “Industry” is interpreted broadly and encompasses useful and practical activity, not merely factory production. This ground is distinct from the question whether an invention is commercially profitable.

5. **Excluded subject matter.** Some subject matter is excluded from patentability even if it is novel and non-obvious. Section 1(2) identifies, among other things, discoveries, scientific theories, mathematical methods, aesthetic creations, schemes or methods for doing business or performing mental acts, programs for computers, and presentations of information. These exclusions apply only to the excluded subject matter “as such.” Additional statutory restrictions include certain methods of medical treatment or diagnosis under section 4A and inventions whose commercial exploitation would be contrary to public policy or morality under section 1(3). The precise test is context-specific, particularly for computer-implemented inventions.
6. **Added matter.** Under section 72(1)(d), a patent may be revoked if the matter disclosed in the granted specification extends beyond the disclosure of the application as filed.
7. **Extension of protection by post-grant amendment.** This ground arises when a patent has been amended after grant. Under sections 72(1)(e) and 76(3)(b), an amendment is impermissible if it extends the protection conferred by the patent as granted. This differs from added matter in what is compared: added matter concerns whether the specification discloses subject matter beyond the application as filed, whereas extension of protection concerns whether the amended patent broadens the scope of protection beyond that conferred at grant.

A claim only needs to fail on one established ground to be invalid. At the same time, the same claim may be attacked on several grounds, and the tribunal may accept some attacks while rejecting others.

Returning to the patent in Example 2.1, LEO Pharma A/S and LEO Laboratories Limited alleged that ointment and cream products sold by Sandoz Limited infringed EP0679154B1, which covered calcipotriol monohydrate and pharmaceutical compositions containing it. In response, Sandoz challenged the validity of claims 1–4. If an asserted claim were held invalid, LEO could not rely on that claim to establish infringement.

Sandoz attacked each of claims 1–4 on two grounds: lack of novelty and obviousness. For novelty, Sandoz relied on patent application WO1991012807A1, which concerns the use of vitamin D analogues, including calcipotriol, to treat acne. Sandoz argued that following Example 4 of this application would inevitably produce a cream containing calcipotriol monohydrate.

For obviousness, Sandoz relied on WO1991012807A1 for one attack and WO1987000834A1 for another. The latter application describes vitamin D analogues, including calcipotriol, their preparation, and pharmaceutical compositions containing them. Sandoz argued that, given these disclosures and common general knowledge, routine formulation and crystallisation work would have led a skilled person to discover and use calcipotriol monohydrate.

Ultimately, all these attacks failed, and the patent was upheld.

3 Data collection and description

3.1 Sample scope and judgment collection

I construct a dataset of cases concerning patentability or validity decided by the Patents Court of the High Court of Justice of England and Wales and the Intellectual Property Enterprise Court (IPEC) between 1995 and 2022. This court sample is used for training and validation. The separate test samples are described in Subsection 3.7.

Scope of the sample. The sample covers three types of proceedings: applications for revocation; infringement proceedings involving a counterclaim challenging patent validity; and appeals to the Patents Court from UKIPO decisions. The third category includes appeals against the refusal of a patent application and appeals against decisions concerning the revocation of a granted patent.

The sample therefore includes both patentability decisions concerning applications and validity decisions concerning granted patents. Patentability and validity are similar in the sense that patentability also uses 6 grounds mentioned in the previous section, with the exception of extension by protection.

I exclude judgments of the Court of Appeal and the UK Supreme Court to avoid counting the same underlying dispute again at a subsequent stage. Including both the lower-court judgment and the appellate judgment would generate repeated observations with substantially overlapping patent and party characteristics, as well as the grounds of attacks. For disputes heard at multiple court levels, I use the Patents Court or IPEC judgment. The sample also excludes proceedings before courts in Scotland and Northern Ireland and proceedings before the European Patent Office.

Sources and inclusion criteria. I identify cases using vLex, a legal research database, and cross-check its listings against BAILII, the British and Irish Legal Information Institute’s publicly accessible repository of legal decisions. Some vLex entries contain the full judgment, while others provide only the case title or bibliographic information. When neither source provides the full judgment, I search CaseMine, another legal research database, and other available sources.

Where necessary, I reconstruct information about the Patents Court or IPEC proceedings from a subsequent Court of Appeal judgment. The appellate judgment is used only as a source of information about the earlier proceedings; the observation and recorded outcome remain those of the Patents Court or IPEC.

A case is retained only if the available material provides the information required to complete the case-level record. This includes the names of the parties—the claimant and defendant, or the appellant and respondent—the identity of the judge issuing the relevant judgment, the identity of the patent or application in dispute, and the outcome. Cases for which the required information cannot be recovered are excluded. For 1995–2003, at most four otherwise eligible cases are missing from the sample in any given year. Coverage of eligible cases is complete for 2004–2022.

A case may satisfy these inclusion criteria even when the available material does not identify the

specific claims challenged. Such cases are retained for patent-level analysis but do not contribute observations to the claim-level sample. Subsections 3.3 and 3.4 describe the observation units and the resulting differences in sample size.

Sample size. The sample contains 361 cases: 326 in the Patents Court and 35 in the IPEC. These cases yield 502 patent-level observations, comprising 461 from the Patents Court and 41 from the IPEC. Each observation corresponds to a patent or application considered in a particular case. The 502 observations cover 490 distinct patents or applications, with 12 patents appearing in two proceedings each. For each of these patents, the two proceedings differ materially in the claims challenged or the prior art relied upon to support an obviousness challenge.

The sample includes both cases in which invalidity was conceded and cases in which patentability or validity was contested and determined by the court. Detailed claim-level sample sizes and the frequency of attacks on each ground are reported in Subsection 3.4.

3.2 LLM-assisted data extraction

Once a judgment has been obtained, I process it using four structured LLM prompts. Judgments contain the judge’s reasoning, the facts of the case, the parties’ arguments, and other information that must be extracted and organised for prediction.. The first prompt extracts case metadata and the information required to construct the numerical features used in the prediction models. These features include patent citation statistics, patent age, the classification of the claimant and defendant as listed companies, private companies, or individuals, whether the claimant is the patent owner, and whether each party is headquartered in the United Kingdom.³ The prompt also records the judgment date, the identity of the judge, the patents considered in the case, and the patent-level outcome. The first prompt is reproduced in Section D.1.

Because some judgments do not report the hearing date, I calculate forward-citation counts and patent age at the judgment date. Backward-citation counts are the same at the hearing and the judgment date. Among the 460 patent-level observations with a fully specified hearing date, forward-citation counts remain unchanged or increase by less than 5% relative to their first-hearing values in 89.8% of observations. The corresponding proportion is 96.1% when the threshold is 10%. These comparisons suggest that forward-citation counts are relatively stable between the first hearing and judgment dates.

The second prompt constructs the claim-level dataset. It identifies the claims considered by the court, records the outcome of each of the seven substantive grounds of invalidity raised against each claim, and identifies the prior art relied upon for novelty and obviousness attacks. It also records the dependency relationships among claims and produces the expanded claim representations described in Subsection 3.5. The second prompt is reproduced in Section D.2.

The second prompt may occasionally omit attacked claims or incorrectly identify claims as having been attacked. To help detect these errors, I use an additional prompt that focuses on

³For the full list and definitions of the numerical features, see Appendix A and Table A.1.

identifying the claims challenged in the judgment and the grounds of challenge applied to each claim. The resulting inventory is used to cross-check the more detailed extraction produced by the second prompt. The LLM is instructed to flag discrepancies between the two outputs, including missing or incorrectly included claims and grounds. I then manually review the relevant passages of the judgment to resolve these discrepancies and correct the extracted records before incorporating them into the dataset. The extra prompt is reproduced in Section D.5.

To give an example, in *Leo Pharma v Sandoz*, involving the patent in Example 2.1, the second extraction prompt would output no attack-level observations for added matter, lack of industrial applicability, excluded subject matter, insufficiency, and extension of protection by post-grant amendment. It would output four novelty attack-level observations and eight obviousness attack-level observations: each of claims 1–4 faced one novelty attack and two obviousness attacks.

The third prompt extracts the English patent text used as an input to the prediction models after verifying that the supplied document is the correct patent publication and version for the case. It uses Google Patents HTML as the primary source when a link is supplied, falls back to the PDF text layer when the HTML is missing or unusable, and uses OCR only when neither source provides usable text. The prompt does not translate the source text. The third prompt is reproduced in Section D.3.

The extracted English description and claims are stored in separate plain-text files. Bibliographic front matter, repeated headers and footers, non-English duplicates, tables, drawings, and unusable formula or chemical-structure text are removed. The substantive patent language is neither summarized nor paraphrased.

The fourth prompt performs the corresponding extraction for prior art. The fourth prompt is reproduced in Section D.4. Prior art may take many forms, including another patent, a catalogue, an academic paper, a book, or a technical guideline. As a general rule, I retain the prior-art document in its entirety unless a portion is clearly irrelevant even without specialized knowledge of the relevant technical field. For example, if the prior art is an academic paper, I retain the entire paper, including its references. If the prior art is a book and a particular chapter is relied upon, I retain the entire chapter. Thus, even if the judgment subsequently identifies only particular passages within a chapter as relevant, I do not restrict the input to those passages. Which portions the judge ultimately regards as relevant is information that would not have been available before the judgment was released and therefore should not be incorporated into the prediction input. The only material excluded is content that is facially unrelated to the subject matter at issue. For example, if the patent concerns a particular process and part of the prior-art document discusses an entirely unrelated process, that portion may be omitted.

At the same time, the judgment is used to identify the precise prior-art document and publication version relied upon by the court. If, for example, the prior art is a book, the prompt retrieves the edition or version submitted as evidence rather than substituting a newer edition simply because it is easier to obtain. More generally, the selected source must correspond as closely as possible to the version considered by the court and must have been publicly available by the relevant date. This

prevents later editions, grants, amendments, or other subsequent versions from being substituted for the document that was actually available and relied upon.

Patent prior art requires an additional source-selection rule. The prompt first attempts to retrieve the exact patent publication identified in the judgment. If that publication cannot be located or does not contain usable text, the corresponding WO publication is used as a fallback, provided that it belongs to the same patent family and was publicly available by the judgment date. As with the patent in suit, priority is given to English text or to an English translation available through Google Patents. When neither is available, the LLM translates the original text as literally as possible, without summarizing the disclosure or altering technical terminology or claim scope.

The complete prompts contain additional source-selection, formatting, and error-handling instructions analogous to those used in the patent-extraction prompt. Bibliographic front matter, repeated headers and footers, non-English duplicates, tables, drawings, and unusable formula or chemical-structure text are removed from the extracted prior-art text.

For the full prompts, see Appendix D.

3.3 Units of observation and outcome coding

At the patent level, an observation corresponds to a patent or application considered in a particular case. I classify the owner or applicant as losing if all challenged claims are invalidated or refused, winning if all challenged claims survive, and drawing if at least one challenged claim survives while at least one fails. These outcomes are coded as 1, 0, and 0.5, respectively. This classification considers all challenged claims, including both independent and dependent claims.

At the claim level, I include a claim when at least one ground of validity or patentability has been raised against it and the judgment determines its outcome. Claims mentioned only as technical background, through dependency references, or in relation to infringement are not included on that basis alone. I do not infer individual claim outcomes solely from the revocation of the patent. A claim fails overall if it is invalidated on at least one ground, even if it survives other challenges. It survives overall if it withstands every ground adjudicated against it. Claim-level outcomes are binary, with 1 indicating a loss for the owner or applicant and 0 indicating a win.

Each separately assessed version of a claim is treated as a distinct claim-level observation. Thus, when the judgment separately determines the validity of an original claim and an amended version, both are recorded with their respective outcomes. If only the amended version is assessed, no separate observation is created for the original version.

For ground-specific analysis, an attack-level observation corresponds to a claim, a ground of challenge, and, for novelty and obviousness, a particular prior-art unit. A claim challenged using several distinct prior-art units therefore generates multiple attack-level observations. Documents assessed jointly in support of the same attack form a single prior-art unit. Although their component documents may occupy separate extraction rows, they represent one attack-level observation. Outcomes are recorded separately for each attack, so a claim may survive one attack while failing another.

Express concessions of invalidity are included as an exception to the requirement of judicial determination. When the owner expressly concedes that an entire patent is invalid, I include all independent claims in the patent version covered by the concession and code them as losses. Dependent claims are included only when they separately satisfy the general inclusion rule or are individually conceded invalid. Withdrawal, amendment, or failure to defend a patent does not by itself establish a concession of invalidity. When a concession identifies a specific ground, the loss is recorded under that ground. The residual category of conceded invalidity is used only when no specific ground is identified.

There are two exceptions to the general inclusion rule. First, express concessions of invalidity are recorded as losses even when the court does not separately adjudicate the relevant claims. When the patentee concedes that an entire patent is invalid, I include all independent claims in the patent version covered by the concession and code them as losses. Dependent claims are included only when they separately satisfy the general inclusion rule or are individually conceded invalid. If the concession identifies a specific ground, the loss is recorded under that ground; otherwise, it is recorded under an eighth category, conceded invalidity. Withdrawal, amendment, or failure to defend a patent does not by itself establish a concession of invalidity.

Second, when the court considers several patents together and determines their validity by reference to one representative patent, I assign the same outcome to the corresponding independent claims in the other patents covered by that determination. Together, claims included under these two exceptions account for approximately 5% of the claim-level sample.

3.4 Sample composition

My training and validation sample contains 361 cases decided between 1995 and 2022: 326 in the Patents Court and 35 in the IPEC. These cases generate 502 patent observations, of which 461 are from the Patents Court and 41 from the IPEC. The sample includes both patents whose invalidity was conceded and patents whose validity was contested and determined by the court.

The claim-level sample contains 1,830 claims. Of these, 50 are classified under conceded invalidity, while the remaining 1,780 were challenged on at least one substantive ground. Table 1 reports the frequency with which each ground was raised. The table excludes claims classified only under conceded invalidity and patent observations for which the challenged claims cannot be identified. It therefore covers 354 cases, 479 patent observations, and 1,780 claims. Because a claim may be challenged on several grounds, the numbers across columns do not sum to 1,780.

Table 1: Frequency of invalidity attacks

	Novelty	Obviousness	Added matter	Industrial applicability	Excluded subject matter	Extension by amendment	Insufficiency
Cases with attack	207	317	119	7	42	13	136
Share of cases (%)	58.5	89.5	33.6	2.0	11.9	3.7	38.4
Patents with attack	241	429	145	7	56	15	172
Share of patents (%)	50.3	89.6	30.3	1.5	11.7	3.1	35.9
Claims with attack	753	1,519	384	35	175	27	626
Share of claims (%)	42.3	85.3	21.6	2.0	9.8	1.5	35.2

Denominators: 354 cases, 479 patent observations, and 1,780 claims.

The same patent is counted separately when considered in different judgments.

3.5 Claim dependencies and expanded claim representations

A dependent claim incorporates the limitations of one or more earlier claims and therefore generally cannot be interpreted from its own text alone. For a given target claim, its dependency structure can be visualized by unfolding the dependency paths into a tree. The target claim forms the root, each branch follows a reference to an earlier claim, and each path terminates at an independent claim. An earlier claim may appear on more than one branch when different dependency paths lead to it.

Definition 3.1 (Maximum dependency depth). A claim’s *maximum dependency depth* is the number of dependency links in the longest path from that claim to an independent claim. An independent claim therefore has maximum dependency depth zero. For a dependent claim, the depth equals one plus the greatest maximum dependency depth among its direct parent claims.

Formally, let $\mathcal{P}(c)$ denote the set of direct parent claims referenced by claim c , including all alternatives specified by references such as “any preceding claim.” The maximum dependency depth of claim c , denoted by $d(c)$, is

$$d(c) = \begin{cases} 0, & \text{if } \mathcal{P}(c) = \emptyset, \\ 1 + \max_{p \in \mathcal{P}(c)} d(p), & \text{if } \mathcal{P}(c) \neq \emptyset. \end{cases}$$

An *expanded claim* is a self-contained representation that explicitly incorporates the limitations inherited through this dependency structure. The claim-level extraction prompt instructs the LLM to trace every dependency path from the target claim to an independent claim and incorporate the relevant wording along that path. Where the dependency references specify alternative paths, the expanded text preserves those alternatives. Each alternative includes both the limitations inherited along its path and the limitations added by the target claim.

The resulting text expresses the claim’s technical content without requiring the reader to consult other claims. For an independent claim, the expanded text is identical to the literal text. For an amended claim, the expansion follows the wording and dependency structure of the relevant amended claim set. The prompt requires checks that every path reaches an independent claim, that the reported depth matches the longest path, and that the expanded text preserves the substantive

limitations and alternative dependencies without leaving unresolved claim references.

In Example 2.1, claim 5 refers to claims 2, 3, and 4. Claims 3 and 4 each depend on claim 2, which in turn depends on independent claim 1. The longest paths therefore contain three dependency links:

$$d(5) = 1 + \max\{d(2), d(3), d(4)\} = 1 + \max\{1, 2, 2\} = 3.$$

Expanding these paths produces three alternatives: the composition defined through claim 2 without an additional dosage-form restriction, the cream specified through claim 3, and the gel specified through claim 4. The concentration limitation introduced by claim 5 applies to all three alternatives. Its standalone expanded text is:

A pharmaceutical composition containing calcipotriol monohydrate: (1 α , 3 β , 5 Z , 7 E , 22 E , 24 S)-24-cyclopropyl-9,10-secochola- 5,7,10(19),22-tetraene-1,3,24-triol, monohydrate, under any one of the following alternatives: (i) no additional limitation as to dosage form; (ii) the pharmaceutical composition is a cream; or (iii) the pharmaceutical composition is a gel; with an active-component content of 1–100 $\mu\text{g/g}$ of the composition.

Figure 1 shows the distribution of maximum dependency depths in the complete claim-level sample.

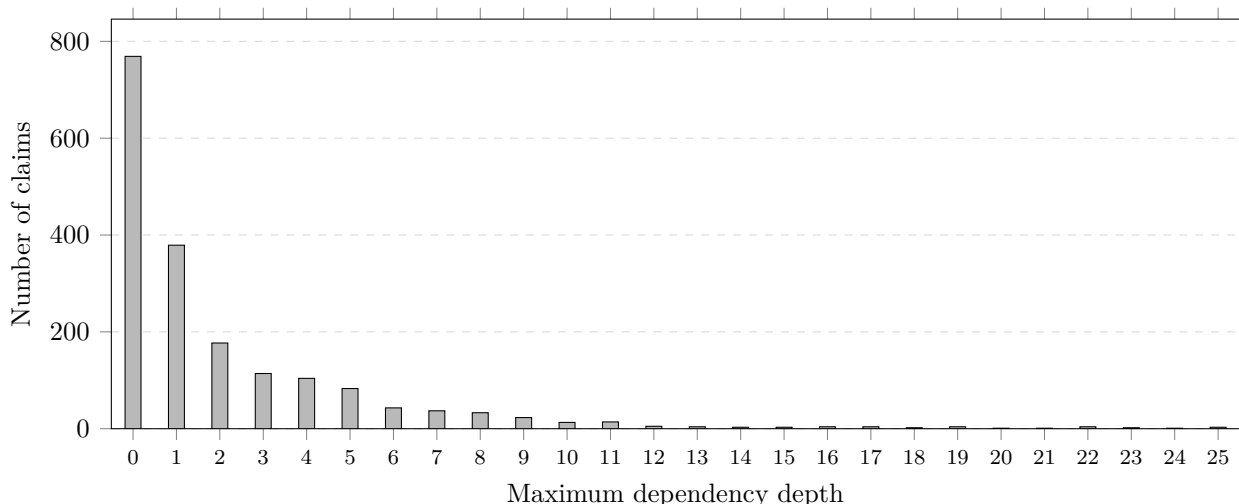


Figure 1: Distribution of maximum claim-dependency depths. The sample contains 1,830 claim-level observations.

The distribution is concentrated at relatively low dependency depths. Level 0 accounts for 42.0% of the observations, while levels 1, 2, and 3 account for 20.7%, 9.7%, and 6.2%, respectively. Taken together, claims at levels 0–2 account for 1,325 observations, or 72.4% of the sample, while claims at levels 0–3 account for 1,439 observations, or 78.6%.

Greater dependency depth requires information to be combined across longer chains of claims, potentially increasing the difficulty of interpretation and prediction. I therefore conduct the main

specification search and architecture selection using claims with a maximum dependency depth of at most two. After selecting the architectures, I evaluate the same model specifications on the full claim sample without imposing a dependency-depth restriction. This exercise assesses whether architectures selected using claims with simpler dependency structures also perform well when deeper claims are included, without conducting a separate architecture search for the unrestricted sample.

3.6 Ground-specific prediction sample

For the ground-specific prediction exercises, I focus on novelty, obviousness, and insufficiency, the three grounds with the largest numbers of challenged claims. Although added matter is raised in nearly as many cases as insufficiency, it produces substantially fewer claim-level observations. At least one of the three focal grounds is raised in 332 of the 354 cases involving a substantive invalidity attack, corresponding to 93.8% of such cases.

These grounds also capture most overall claim-level outcomes. Among the 1,702 claims challenged on at least one of the three focal grounds, considering the outcomes on these grounds alone yields the same overall claim-level classification for 1,670 claims, or 98.1%. Only 32 claims have an overall outcome different from that implied by the focal grounds alone. Each of these claims survives the challenges brought against it on the focal grounds but is invalidated for added matter.

3.7 Training, validation, and evaluation protocol

Within the 1995–2022 court sample, I construct the training and validation sets at the case level. All patents, claims, and prior-art attack observations belonging to the same case are assigned together to either the training set or the validation set. This prevents observations from the same case from appearing in both sets within a given split.

Model and architecture selection is initially conducted using a designated training–validation split, with approximately 80% of observations assigned to training and 20% to validation. Once the shortlisted specifications and their hyperparameters have been fixed, I assess their stability through repeated validation before evaluating them on the two holdout test sets. For each selected architecture, I also evaluate a version without numerical features to assess how well the model performs using textual information alone.

I use Monte Carlo cross-validation (MCCV), a procedure that repeatedly partitions the available sample at random into training and validation sets, fits a model on each training set, and evaluates it on the corresponding validation set (Arlot and Celisse, 2010). I implement this procedure using 25 or 50 random case-level splits, depending on the prediction task, each assigning approximately 80% of cases to training and 20% to validation. For every split, I fit each shortlisted model anew. Architectures, model specifications, and hyperparameters remain fixed across repetitions, while model parameters are re-estimated.

I use MCCV rather than a single run of k -fold cross-validation to examine sensitivity to the allocation of cases between training and validation. In NLP experiments, Moss et al. (2018) show

that performance estimates and hyperparameter choices obtained from k -fold cross-validation can depend substantially on the random partition into folds, and that repeating the partitioning reduces this variability. These findings motivate evaluating the shortlisted specifications across multiple case allocations. Varoquaux et al. (2017) recommend repeated random splits as a way to increase the number of evaluations while maintaining a fixed training–validation ratio. My MCCV exercise follows this rationale: it assesses performance across many alternative splits while retaining 20% of cases for validation in each repetition. The resulting distributions characterize the stability of model performance and rankings conditional on the initial model selection.

After completing the MCCV exercise, I evaluate the shortlisted specifications on two distinct holdout test sets. The first holdout consists of cases decided in the Patents Court and the IPEC between 2023 and 2026. This temporal test set approximates the intended prediction setting: models trained on past decisions are used to predict the outcomes of future disputes. A random split over the full period could instead allow information from later judgments to influence predictions for earlier cases.

The second holdout consists of decisions from the UK Intellectual Property Office (UKIPO) and serves as an institutional out-of-domain test set. UKIPO proceedings are decided by hearing officers rather than judges. This sample therefore tests whether the models generalize across decision-makers, institutional settings, and procedures.

The temporal court holdout contains 59 patent observations from 44 cases: 38 in the Patents Court and 6 in the IPEC. The UKIPO sample initially contained 93 patent observations from 90 cases. I exclude 8 patent observations corresponding to 8 cases because these disputes were subsequently appealed to the Patents Court, and the appeal decisions are already included in my training and validation sample. The final UKIPO holdout therefore contains 85 patent observations from 82 cases. Together, the two final holdout samples contain 144 patent observations from 126 cases.

Although the temporal and institutional test sets provide genuine holdout evaluations, each consists of a single finite collection of cases. Measured test performance may therefore depend partly on the particular cases included. I consequently interpret the final test results alongside the performance distributions obtained in the MCCV exercise, which provide complementary evidence on stability within the 1995–2022 court sample.

4 Models and evaluation metrics

I compare three groups of methods: classical models, models using frozen transformer embeddings, and transformer models fine-tuned with low-rank adaptation (LoRA). In the main text, I explain the intuition behind these models. Additional mathematical details are provided in Appendix ??.

4.1 Classification heads

A model first maps each document into a numerical representation vector. The resulting representation may also be combined with additional numerical features before being passed to a task-specific prediction head. For binary classification, this paper considers three main families of classification heads: logistic regression, support vector machines (SVMs), and L_2 -regularized least-squares classifiers.

These classifiers can be described using the generic regularized training objective

$$(\hat{w}, \hat{b}) = \arg \min_{w, b} \left\{ \sum_{i=1}^n a_{y_i} s_i \ell_i(w, b) + \lambda R(w) \right\},$$

where w is the coefficient vector, b is the intercept, $\ell_i(w, b)$ is the classifier-specific loss for observation i , $R(w)$ is a regularization penalty applied to the coefficient vector, a_{y_i} is the weight assigned to the outcome class of observation i , and s_i is its sample weight. The classifier-specific loss functions and regularization penalties are provided in the Appendix. The patent-size weighting scheme is defined in Equation 1.

Class weights control the relative importance assigned to the two outcome classes. I consider either uniform class weights or balanced class weights, under which observations from less frequent classes receive greater weight. Because the outcome classes are substantially imbalanced in many of my prediction tasks, balanced class weighting is included among the candidate specifications.

Sample weights determine the relative influence of observations according to the patent from which they originate. In addition to uniform weighting, I consider patent-size-based weighting, which assigns less weight to individual observations from patents that generate more observations. This reduces the influence of patents with many observations on model fitting.

Regularization discourages the classifier from fitting the training data too closely and can therefore improve out-of-sample performance. The parameter λ controls the strength of regularization: a larger λ places greater emphasis on the regularization penalty, whereas a smaller λ places greater emphasis on fitting the training observations.

Logistic regression. Logistic regression maps a linear combination of the input features to the probability of an adverse outcome for the patent owner or applicant. The coefficients are estimated by maximizing the weighted log-likelihood of the observed outcomes, with regularization to limit overfitting.

Support vector machines. Unlike logistic regression, a support vector machine does not directly model class probabilities. Instead, it seeks a decision boundary that balances a wide margin between the two classes against penalties for observations that fall within the margin or on the wrong side of the boundary. I consider both linear SVMs and SVMs with a radial basis function (RBF) kernel, which allows for nonlinear decision boundaries.

Ridge-regularized least-squares classifier. The ridge classifier fits a linear model to the class labels by minimizing weighted squared prediction errors with L_2 regularization. Classification is based on whether the resulting decision score exceeds zero.

4.2 Classical/ non neural models

Numerical-only models and Extra tree I also investigate how accurately outcomes can be predicted without textual features. For this exercise, I use only the numerical features and apply the classification heads described above. In addition, I consider the Extremely Randomized Trees classifier, commonly referred to as Extra Trees.

Extra Trees is an ensemble of randomized decision trees. For each tree, I first specify a maximum depth. A greater depth allows the tree to represent more complex relationships and generally fit the training data more closely, but it can also increase the risk of overfitting.

At each decision node, the algorithm randomly selects a subset of the d features. For each selected feature j , it then randomly generates a candidate threshold τ . For each candidate pair (j, τ) , observations whose values of feature j exceed the threshold go to the right branch, while those whose values are at or below the threshold go to the left branch. Among these candidate pairs, I choose the one that maximizes the reduction in impurity, which is defined in the Appendix.

This procedure is repeated recursively until a stopping condition is reached, such as the maximum tree depth, a minimum number of observations in a node, or a node containing observations from only one class. A terminal node, or leaf, predicts according to the class distribution of the training observations it contains. Extra Trees constructs many such randomized trees and obtains its final prediction by aggregating their predictions. Randomizing both the candidate features and their thresholds reduces the dependence of the final prediction on any single fitted tree.

At prediction time, each tree routes a new observation through its nodes according to the observation’s feature values and the fitted threshold rules, following the right branch when the selected feature exceeds the threshold and the left branch otherwise, until a terminal node is reached.

TF-IDF representation. Before the widespread adoption of transformer-based representations, bag-of-words models weighted by term frequency-inverse document frequency (TF-IDF) were a standard approach to representing text in empirical economics and the broader text-as-data literature (Gentzkow et al., 2019; Eberhardt et al., 2023). TF-IDF remains a useful baseline because it is computationally inexpensive, transparent, and can perform competitively in text-classification tasks despite ignoring word order and context (Reusens et al., 2024). TF-IDF represents each document as a numerical vector, with one component for each term in the vocabulary. Each component reflects the frequency of that term in the document, multiplied by an inverse document frequency weight. This weight discounts terms that appear in many documents and gives greater weight to terms that appear in fewer documents.

The resulting vector is concatenated with the numerical features and passed to one of the classification heads described in the previous subsection for training and prediction.

The intuition is that a document can be characterized by its pattern of term use. Documents with similar patterns receive similar representations, allowing the classifier to learn which patterns are associated with each outcome. Inverse document frequency weighting reduces the contribution of words that are common across the corpus and therefore provide less information for distinguishing documents.

Complement Naive Bayes. Complement Naive Bayes uses the TF-IDF representation described above. For each outcome class, it constructs a profile of term use from documents outside that class, referred to as its complement. In my binary setting, the complement of class 1 consists of documents from class 0, and vice versa.

To construct each profile, the classifier sums the TF-IDF weights of each term across documents in the complement. It adds a positive smoothing constant to each total to avoid zero weights, then normalizes these totals so that they sum to one across the vocabulary. Let q_{cj} denote the resulting weight of term j in the complement of class c .

At prediction time, the classifier scores a new document against each complement profile and assigns it to the class

$$\hat{y} = \arg \min_c \sum_{j=1}^d x_j \log q_{cj},$$

where x_j is the document's TF-IDF weight for term j and d is the vocabulary size. The score is lower when the document places greater weight on terms that are uncommon in the complement. Thus, the classifier selects the class whose complement provides the poorest explanation for the document's pattern of term use.

Word2Vec. Unlike TF-IDF, which directly represents each document as a vector, Word2Vec first learns vector representations for individual words. I train these representations from scratch using two variants of Word2Vec: the continuous bag-of-words (CBOW) model and the skip-gram model. In both variants, each word has two vectors during training: an input vector, used when the word helps predict another word, and an output vector, used when the word is itself the target of prediction.

The two variants differ in the direction of prediction. In CBOW, for each occurrence of a target word w , the model uses the surrounding words within m positions on either side to predict w . It averages the input vectors of these surrounding words and uses the resulting vector, together with the output vectors, to predict the identity of the target word.

Skip-gram reverses this direction. For each occurrence of a central word w , the model predicts the surrounding words within m positions on either side. It uses the input vector of w , together with the output vectors, to predict each surrounding word separately. In both variants, training adjusts both sets of vectors to minimize the prediction loss.

After training, I represent each document by averaging the input vectors of the words it contains. I then concatenate this document vector with the numerical features and pass the combined vector

to the classification heads to predict litigation outcomes.

4.3 Frozen embeddings

The second main family of representations considered in this paper consists of frozen embeddings from a pretrained language model. Modern language models generally process text using tokens rather than words. A token may be a complete word, a subword, or a punctuation mark. For example, a tokenizer might split the word `running` into two tokens, `run` and `ning`. A tokenizer is the preprocessing component that performs this division and maps the resulting tokens into numerical identifiers.

For all results in this paper, I use the tokenizer and embedding model from Qwen3-Embedding-8B (Zhang et al., 2025). Unlike Word2Vec, which assigns a fixed vector to each word, a pretrained language model produces a contextual representation: the representation assigned to a token depends on the other tokens in the document. Following the pooling procedure recommended for Qwen3-Embedding, I append a special end-of-sequence token and use its final hidden representation as the document embedding. I then normalize this embedding to have unit Euclidean norm,

$$e_i = \frac{\tilde{e}_i}{\|\tilde{e}_i\|_2},$$

where $\tilde{e}_i$ is the representation of the final token.

The Qwen parameters are held fixed throughout downstream training, which is why I refer to these representations as *frozen embeddings*. The document embeddings are computed before training, and only the subsequent projection layers and classification head, when present, are estimated using the outcome labels.

My applications frequently involve several related documents. For example, predicting whether a patent claim is obvious may require information from both the patent and the relevant prior art. In addition to representing each document separately, it may therefore be useful to explicitly represent the relationship between them. I consider three methods for combining the resulting embeddings.

Simple concatenation. The simplest method concatenates all relevant text embeddings with the numerical features. Let $e_{i1}, \dots, e_{iK}$ denote the embeddings associated with observation i , and let z_i denote its numerical-feature vector. The combined representation is

$$r_i^{\text{concat}} = [e_{i1}; \dots; e_{iK}; z_i],$$

where semicolons denote concatenation. This representation is passed to one of the classification heads described in Section 4.1.

I also consider adding a trainable hidden layer,

$$h_i = \phi\left(W r_i^{\text{concat}} + b\right),$$

before the classification head, where $\phi(\cdot)$ is a nonlinear activation function. I consider hidden dimensions ranging from 32 to 128. This allows the model to learn a task-specific combination of the frozen embeddings and numerical features.

Interaction features. Simple concatenation does not explicitly describe how two documents relate to each other. I therefore construct interaction features for selected pairs of embeddings. For embeddings e_{ia} and e_{ib} , these features include

$$e_{ia} \odot e_{ib}, \quad |e_{ia} - e_{ib}|,$$

together with either cosine similarity alone or a set of five scalar measures: cosine similarity, mean absolute difference, Euclidean distance, maximum absolute difference, and mean element-wise product.

The element-wise product captures dimensions on which the two embeddings are jointly aligned, whereas the absolute difference captures dimensions on which they differ. Similar combinations have been widely used to represent relationships between pairs of texts (Conneau et al., 2017; Reimers and Gurevych, 2019). I write the full interaction representation as

$$r_{iab}^{\text{int}} = [e_{ia} \odot e_{ib}; |e_{ia} - e_{ib}|; s(e_{ia}, e_{ib})],$$

where $s(e_{ia}, e_{ib}) \in \mathbb{R}^5$ contains the five scalar measures listed above. Depending on the specification, all or a subset of these interaction features are concatenated with the original embeddings and numerical features before being passed directly to the classification head or through another hidden layer.

Projected fusion. Because the original embeddings are high-dimensional, I also consider mapping each frozen embedding into a lower-dimensional, task-specific representation. For text field k and sample i ,

$$\mathbf{u}_{ik} = \tanh(W_k \mathbf{e}_{ik} + \mathbf{b}_k), \quad \mathbf{p}_{ik} = \frac{\mathbf{u}_{ik}}{\|\mathbf{u}_{ik}\|_2}, \quad k = 1, \dots, K.$$

The Qwen embeddings $\mathbf{e}_{ik}$ remain frozen, while the projection parameters W_k and $\mathbf{b}_k$, together with the downstream classifier, are learned during training. Separate projections allow the model to retain information that is useful for each type of document while reducing the dimension of the combined representation. This form of intermediate feature fusion is commonly used to combine information from multiple sources (Ngiam et al., 2011).

The projected embeddings may be combined by simple concatenation or together with pairwise interaction features. The general interaction-fusion representation is

$$\mathbf{r}_i^{\text{fusion}} = \left[\mathbf{p}_{i1}; \dots; \mathbf{p}_{iK}; \bigoplus_{1 \leq k < \ell \leq K} g(\mathbf{p}_{ik}, \mathbf{p}_{i\ell}); \tilde{\mathbf{z}}_i \right],$$

where $\oplus$ denotes concatenation across all unordered pairs of text fields and

$$g(\mathbf{p}_{ik}, \mathbf{p}_{i\ell}) = [|\mathbf{p}_{ik} - \mathbf{p}_{i\ell}|; \mathbf{p}_{ik} \odot \mathbf{p}_{i\ell}; \cos(\mathbf{p}_{ik}, \mathbf{p}_{i\ell})].$$

The vector $\tilde{\mathbf{z}}_i$ denotes either the numerical-feature vector itself or a learned projection of that vector. When a specification contains only one text field, the pairwise-interaction block is empty. When explicit interactions are omitted, the representation reduces to concatenation of the projected embeddings and, where applicable, the numerical features. The dimensions of the projections and classifier layers are treated as specification-specific hyperparameters.

4.4 LoRA tuning

LoRA fine-tuning. The final family of models considered in this paper uses Low-Rank Adaptation (LoRA). It is useful first to distinguish fine-tuning from training a language model from scratch. A pretrained language model has already learned a general representation of language from a large text corpus. Fine-tuning continues training this pretrained model using task-specific labeled data, allowing its representations to adjust to the downstream prediction problem. In full fine-tuning, all parameters of the pretrained model are updated together with the classification head. For Qwen3-Embedding-8B, this would require updating approximately eight billion parameters.

This approach is computationally expensive and may be prone to overfitting in my setting because the number of labeled observations is small relative to the number of model parameters. I therefore use LoRA, which keeps the pretrained parameters fixed and learns only low-rank adjustments to selected weight matrices (Hu et al., 2022).

To illustrate, consider a pretrained weight matrix

$$W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}.$$

Full fine-tuning would update all

$$d_{\text{out}} d_{\text{in}}$$

entries of this matrix. LoRA instead writes the fine-tuned matrix as

$$W = W_0 + \frac{\alpha}{r} BA,$$

where

$$A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad B \in \mathbb{R}^{d_{\text{out}} \times r},$$

and

$$r \ll \min\{d_{\text{in}}, d_{\text{out}}\}$$

is the LoRA rank. The original matrix W_0 remains fixed, and only A and B are estimated. LoRA therefore trains

$$r(d_{\text{in}} + d_{\text{out}})$$

parameters in place of the original

$$d_{\text{out}}d_{\text{in}}$$

parameters.

For example, a 4096×4096 weight matrix contains

$$4096^2 = 16,777,216$$

parameters. With $r = 2$, the corresponding LoRA matrices contain only

$$2(4096 + 4096) = 16,384$$

trainable parameters, approximately 0.098% of the parameters in the original matrix. More generally, for a square $d \times d$ matrix, the fraction of trainable parameters is

$$\frac{2dr}{d^2} = \frac{2r}{d}.$$

The rank r controls both the number of trainable parameters and the flexibility of the adaptation. A larger rank reduces the scaling factor because r appears in the denominator, but also introduces more trainable parameters and increases the expressive capacity of the adaptation.

On the other hand, the parameter α controls the magnitude of the LoRA adjustment relative to the pretrained weights only and does not change the number of trainable parameters. I select both r and α using validation performance.

I use a separate LoRA adapter for each document field, such as the patent description, claim text, and prior-art text. The same adapter is shared across all documents belonging to a given field; individual documents do not receive separate sets of parameters. After obtaining a representation for each field, I combine them using one of the interaction specifications described in Section 4.3.

4.5 Evaluation metrics

Because the outcome classes are frequently unbalanced, no single metric provides a complete measure of predictive performance. I use three primary metrics for model evaluation: MCC, macro-F1, and accuracy, in that order of importance. I additionally report balanced accuracy, ROC-AUC, claim-weighted accuracy, and patent-weighted accuracy as supplementary diagnostics. These four supplementary metrics are reported for reference but are not used for epoch or model selection.

Let TP , TN , FP , and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

4.5.1 Primary metrics

Matthews correlation coefficient. MCC measures the correlation between the predicted and observed binary outcomes while incorporating all four cells of the confusion matrix:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}.$$

MCC equals 1 for perfect prediction, 0 when the predictions have no overall association with the observed outcomes, and -1 for perfect disagreement. Because it jointly accounts for performance on both classes, MCC is my principal evaluation metric.

Macro-F1. For class c , the F1 score is the harmonic mean of precision and recall and can be written as

$$\text{F1}_c = \frac{2TP_c}{2TP_c + FP_c + FN_c},$$

where TP_c , FP_c , and FN_c are computed by treating class c as the positive class. I report macro-F1,

$$\text{F1}_{\text{macro}} = \frac{1}{2} (\text{F1}_0 + \text{F1}_1),$$

which gives equal weight to predictive performance on the two outcome classes. Unless otherwise stated, “F1” in the remainder of the paper refers to macro-F1.

Macro-F1 is an important metric since just like MCC, it takes into account all four cells of the confusion matrix.

Accuracy. Accuracy is the fraction of observations assigned to the correct class:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Accuracy provides an intuitive measure of the overall proportion of correct predictions. However, it can be misleading when one class is substantially more frequent because a classifier can obtain relatively high accuracy by predominantly predicting the majority class. I therefore treat accuracy as the third primary metric, after MCC and macro-F1.

Epoch and model selection. During training, I use validation-set MCC to select the best epoch and, where specified, implement early stopping. For the insufficiency validation runs in Subsection 6.2.2, all 20 epochs are completed and the best-MCC checkpoint is selected afterward; external refits instead use the fixed stopping epochs stated there. Final model selection is based primarily on validation MCC and macro-F1. Accuracy is considered only as a secondary criterion when the leading models exhibit very similar performance on the first two metrics. This ordering reduces the influence of the majority class while retaining accuracy as an interpretable measure of the overall frequency of correct predictions.

4.5.2 Supplementary metrics

Precision for class 1. Precision measures how often the model is correct when it predicts that the owner loses:

$$\text{Precision}_1 = \frac{TP}{TP + FP}.$$

Recall for class 1. Recall measures how often the model successfully identifies cases in which the owner actually loses:

$$\text{Recall}_1 = \frac{TP}{TP + FN}.$$

Balanced accuracy. Balanced accuracy gives equal importance to accuracy within each of the two classes:

$$\text{Balanced Accuracy} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right).$$

The two terms are the true-positive and true-negative rates. Balanced accuracy therefore indicates whether the classifier performs comparably across the two classes, rather than obtaining its accuracy primarily from the majority class.

ROC-AUC. ROC-AUC measures how well a classifier ranks positive observations above negative observations across all possible classification thresholds. Let s_i denote the prediction score for observation i . ROC-AUC can be written as

$$\text{ROC-AUC} = P(s_i > s_j \mid y_i = 1, y_j = 0) + \frac{1}{2}P(s_i = s_j \mid y_i = 1, y_j = 0).$$

It is therefore the probability that a randomly selected positive observation receives a higher score than a randomly selected negative observation, with half credit assigned to ties. A value of 0.5 corresponds to random ranking, while a value of 1 corresponds to perfect ranking.

Group-weighted accuracy. The preceding metrics do not account for unequal numbers of observations across patents. Ordinary claim-level accuracy gives greater aggregate influence to patents for which more claims are observed. I therefore report patent-weighted accuracy, which assigns equal aggregate weight to each patent regardless of the number of challenged claims it contains. This adjustment is particularly important because claim outcomes within the same patent are highly correlated: correctly predicting one claim often coincides with correctly predicting other claims from the same patent. Without patent weighting, patents containing many observed claims may therefore exert disproportionate influence on the reported performance.

For novelty and obviousness, an additional issue arises. A single claim may face several distinct attacks, each based on a different prior-art unit. A prior-art unit may consist of one document or several documents assessed jointly, and each claim-prior-art-unit pair constitutes one attack-level observation. Ordinary attack-level accuracy therefore gives greater aggregate weight to claims facing

more attacks. I also report claim-weighted accuracy, which assigns equal aggregate weight to each claim while continuing to evaluate predictions at the attack level.

Let

$$c_i = \mathbf{1}\{\hat{y}_i = y_i\}$$

indicate whether observation i is classified correctly. Let $n_{g(i)}$ denote the number of observations associated with the same claim as observation i , and let $n_{p(i)}$ denote the total number of observations associated with the same patent and case. The two measures are

$$\text{Acc}_{\text{claim}} = \frac{\sum_i c_i / n_{g(i)}}{\sum_i 1 / n_{g(i)}}, \quad \text{Acc}_{\text{patent}} = \frac{\sum_i c_i / n_{p(i)}}{\sum_i 1 / n_{p(i)}}.$$

5 Analysis

I conduct my analysis using three information sets: ex ante, semi-ex-ante, and post-hearing. The ex-ante level is important for a patent owner because it indicates broadly which parts of their patent portfolio are weak and susceptible to attack. Since the owner also has to pay renewal costs, this information also enters into their calculation of which patents are worth retaining. For a trader, ex-ante prediction helps them trade on information available at the time of the first hearing.

The semi-ex-ante level helps the owner look at a deeper level: which claims are particularly susceptible to attack, and which amendments, if needed, could improve them.

Finally, at the post-hearing level, the owner would know which obviousness attacks may be dangerous and how to prepare for them. For the challenger, it indicates which attacks are likely to be successful. For the trader, it allows them to trade on post-hearing information by taking a position shortly before the judgment is issued.

At the ex-ante level, I use only information that an outsider could observe before the hearing. This includes the patent, the identities of the parties, and the identity of the judge. The unit of analysis at this stage is the patent rather than the individual claim. Although the complaint may identify the claims initially challenged, these may not be the claims ultimately decided by the court. A claim may be amended during the proceedings, including by combining it with parts of one or more other claims. The final judgment may therefore concern a claim that is different from the original claim. Assigning the final outcome to the original claim would give that claim an inaccurate outcome. For this reason, the ex-ante analysis predicts whether the patent as a whole will survive.

At the semi-ex-ante level, I move from the patent to the claim. I use the judgment to identify the claims that were ultimately decided and whether each claim survived. However, I restrict the predictors to information that was available before the hearing. In particular, I do not use the grounds actually raised against the claim, or the arguments and evidence presented during the hearing. The sample and the outcome are therefore identified ex post, while the information used to make the prediction remains ex ante.

This analysis is still useful for future prediction. The model can be trained using historical claims with observed outcomes and then applied to the claims of a new patent before a hearing.

The claim-level predictions can also be combined to measure the overall risk that the patent will be held invalid.

Some possible weaknesses can be approximated using information available before the hearing. For example, the claim can be compared with the patent description to examine whether the description provides enough information to carry out the claimed invention. This helps me to study how strong a claim is under the insufficiency attack. Novelty and obviousness are harder to assess at this stage because an outsider does not know which prior art the parties will rely on or how that prior art will be used in court. The semi-ex-ante analysis therefore asks whether claim outcomes can be predicted without knowing the grounds that were actually used in the case.

Finally, I consider the ex-post hearing information level, which corresponds to the ground-of-attack level in my analysis. I use the judgment only to identify the claims, grounds, and documents involved in the litigation. I then retrieve the underlying patent and prior-art materials and provide the complete materials to the model. This choice is different from what has been done in the literature.

PILOT-Bench begins with the written PTAB decision and uses an LLM to separate its sentences into appellant arguments, examiner findings, and the Board’s own opinion (Jang et al., 2025). Although the sentences classified as the Board’s opinion are excluded, the retained inputs are still selected and reconstructed from the written decision. FiNE-Patents conditions its analysis on the closest prior-art document identified by the examiner in the European Search Opinion (Knappich et al., 2026). In one specification, the final prediction is also supplied directly with the examiner’s feature-level passage references: the precise passages that the examiner identified as relevant.

The post-hearing model should therefore not be interpreted as observing everything that was available at the hearing. An analyst present in the courtroom might infer from the judge’s questions and reactions which evidence the judge considers important and which attacks or defenses are likely to be rejected. My model does not observe these signals. It can therefore be interpreted as a worst-case scenario for an analyst present at the hearing: the analyst observes the claims, grounds, and documents relied upon by the parties but learns nothing from the judge’s behavior about the likely outcome.

6 Results

6.1 Ex-ante level

Across 502 patent observations, patent owners lost on all challenged claims in 303 observations, representing 60.4% of the sample. All challenged claims survived in 163 observations, or 32.5%, while the remaining 36 observations, or 7.2%, had a partial outcome. A partial outcome means that at least one challenged claim survived while at least one was invalidated.

A follow-up check using ChatGPT indicated that every patent with a partial outcome remained in force following the dispute. For the binary analysis in this section, I therefore group partial outcomes with owner wins, recoding 0.5 as 0. Under this coding, class 0 denotes full or partial

survival, while class 1 denotes the loss of all challenged claims. The patent-level extraction prompt is reproduced in Section D.1; the separate follow-up query is not reproduced in the appendix.

The resulting binary sample contains 303 owner losses and 199 owner wins or survivals, representing 60.4% and 39.6% of the sample, respectively. Owner loss is the majority class, so a classifier that always predicts an owner loss would achieve an accuracy of 60.4%. I construct the training and validation sets using an 80:20 split conducted at the case level. All patents belonging to the same case are therefore assigned to the same set. Because cases contribute different numbers of patent observations, a case-level split does not automatically preserve the overall outcome distribution. I therefore make sure in the split the owner-loss rate in both the training and validation sets lies within five percentage points of the full-sample rate of 60.4%.

I begin with neural language models. For each encoder, I consider three combinations of textual inputs: the patent description alone, the complete claim set alone, and both the description and full claim set. When both fields are used, they are encoded separately before their representations are used for further analysis. The resulting textual representation is then combined with the numerical features reported in Table A.1. I evaluate both frozen-embedding architectures and LoRA-tuned architectures using the procedures described in the preceding section.

Some neural specifications collapse to the majority-class solution and predict an owner loss for every observation. Such a classifier achieves an accuracy of 60.4% but a balanced accuracy of only 50.0%, demonstrating that its apparently high raw accuracy contains no discriminatory information. Among the neural specifications that produce predictions for both classes, raw accuracy remains below the 60.4% majority-class benchmark.

There are two plausible explanations for this weak performance. First, 502 patent observations may be insufficient to estimate high-capacity neural models reliably, even when most model parameters are frozen or adaptation is restricted through LoRA. Second, the ex-ante task excludes information that is central to the legal determination. As shown in Section 3, obviousness is by far the most frequently raised ground of invalidity, followed by novelty. Both grounds depend critically on the relationship between the claimed invention and the relevant prior art. At the ex-ante stage, however, I do not know which prior-art references will subsequently be relied upon in litigation. Patent descriptions and claims alone may therefore contain only limited information about the eventual outcome. The weak performance of the neural models may consequently reflect both the small sample and the absence of this legally important comparative information.

I then turn to the classical models in Section 4.2. My shortlist comprises two classical models: a TF-IDF model and a Word2Vec model. The TF-IDF model represents the concatenated patent description and claims and uses an L_2 -regularized logistic regression classifier. The Word2Vec model uses representations of the description and claims with an L_1 -regularized linear SVM. As mentioned in Section 3.7, I also retain a version of each model without numerical features to assess how well it performs using textual information alone. These paired specifications allow me to assess the contribution of numerical features while holding the text representation, classifier, and hyperparameters fixed.

The table below reports performance on the original model-selection validation split. Precision and recall are reported for owner loss (class 1), which is the positive class throughout this binary exercise. On this split, the Word2Vec L_1 model with numerical features achieves the highest MCC, macro- F_1 , accuracy, and balanced accuracy. The text-only Word2Vec model achieves the highest precision, while the text-only TF-IDF model achieves the highest recall.

Table 2: Performance of the shortlisted classical models on the original model-selection validation split

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Precision (owner loss)	Recall (owner loss)
TF-IDF, without numerical features	0.655	0.612	0.601	0.213	0.595	0.609	0.764
TF-IDF, with numerical features	0.608	0.534	0.529	0.059	0.529	0.559	0.600
Word2Vec + linear SVM (L_1), without numerical features	0.673	0.660	0.658	0.316	0.658	0.679	0.691
Word2Vec + linear SVM (L_1), with numerical features	0.692	0.670	0.664	0.334	0.665	0.672	0.745

Precision and recall treat owner loss (class 1) as the positive class.

I next evaluate the shortlisted architectures through three exercises: Monte Carlo cross-validation within the training and validation sample; evaluation on a temporal holdout of Patents Court and IPEC cases decided after the training and validation period; and evaluation on an institutional holdout of UKIPO decisions, which tests generalization to a different institutional and procedural setting.

Monte Carlo cross-validation. I first assess the stability of the shortlisted architectures using Monte Carlo cross-validation. I randomly partition the sample at the case level 50 times, assigning approximately 80% of cases to training and 20% to validation in each split. For every split, I train each model anew and evaluate its performance on the corresponding validation set, while keeping the architecture and hyperparameters fixed. The results below summarize performance across these 50 splits.

Table 3: Median performance of the shortlisted classical models across 50 case-grouped validation splits

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Precision (owner loss)	Recall (owner loss)
	0.631	0.640	0.597	0.212	0.598	0.676	0.767
TF-IDF, without numerical features	(0.062)	(0.055)	(0.056)	(0.115)	(0.059)	(0.042)	(0.072)
TF-IDF, with numerical features	0.619	0.600	0.582	0.164	0.581	0.667	0.683
	(0.056)	(0.049)	(0.050)	(0.102)	(0.051)	(0.039)	(0.074)
Word2Vec + linear SVM (L_1), without numerical features	0.590	0.580	0.572	0.150	0.565	0.667	0.583
	(0.068)	(0.061)	(0.061)	(0.122)	(0.059)	(0.060)	(0.100)
Word2Vec + linear SVM (L_1), with numerical features	0.600	0.580	0.574	0.157	0.568	0.670	0.615
	(0.064)	(0.061)	(0.060)	(0.120)	(0.059)	(0.053)	(0.091)

Entries report medians; standard deviations across the same 50 splits are shown in parentheses below.

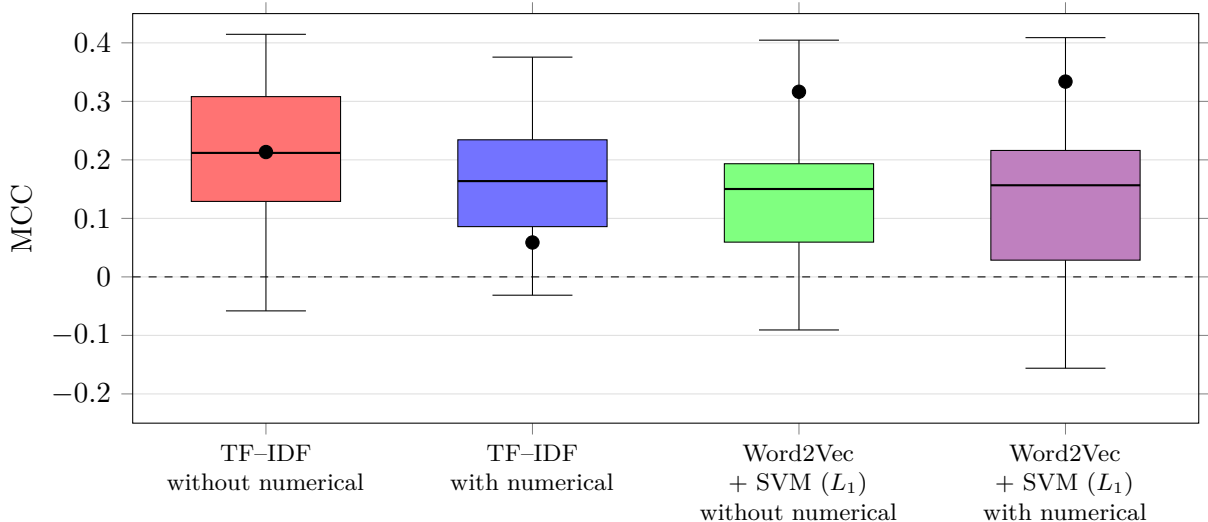


Figure 2: Distribution of MCC across 50 case-grouped validation splits. Boxes show the interquartile range, horizontal lines show medians, whiskers show the observed minimum and maximum, and black dots show performance on the original validation split.

External test performance. Using these four specifications, I fit each model using the full training sample and evaluate it without further model selection on three reported views of the held-out data. The court test set combines the Patents Court and the IPEC ($N = 59$); the institutional out-of-domain test set contains the two UKIPO samples ($N = 85$); and the overall test set combines all of these observations ($N = 144$). After binary coding, the court sample contains 28 owner wins/survivals and 31 owner losses, the UKIPO sample contains 45 wins/survivals and 40 losses, and the combined sample contains 73 wins/survivals and 71 losses. Table 4 reports the court results using the same seven performance metrics as the validation tables. The UKIPO and combined results appear in Tables C.1 and C.2 in Appendix C. I also evaluate ChatGPT 5.6 Sol as a zero-shot benchmark. The prompt is reproduced in Section E.1.

Table 4: External test performance: Patents Court and IPEC

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Precision (owner loss)	Recall (owner loss)
TF-IDF, without numerical features	0.584	0.627	0.624	0.250	0.624	0.636	0.677
TF-IDF, with numerical features	0.540	0.610	0.608	0.217	0.608	0.625	0.645
Word2Vec + linear SVM (L_1), without numerical features	0.616	0.610	0.608	0.217	0.608	0.625	0.645
Word2Vec + linear SVM (L_1), with numerical features	0.705	0.678	0.678	0.356	0.678	0.700	0.677
ChatGPT (GPT-5.6 Sol), zero-shot*	0.684	0.542	0.523	0.071	0.452	0.538	0.903

$N = 59$: 28 owner wins/survivals and 31 owner losses after binary coding.

Precision and recall treat owner loss as the positive outcome. Bold indicates the highest value in each column.

*Includes one manually obtained ChatGPT prediction reported by the user; all other predictions are from the API.

ChatGPT ROC-AUC uses 58 observations; the manual prediction has no recorded probability.

The ChatGPT prompt is reproduced in Section E.1.

The court sample produces the strongest external performance for the Word2Vec L_1 model with numerical features, with an MCC of 0.356 and macro- F_1 of 0.678, compared with 0.071 and 0.452 for GPT. Performance is weaker on the UKIPO sample: TF-IDF without numerical features achieves the highest MCC (0.093) and macro- F_1 (0.540), compared with 0.073 and 0.450 for GPT.

On the combined test set, the Word2Vec L_1 model with numerical features again has the highest MCC (0.183) and macro- F_1 (0.589), compared with 0.075 and 0.452 for GPT. Thus, the strongest supervised specification outperforms GPT on both measures in each reported test sample.

6.2 Semi ex-ante level

6.2.1 Claim level prediction without ground of attack

This subsection considers claim-level prediction without conditioning on a particular ground of invalidity. After restricting the sample to claims with a maximum dependency depth of two, the dataset contains 1,325 claim observations. Patent owners prevailed on 406 claims, representing 30.6% of the sample, and lost on the remaining 919 claims, representing 69.4%. A classifier that predicts an owner loss for every claim would therefore achieve an accuracy of 69.4%.

For the textual inputs, I consider four representations. The *literal claim* contains the disputed claim exactly as written. The *expanded claim* is the standalone representation constructed by incorporating the limitations inherited through every applicable dependency path, as described in Subsection 3.5. The *complete claim set* contains all claims in the patent, while the *description* contains the patent description. The numerical features are reported in Table A.1.

I evaluate different combinations of these inputs. I do not require the architecture to include the complete claim set or the description, but every specification must include either the literal or the expanded version of the disputed claim. Without either of these claim-specific inputs, all claims belonging to the same patent would receive essentially the same textual representation, and the exercise would reduce to the patent-level prediction problem considered in the ex-ante analysis. Depending on the specification, the selected textual representations are used alone or combined with one another and with the numerical features.

Across the classical, frozen-embedding, and LoRA-tuned specifications, no model improves raw accuracy by more than one percentage point relative to the 69.4% majority-class benchmark, and no model achieves an MCC above 0.2. I therefore find little evidence of reliable out-of-sample discrimination in the general claim-level task.

This result contrasts with the ex-ante patent-level analysis, in which the models exhibit modest but detectable predictive power. Although the claim-level models receive more granular information, they also face a more difficult prediction problem.

The contrast with the ex-ante results may partly reflect the difference between the two prediction targets. In the ex-ante analysis, partial patent-level outcomes are coded as owner wins. Consequently, the patent owner is classified as winning whenever at least one independent claim survives, even if other claims are invalidated. The task therefore asks whether the patent retains some valid independent protection, rather than which particular claims will survive. The patent description and complete claim set may contain information about the overall strength of the patent and may therefore be useful for this broader outcome.

The claim-level task requires a much finer distinction. Claims belonging to the same patent share the same description, complete claim set, and most numerical features, so these inputs provide little

information for distinguishing the outcomes of different claims within that patent. The literal and expanded representations supply claim-specific language, but that language alone may add limited information about whether a particular claim will fail. Claim validity often depends on the specific ground advanced, the relevant prior art, and the legal and technical arguments made against the claim, none of which is observed in this general claim-level exercise. The weak claim-level results are therefore consistent with the possibility that patent-wide text helps predict whether some protection will survive, while claim language alone is insufficient to identify precisely which claims will survive. This interpretation is suggestive rather than a separately identified explanation for the difference in performance.

6.2.2 Insufficiency attack

In this subsection, I predict the outcomes of insufficiency attacks. Unlike novelty and obviousness, which depend heavily on the relevant prior art, insufficiency is more amenable to prediction from the patent itself. An insufficiency attack succeeds when the specification does not disclose the invention clearly and completely enough for it to be performed by a person skilled in the art. Its outcome may therefore be predictable from the relationship between the language of the challenged claim and the corresponding patent description.

Restricting the modeling sample to claims with a maximum dependency depth of two yields 446 claim-level observations from 133 cases. Patent owners prevailed in 276 observations (61.9%) and lost in 170 observations (38.1%). Accordingly, an always-owner-win classifier would achieve an accuracy of 61.9% in the sample used for model training and evaluation. For reference, the unrestricted sample contains 626 observations from 136 cases, comprising 384 owner wins (61.3%) and 242 owner losses (38.7%); its corresponding majority-class accuracy is therefore 61.3%.

As in the preceding subsection, I consider four textual inputs: the literal claim, the expanded claim, the complete claim set, and the patent description. Every specification includes either the literal or expanded version of the challenged claim, while the complete claim set and description are included or excluded depending on the architecture.

The legal structure of insufficiency suggests that the description should be particularly important. The relevant question is not whether the claim appears strong in isolation, but whether the description provides sufficient information to perform the invention defined by that claim. I therefore expect the strongest architectures to compare the description with the challenged claim.

For an independent claim, the literal text may be enough on its own. For a dependent claim, however, the literal text omits the limitations inherited from its parent claims. Those limitations can be supplied either directly through the expanded claim (Subsection 3.5) or indirectly by providing both the literal claim and the complete claim set. This reasoning motivates three principal input combinations: the expanded claim and description; the literal claim, complete claim set, and description; and the expanded claim, complete claim set, and description.

The initial architecture search supports this intuition. The leading input combinations use either the expanded claim and description or the expanded claim, complete claim set, and description.

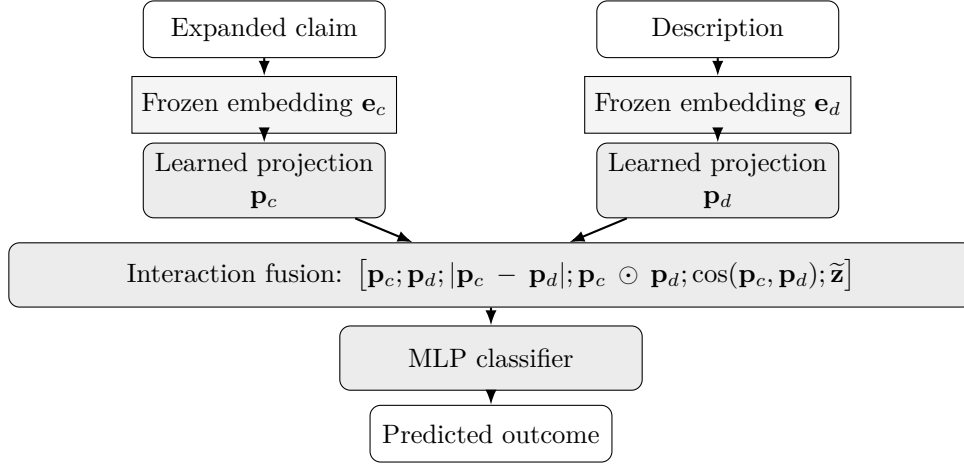
Specifications that exclude the description fail to match the 61.9% majority-class accuracy benchmark. Models using the literal claim and description exceed that benchmark, but perform substantially worse than the two leading expanded-claim families. These results suggest that the description contains essential information for predicting insufficiency and that representing the complete scope of a dependent claim provides additional predictive value.

Selected architectures. I retain two frozen-embedding architectures, denoted INS-A1 and INS-A2. INS-A1 uses exactly two textual inputs: the expanded claim and the patent description. The two texts are embedded separately, and each embedding is independently projected into a 32-dimensional space. The projected embeddings, their interaction features, and the numerical features are then concatenated. Specifically, these interactions comprise the element-wise absolute difference and product of the projected embeddings, together with their cosine similarity, as in the projected-fusion specification in Subsection 4.3. The resulting vector is passed to an MLP classifier.

INS-A2 uses three textual inputs: the expanded claim, the complete claim set, and the patent description. The three texts are embedded separately, after which their embeddings are concatenated directly with the numerical features and passed to an MLP classifier with dimension 128.

Both INS-A1 and INS-A2 are trained on 20 epochs with initial learning rate 0.0003 and 0.001, respectively. I also reported their non-numerical versions for comparison. Figure 3 illustrates the two architectures.

(a) Projected interaction fusion, illustrated using INS-A1



(b) Simple concatenation, illustrated using INS-A2

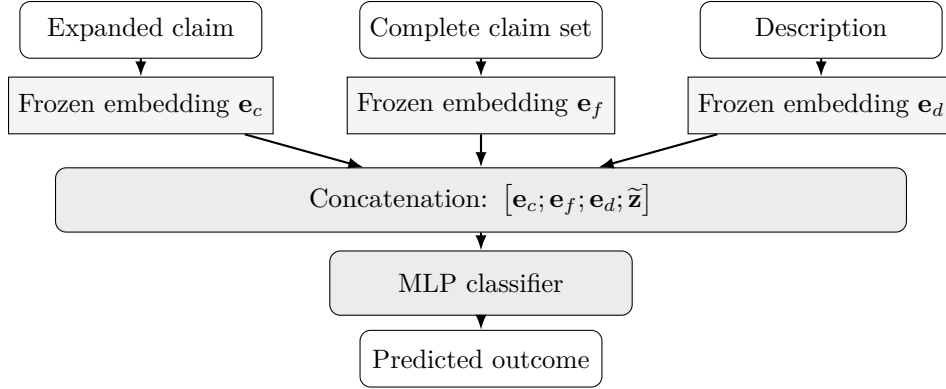


Figure 3: Frozen-embedding architecture families. Panel (a) illustrates projected interaction fusion using the two text fields in INS-A1. Each frozen embedding is mapped into a learned, normalized, task-specific representation before the projected embeddings and their interactions are passed to an MLP classifier. The same construction extends to specifications containing one or more than two text fields by including all available projected embeddings and all unordered pairs. Panel (b) illustrates the simple-concatenation architecture using INS-A2. Here the frozen embeddings are concatenated directly, without learned text projections or explicit interaction features. In both panels, $\tilde{\mathbf{z}}$ denotes either the numerical-feature vector or a learned projection of that vector. Projection dimensions, included text fields, and classifier dimensions vary across model specifications.

Sample weighting. To control the influence of patents contributing different numbers of challenged claims, let $n_{p(i)}$ denote the number of training observations associated with the patent containing observation i . A patent group is identified by the case and patent together. I write the observation weight and normalized training loss as

$$w_i = n_{p(i)}^{-\beta}, \quad \mathcal{L} = \frac{1}{n\bar{w}} \sum_{i=1}^n w_i \ell_i, \quad \bar{w} = \frac{1}{n} \sum_{i=1}^n w_i, \quad (1)$$

where ℓ_i is the cross-entropy loss. With $\beta = 0$, every observation receives the same weight. With $\beta = 1$, each observation receives weight $1/n_{p(i)}$, so the weights sum to one within each patent group and each patent has equal aggregate influence. I tested for $\beta = 0, 1$ and 2 random β between 0 and 1 and found that $\beta = 1$ had the best result. Hence my insufficiency specification uses $\beta = 1$.

Original validation performance. The table below reports performance on the original model-selection validation split. Precision and recall are reported for owner loss (class 1), which is the positive class throughout this binary exercise. Both Ins-A1 and Ins-A2 numerical and nonnumerical versions reported good performance with MCC above 0.6, Macro-F1 close or above 0.8, and accuracy all above 0.8.

Table 5: Insufficiency: performance on the original depth- ≤ 2 validation split

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
INS-A1, text only	0.858	0.812	0.806	0.605	0.802	0.816	0.735	0.781
INS-A1, text + numerical	0.840	0.824	0.772	0.632	0.790	0.838	0.947	0.562
INS-A2, text only	0.830	0.835	0.794	0.650	0.810	0.830	0.909	0.625
INS-A2, text + numerical	0.848	0.835	0.831	0.654	0.827	0.847	0.765	0.812

INS-A1: expanded claim and description. INS-A2: expanded claim, complete claim set, and description. All metrics use the checkpoint with the highest validation MCC within each run; ties select the earliest epoch. Precision and recall refer to owner loss on the insufficiency ground. Bold indicates column maxima.

I next evaluate the shortlisted architectures through three exercises: Monte Carlo cross-validation within the training and validation sample; evaluation on a temporal holdout of Patents Court and IPEC cases decided after the training and validation period; and evaluation on an institutional holdout of UKIPO decisions, which tests generalization to a different institutional and procedural setting.

Monte Carlo cross-validation. I first assess the stability of the shortlisted architectures using Monte Carlo cross-validation under two dependency-depth regimes: depth ≤ 2 and depth ≤ 100 . For each regime, I construct 25 random case-level splits, assigning approximately 80% of cases to training and 20% to validation in each split. For every split, I train each model anew and evaluate its performance on the corresponding validation set, while keeping the architecture and hyperparameters fixed. The results below summarize performance separately across the 25 splits within each depth regime.

Table 6 reports the depth- ≤ 2 results in Panel A and the full-depth results in Panel B.

Table 6: Insufficiency: median performance across validation splits

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 (25 splits)								
INS-A1, text only	0.711 (0.108)	0.695 (0.081)	0.647 (0.101)	0.349 (0.213)	0.645 (0.127)	0.713 (0.070)	0.667 (0.242)	0.415 (0.226)
INS-A1, text + numerical	0.708 (0.111)	0.663 (0.090)	0.606 (0.105)	0.305 (0.236)	0.590 (0.127)	0.715 (0.075)	0.621 (0.225)	0.310 (0.205)
INS-A2, text only	0.697 (0.119)	0.663 (0.084)	0.645 (0.090)	0.315 (0.198)	0.643 (0.111)	0.730 (0.064)	0.647 (0.224)	0.412 (0.192)
INS-A2, text + numerical	0.693 (0.117)	0.663 (0.074)	0.643 (0.089)	0.315 (0.195)	0.636 (0.115)	0.735 (0.071)	0.615 (0.256)	0.367 (0.202)
Panel B: Full depth (25 splits)								
INS-A1, text only	0.688 (0.079)	0.723 (0.067)	0.667 (0.075)	0.420 (0.159)	0.672 (0.087)	0.728 (0.058)	0.682 (0.154)	0.481 (0.183)
INS-A1, text + numerical	0.713 (0.080)	0.718 (0.067)	0.669 (0.078)	0.380 (0.171)	0.675 (0.094)	0.711 (0.052)	0.700 (0.174)	0.462 (0.173)
INS-A2, text only	0.707 (0.075)	0.706 (0.069)	0.683 (0.083)	0.385 (0.170)	0.678 (0.101)	0.729 (0.056)	0.667 (0.152)	0.521 (0.189)
INS-A2, text + numerical	0.723 (0.073)	0.732 (0.065)	0.690 (0.079)	0.414 (0.159)	0.700 (0.091)	0.718 (0.056)	0.680 (0.155)	0.500 (0.182)

INS-A1: expanded claim and description. INS-A2: expanded claim, complete claim set, and description. Entries report medians; sample standard deviations across the 25 splits are shown in parentheses. All metrics use the checkpoint with the highest validation MCC within each run; ties select the earliest epoch. Precision and recall refer to owner loss on the insufficiency ground. Bold indicates column maxima within each panel.

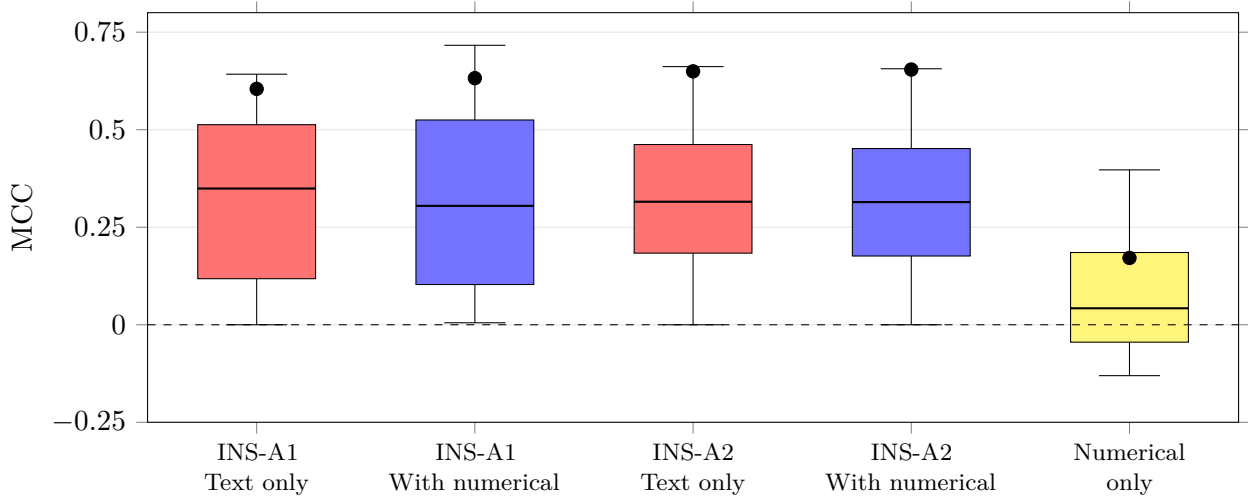


Figure 4: Insufficiency: distribution of MCC across 25 depth- ≤ 2 case-grouped validation splits. Boxes show the interquartile range, horizontal lines show medians, and whiskers show the observed minimum and maximum. Black dots show the corresponding original validation split. Each run uses its best-validation-MCC checkpoint. Red denotes text without numerical features, blue denotes text with numerical features, and yellow denotes numerical features alone.

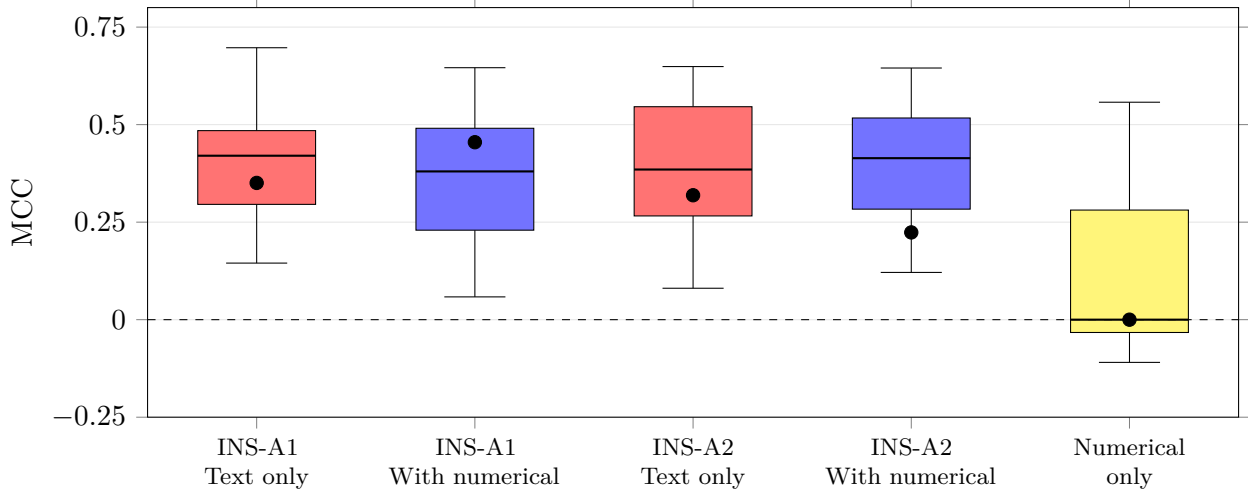


Figure 5: Insufficiency: distribution of MCC across 25 full-depth (≤ 100) case-grouped validation splits. Boxes show the interquartile range, horizontal lines show medians, and whiskers show the observed minimum and maximum. Black dots show the corresponding original validation split. Each run uses its best-validation-MCC checkpoint. Red denotes text without numerical features, blue denotes text with numerical features, and yellow denotes numerical features alone.

Under the depth- ≤ 2 regime, median MCC ranges from 0.305 to 0.349 among the text-based specifications. Under the full-depth regime, it ranges from 0.380 to 0.420. Adding numerical features does not consistently improve performance: its effect depends on the architecture and depth regime. It also shows that counterintuitively, increasing depth does not reduce prediction power.

External test performance. Table 7 reports the court results, with depth ≤ 2 in Panel A and full depth in Panel B. The corresponding UKIPO and combined results are reported in Tables C.3 and C.4 in Appendix C.

I refit INS-A1 and INS-A2, each with and without numerical features, using the entire training and validation sample for the relevant depth regime. The learning-rate scheduler retains a horizon of 20 epochs. Training stops at epoch 7 for INS-A1 and epoch 9 for INS-A2, and the corresponding checkpoint is used for test prediction.

At depth ≤ 2 , the court test sample contains 46 claim observations, of which patent owners won 25 (54.3%). The UKIPO test sample contains 37 observations, with 23 owner wins (62.2%). The combined sample therefore contains 83 observations, with 48 owner wins (57.8%).

At depth ≤ 100 , the court test sample contains 64 claim observations, of which patent owners won 35 (54.7%). The UKIPO test sample contains 60 observations, with 34 owner wins (56.7%). The combined sample therefore contains 124 observations, with 69 owner wins (55.6%).

I also evaluate ChatGPT 5.6 Sol as a zero-shot benchmark. The prompt is reproduced in Section E.2.

Table 7: Insufficiency: external-test performance on court observations

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 46$)								
INS-A1, text only	0.644	0.500	0.502	0.004	0.500	0.552	0.458	0.524
INS-A1, text + numerical	0.669	0.565	0.570	0.139	0.565	0.601	0.520	0.619
INS-A2, text only	0.657	0.543	0.550	0.100	0.543	0.563	0.500	0.619
INS-A2, text + numerical	0.705	0.630	0.630	0.258	0.629	0.688	0.591	0.619
GPT-5.6 Sol, zero-shot	0.704	0.587	0.559	0.155	0.522	0.642	0.625	0.238
Panel B: Full depth ($n = 64$)								
INS-A1, text only	0.720	0.609	0.622	0.251	0.607	0.608	0.550	0.759
INS-A1, text + numerical	0.705	0.625	0.631	0.261	0.625	0.616	0.571	0.690
INS-A2, text only	0.708	0.625	0.633	0.269	0.625	0.620	0.568	0.724
INS-A2, text + numerical	0.743	0.656	0.662	0.324	0.656	0.640	0.600	0.724
GPT-5.6 Sol, zero-shot	0.611	0.516	0.489	-0.026	0.457	0.631	0.429	0.207

INS-A1: expanded claim and description. INS-A2: expanded claim, complete claim set, and description. Supervised models are refitted on all training and validation data at the matching depth, using a 20-epoch schedule and stopping at epoch 7 (INS-A1) or 9 (INS-A2). GPT is zero-shot; its predictions are subsetting by depth. Precision and recall refer to owner loss on the insufficiency ground; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel. The GPT prompt is reproduced in Section E.2.

On the court holdout, INS-A2 with numerical features achieves the highest MCC at both depths: 0.258 at depth ≤ 2 and 0.324 at full depth, compared with 0.155 and -0.026 for GPT. Its macro- F_1 scores are 0.629 and 0.656, compared with 0.522 and 0.457 for GPT. On the combined sample, the same specification attains MCCs of 0.206 and 0.197, compared with 0.095 and -0.033 for GPT.

Generalization to UKIPO is weaker. At depth ≤ 2 , all four supervised specifications predict that every claim survives the insufficiency challenge, producing an MCC of zero. At full depth, INS-A1 with numerical features attains an MCC of 0.149, but identifies only 3.8% of owner losses. GPT also has low owner-loss recall, with no detected losses in either UKIPO depth subset.

Overall, my architectures perform well within the same regime, but when the regime changes, it could not do as well. This is similar to the test result for patent level prediction. Further, it appears that increasing depth level does not affect predictive power. Finally, the inclusion of numerical feature greatly increases the performance.

Separate versus combined encoding. I next examine whether the texts should be encoded separately or concatenated before encoding. I consider three configurations: the literal claim, complete claim set, and description (INS-X1); the expanded claim, complete claim set, and description (INS-X2); and the expanded claim and description (INS-X3). For each configuration, I concatenate the selected texts and encode them jointly using a frozen Qwen-8B encoder. The resulting single embedding is combined with numerical features and passed to an MLP classification head with the same design as INS-A2. The encoder is not fine-tuned. I use equal patent weights ($\beta = 1$) and a 20-epoch learning-rate schedule, stopping training at epoch 9.

Table 8 compares these configurations with INS-A2 using separate encoding and numerical features. Panel A reports results when both training and evaluation are restricted to depth ≤ 2 ; Panel B reports the corresponding full-depth results. INS-X2 and INS-A2 use the same three textual sources, providing the most direct comparison of the two encoding strategies among these

specifications.

Table 8: Insufficiency: separate versus combined encoding on court observations

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 46$)								
INS-A2, separate	0.705	0.630	0.630	0.258	0.629	0.688	0.591	0.619
INS-X1, combined	0.625	0.500	0.483	-0.037	0.470	0.528	0.429	0.286
INS-X2, combined	0.617	0.478	0.463	-0.079	0.452	0.507	0.400	0.286
INS-X3, combined	0.632	0.500	0.479	-0.048	0.457	0.535	0.417	0.238
Panel B: Full depth ($n = 64$)								
INS-A2, separate	0.743	0.656	0.662	0.324	0.656	0.640	0.600	0.724
INS-X1, combined	0.726	0.594	0.569	0.163	0.550	0.568	0.600	0.310
INS-X2, combined	0.700	0.594	0.569	0.163	0.550	0.568	0.600	0.310
INS-X3, combined	0.716	0.609	0.587	0.199	0.572	0.581	0.625	0.345

INS-A2 encodes the expanded claim, complete claim set, and description separately. INS-X1 concatenates the literal claim, complete claim set, and description; INS-X2 concatenates the expanded claim, complete claim set, and description; INS-X3 concatenates the expanded claim and description. Each INS-X specification uses one frozen Qwen-8B encoding of the concatenated text and the same MLP classification-head design as INS-A2. All specifications include numerical features, use equal patent weights ($\beta = 1$), and use a 20-epoch learning-rate schedule with training stopped at epoch 9. Training and evaluation depths match within each panel. The classification threshold is 0.5. The INS-A2 results are reproduced from Table 7. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

On the court test set, all three combined-encoding configurations have lower MCC and macro- F_1 than INS-A2 at both depths. At depth ≤ 2 , their MCCs range from -0.079 to -0.037 , and their macro- F_1 scores range from 0.452 to 0.470, compared with 0.258 and 0.629 for separate encoding. At full depth, combined encoding achieves positive MCCs of 0.163–0.199 and macro- F_1 scores of 0.550–0.572, compared with 0.324 and 0.656 for INS-A2. In the comparison using the same textual sources, INS-X2 achieves an MCC of 0.163 and macro- F_1 of 0.550 at full depth. These results favor separate encoding in this setting, while the positive full-depth MCCs show that combined encoding retains predictive information.

Tables C.7 and C.8 in Appendix C report the UKIPO and combined-sample results. All three combined-encoding configurations have negative MCCs on both samples at both depths.

Overall, my results suggest that encoding the texts separately performs better than encoding them together for predicting insufficiency outcomes. One possible explanation is that the claim text provides information that is as important as the description. When the texts are encoded together, the much longer description may dominate the resulting representation, making it harder for the model to retain information specific to the claim.

6.3 Ex-post level

6.4 Obviousness

In this section, the unit of analysis is an obviousness attack against a claim. Each attack may rely on a single prior-art reference or a combination of prior art references. Within a claim, references belonging to the same attack are grouped into one observation. A claim may therefore generate multiple observations when it faces distinct attacks, and the outcome may differ across those attacks.

I exclude relying on more than three prior-art documents since Qwen model only allows up to approximately 32k tokens. This only reduces my sample size by 2 observations.

After this exclusion, restricting the sample to claims with a maximum dependency depth of two yields 1,342 attack-level observations, representing 853 distinct claims from 259 cases. Patent owners prevailed in 747 observations (55.7%) and lost in 595 observations (44.3%). An always-owner-win classifier would therefore achieve an accuracy of 55.7%.

The full-depth sample contains 1,789 attack-level observations, representing 1,157 distinct claims from 262 cases. Patent owners prevailed in 954 observations (53.3%) and lost in 835 observations (46.7%), giving a corresponding majority-class accuracy of 53.3%. The number of cases is lower than the 318 obviousness cases reported in Table 1 because the full text of some prior-art references could not be located.

For the obviousness analysis, I consider five potential textual inputs: the literal claim, the expanded claim, the complete claim set, the patent description, and the prior art. Every specification includes either the literal or expanded version of the challenged claim, while the inclusion of the other textual inputs varies across architectures.

The legal structure of obviousness suggests that the prior art should be present in any well-specified architecture. A claim is obvious if the claimed invention would have been obvious to a skilled person in light of the prior art. I therefore expect the strongest architectures to compare the challenged claim with the relevant prior art.

The patent description may also be informative because some claim terms can be understood only by reference to the description. Its inclusion creates a trade-off, however. For claims that can be understood without such reference, the lengthy description may add noise and increase the risk of overfitting.

Finally, as in the previous section, I do not expect the literal text of a dependent claim to be sufficient without the complete claim set. I therefore consider three representations of the challenged claim: the expanded claim; the literal claim together with the complete claim set; and the expanded claim together with the complete claim set.

The results are consistent with this intuition. The two best-performing architectures use, respectively, the literal claim and the complete claim set together with the prior art, and the expanded claim and the complete claim set together with the prior art. Models that use the literal claim and prior art without the complete claim set, as well as architectures that exclude the prior art, perform only slightly better than the majority-class baseline.

Selected architectures. In both selected architectures, the prior-art encoder is frozen, while the encoders for the challenged claim and the complete claim set are LoRA-tuned. The LoRA adaptation is mild, with $r = 2$ and $\alpha = 4$. The resulting representation of either the literal or expanded claim is concatenated with the LoRA-tuned representation of the complete claim set, the frozen prior-art embedding, and the numerical features. This combined representation is then passed directly to a linear classification head, without explicit interaction features or an additional

MLP layer.

I use the normalized patent-size weighting scheme in Equation 1. Whereas the insufficiency specification uses $\beta = 1$, the selected obviousness specification uses $\beta = 1/16$, a milder adjustment for the number of observations contributed by each patent. This choice was not arbitrary but through a tuning of β from 0 to 0.5 with grid of $1/32$.

The OBV-A1 model LoRA-tuned the literal claim and the complete claim set while the OBV-A2 model LoRA-tuned the expanded claim and the complete claim set.

Table 9: Performance of the shortlisted models on the original validation split

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy
OBV-A1, without numerical features	0.596	0.579	0.537	0.141	0.455	0.508	0.555
OBV-A1, with numerical features	0.695	0.725	0.719	0.442	0.720	0.733	0.639
OBV-A2, without numerical features	0.593	0.579	0.536	0.145	0.450	0.508	0.556
OBV-A2, with numerical features	0.640	0.681	0.687	0.374	0.681	0.693	0.573

Monte Carlo cross-validation. Again, we do 50 random splits across depth 2 and 50 across full depth. Table 10 reports the depth- ≤ 2 results in Panel A and the full-depth results in Panel B.

Table 10: Obviousness: median performance across validation splits

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy
Panel A: Depth ≤ 2 (50 splits)							
OBV-A1, with numerical features	0.622	0.611	0.605	0.245	0.600	0.604	0.600
OBV-A1, without numerical features	0.571	0.559	0.510	0.075	0.383	0.507	0.564
OBV-A2, with numerical features	0.630	0.623	0.623	0.260	0.617	0.609	0.591
OBV-A2, without numerical features	0.566	0.564	0.519	0.076	0.425	0.514	0.550
Panel B: Full depth (50 splits)							
OBV-A1, with numerical features	0.584	0.571	0.582	0.169	0.568	0.587	0.572
OBV-A2, with numerical features	0.585	0.573	0.577	0.181	0.565	0.603	0.576

Bold indicates column maxima within each panel.

As I observe, without numerical features, model performance drops significantly. Further, unlike in the insufficiency analysis, performance also decreases when I move from depth- ≤ 2 validation to full-depth validation.

External test performance. Table 11 reports the court results, with depth ≤ 2 in Panel A and full depth in Panel B. The corresponding UKIPO and combined results are reported in Tables C.5 and C.6 in Appendix C.

I refit OBV-A1 and OBV-A2, each with and without numerical features, using the entire training and validation sample for the relevant depth regime. The learning-rate scheduler retains a horizon of 6 epochs. Training stops at epoch 2 for both configurations, and the corresponding checkpoint is used for test prediction.

I then evaluate these models on the court and UKIPO test samples. For each depth regime, I report results from models trained at that same depth. Each observation represents an obviousness attack against a claim, potentially relying on several prior-art documents.

At depth ≤ 2 , the court test sample contains 228 attack-level observations, of which patent owners won 154 (67.5%). The UKIPO test sample contains 232 observations, with 189 owner wins (81.5%). The combined sample therefore contains 460 observations, with 343 owner wins (74.6%).

At depth ≤ 100 , the court test sample contains 306 attack-level observations, of which patent owners won 180 (58.8%). The UKIPO test sample contains 379 observations, with 324 owner wins (85.5%). The combined sample therefore contains 685 observations, with 504 owner wins (73.6%). These counts exclude attacks involving more than three prior-art documents. This restriction removes 24 attack-level observations from the full-depth external test sample, all from UKIPO; 23 of these also belong to the depth- ≤ 2 sample. Overall, this implies a massive difference between the two regimes. At UKIPO, the owner won with overwhelming probability.

I also evaluate GPT-5.6 Sol as a zero-shot benchmark. The prompt is reproduced in Section E.3.

Table 11: Obviousness: external-test performance on court observations

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 228$)									
OBV-A1, text only	0.415	0.675	0.500	0.000	0.403	0.677	0.711	0.000	0.000
OBV-A1, text + numerical	0.647	0.614	0.535	0.075	0.535	0.610	0.698	0.383	0.311
OBV-A2, text only	0.417	0.675	0.500	0.000	0.403	0.677	0.711	0.000	0.000
OBV-A2, text + numerical	0.654	0.636	0.559	0.125	0.559	0.642	0.709	0.424	0.338
GPT-5.6 Sol, zero-shot	0.676	0.500	0.598	0.211	0.498	0.499	0.448	0.382	0.878
Panel B: Full depth ($n = 306$)									
OBV-A1, text only	0.442	0.582	0.494	-0.068	0.368	0.577	0.668	0.000	0.000
OBV-A1, text + numerical	0.738	0.618	0.654	0.323	0.615	0.611	0.615	0.522	0.857
OBV-A2, text only	0.452	0.585	0.497	-0.048	0.369	0.580	0.678	0.000	0.000
OBV-A2, text + numerical	0.738	0.588	0.624	0.261	0.585	0.570	0.586	0.500	0.825
GPT-5.6 Sol, zero-shot	0.690	0.578	0.627	0.291	0.566	0.580	0.471	0.494	0.905

OBV-A1: literal claim, complete claim set, and prior-art material. OBV-A2: expanded claim, complete claim set, and prior-art material. Supervised models are trained at the same depth as the evaluation sample. GPT is zero-shot; its predictions are subsetting by depth. Metrics are computed at the attack level. Patent-weighted accuracy is the mean of within-patent attack accuracies, with patents grouped within proceedings. Precision and recall refer to owner loss on the obviousness attack; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel at the reported precision. The GPT prompt is reproduced in Section E.3.

On the court holdout at depth ≤ 2 , GPT achieves the highest MCC (0.211), while OBV-A2 with numerical features achieves the highest macro- F_1 (0.560), compared with 0.497 for GPT. Among the supervised models, OBV-A2 with numerical features performs best on both measures, with an MCC of 0.125 and macro- F_1 of 0.560, compared with 0.075 and 0.535 for OBV-A1 with numerical features. At full depth, OBV-A1 with numerical features achieves the highest MCC (0.323) and macro- F_1 (0.615). OBV-A2 with numerical features attains an MCC of 0.261 and macro- F_1 of 0.585, compared with 0.291 and 0.566 for GPT.

Generalization to UKIPO is weaker for the supervised models. All four specifications have negative MCCs at both depths. GPT has positive MCCs of 0.195 at depth ≤ 2 and 0.119 at full depth, with corresponding macro- F_1 scores of 0.403 and 0.324. Its high owner-loss recall comes with low precision: at full depth, it detects 90.9% of owner losses, but only 16.7% of its predicted losses are correct. The strong imbalance toward owner wins also makes raw accuracy difficult to interpret: an always-owner-win classifier achieves 81.5% accuracy at depth ≤ 2 and 85.5% at full depth on UKIPO.

On the combined sample, GPT achieves the highest MCC at both depths, with scores of 0.199 and 0.188. Its corresponding macro- F_1 scores are 0.452 and 0.439. At depth ≤ 2 , OBV-A1 with numerical features has the highest macro- F_1 (0.456), slightly exceeding GPT despite its negative MCC. At full depth, GPT leads on both measures: OBV-A1 and OBV-A2 with numerical features attain MCCs of 0.110 and 0.080 and macro- F_1 scores of 0.425 and 0.410, respectively.

Overall, numerical features improve both MCC and macro- F_1 on the court holdout for both architectures and depth regimes. The text-only models predict almost exclusively owner wins and detect no owner losses in any test subset. Their relatively high accuracy therefore reflects the prevalence of owner wins.

As in the previous sections, predictive performance is stronger on the court test set than on the UKIPO test set. With numerical features, both architectures achieve positive MCCs on the court holdout, but neither achieves a positive MCC on UKIPO at either depth.

Finally, performance does not deteriorate when deeper claims are included. On the court holdout, both architectures with numerical features achieve higher MCC and macro- F_1 in the full-depth setting than at depth ≤ 2 .

Separate versus combined encoding. I also examine combined encoding for obviousness. OBV-X1 combines the literal claim, complete claim set, and prior-art material, while OBV-X2 replaces the literal claim with the expanded claim. Both specifications include numerical features. Table 12 compares these specifications with OBV-A1 and OBV-A2 on the court holdout. Tables C.9 and C.10 in Appendix C report the UKIPO and combined-sample results.

Table 12: Obviousness: separate versus combined encoding on court observations

Model	Training depth	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 228$)										
OBV-A1, separate	2	0.647	0.614	0.535	0.075	0.535	0.610	0.698	0.383	0.311
OBV-A2, separate	2	0.654	0.636	0.559	0.125	0.559	0.642	0.709	0.424	0.338
OBV-X1, combined	2	0.229	0.412	0.347	-0.296	0.351	0.397	0.444	0.143	0.162
OBV-X2, combined	2	0.226	0.412	0.347	-0.296	0.351	0.397	0.444	0.143	0.162
Panel B: Full depth ($n = 306$)										
OBV-A1, separate	100	0.738	0.618	0.654	0.323	0.615	0.611	0.615	0.522	0.857
OBV-A2, separate	100	0.738	0.588	0.624	0.261	0.585	0.570	0.586	0.500	0.825
OBV-X1, combined	100	0.198	0.343	0.315	-0.375	0.312	0.303	0.369	0.174	0.159
OBV-X2, combined	100	0.197	0.340	0.313	-0.380	0.310	0.300	0.366	0.172	0.159

OBV-X1 combines the literal claim, complete claim set, and prior-art material; OBV-X2 uses the expanded claim in place of the literal claim. All four specifications include numerical features. The separate-encoding models are trained at the evaluation depth and use patent-size weighting with $\beta = 1/16$. All models in Panel A are trained and evaluated at depth ≤ 2 ; all models in Panel B are trained and evaluated at full depth (≤ 100). Both combined-encoding models use uniform loss weights and stop at epoch 2. Training and evaluation depths therefore match within each panel, although the separate- and combined-encoding specifications differ in loss weighting. The OBV-A1 and OBV-A2 results are reproduced from Table 11. Metrics are computed at the attack level. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

On the court holdout at depth ≤ 2 , both OBV-X1 and OBV-X2 attain an MCC of -0.296 and a macro- F_1 score of 0.351. At full depth, their MCCs are -0.375 and -0.380 , with macro- F_1 scores of 0.312 and 0.310, respectively. Both combined-encoding specifications therefore perform worse than the separate-encoding models on these measures when training and evaluation depths are matched.

These results favour the separate-encoding specifications tested here. The intuition is the same as in the insufficiency section: When the texts are encoded together, the much longer description may dominate the resulting representation, making it harder for the model to retain information specific to the claim.

6.4.1 Novelty

As in the obviousness analysis, I define observations at the attack level, with each claim–prior-art-unit pair constituting a separate observation. Restricting the novelty sample to claims with a maximum dependency depth of two yields 637 attack-level observations, representing 450 distinct claims from 165 cases. Patent owners prevailed in 451 observations (70.8%) and lost in 186 observations (29.2%). Accordingly, an always-owner-win classifier would achieve an accuracy of 70.8% in the sample used for model training and evaluation. For reference, the unrestricted sample contains 863 attack-level observations, representing 625 distinct claims from 167 cases. Patent owners prevailed in 613 observations (71.0%) and lost in 250 observations (29.0%), giving a corresponding majority-class accuracy of 71.0%. The number of cases is substantially lower than the 207 novelty cases reported in Table 1 because the full text of many cited prior-art references, such as product catalogues, could not be located.

As in the obviousness analysis, I consider five possible textual inputs: the literal claim, the expanded claim, the complete claim set, the patent description, and the prior art. Every model specification includes either the literal or the expanded version of the challenged claim. The complete claim set, patent description, and prior-art text are then included or excluded depending on the architecture.

I expected the prior-art text to be the most important additional input. Conditional on its inclusion, I expected one of three claim representations to be sufficient: the expanded claim alone; the literal claim together with the complete claim set; or the expanded claim together with the complete claim set. I did not expect the patent description to be necessary.

Selected architectures The results do not fully conform to these expectations. The three best-performing models all use frozen text encoders. In descending order of validation performance in MCC, their inputs are: prior art, the literal claim, and the patent description (A1); prior art, the literal claim, and the complete claim set (A2); and prior art, the expanded claim, and the patent description (A3). Although all three architectures include the prior art, A1 performs best using the literal claim without support from the complete claim set.

A1 and A2 use an architecture analogous to INS-A1 in the insufficiency analysis (Subsection 6.2.2), except that they combine three embeddings rather than two. A3 instead uses simple embedding concatenation, as in INS-A2 of the insufficiency analysis, but without an additional MLP layer.

Table 13: Performance of the shortlisted novelty models on the original validation split

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy
A1, without numerical features	0.696	0.720	0.686	0.359	0.678	0.710	0.667
A1, with numerical features	0.719	0.810	0.750	0.529	0.762	0.801	0.779
A2, without numerical features	0.712	0.730	0.655	0.328	0.661	0.716	0.676
A2, with numerical features	0.721	0.800	0.724	0.495	0.740	0.790	0.733
A3, without numerical features	0.645	0.320	0.495	-0.018	0.278	0.358	0.363
A3, with numerical features	0.650	0.780	0.738	0.476	0.738	0.767	0.667

Monte Carlo cross-validation Despite their strong performance on the initial validation set, the selected models fail to outperform the majority-class baseline when evaluated across 50 random splits of either the depth-restricted sample or the unrestricted full-depth sample.

Table 14 reports the depth- ≤ 2 results in Panel A and the full-depth results in Panel B.

Table 14: Novelty: median performance across validation splits

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy
Panel A: Depth ≤ 2 (50 splits)							
A1, without numerical features	0.569	0.630	0.517	0.042	0.493	0.613	0.653
A1, with numerical features	0.588	0.646	0.522	0.064	0.508	0.621	0.651
A2, without numerical features	0.591	0.650	0.523	0.067	0.513	0.638	0.648
A2, with numerical features	0.589	0.642	0.516	0.049	0.503	0.627	0.667
A3, without numerical features	0.577	0.366	0.504	0.014	0.353	0.383	0.409
A3, with numerical features	0.623	0.591	0.575	0.148	0.552	0.582	0.573
Panel B: Full depth (50 splits)							
A1, without numerical features	0.558	0.675	0.507	0.019	0.478	0.665	0.665
A1, with numerical features	0.571	0.687	0.517	0.058	0.496	0.673	0.667
A2, without numerical features	0.569	0.682	0.522	0.059	0.506	0.674	0.660
A2, with numerical features	0.573	0.684	0.533	0.085	0.504	0.680	0.664
A3, without numerical features	0.573	0.713	0.500	0.000	0.422	0.693	0.697
A3, with numerical features	0.569	0.620	0.559	0.123	0.551	0.618	0.611

Bold indicates column maxima within each panel.

In light of the results from the insufficiency analysis, this finding is not surprising. The novelty analysis must consider five possible textual inputs rather than four, despite having slightly fewer claims and cases available for training and validation.

Since the novelty performance was very weak in the MCCV tests, I do not evaluate the novelty models on the test set.

7 Conclusion

In this paper, I investigate whether patent texts and numerical features can predict outcomes in UK patent litigation. I construct a dataset linking judicial decisions to patents, individual claims, grounds of challenge, and prior-art references, and examine prediction under different information sets. The analysis addresses three questions: which texts are informative for each prediction task, how dependent claims should be represented, and whether different texts should be encoded separately or combined before encoding.

The results show that predictive performance depends on both the legal ground and the representation of the available information. Classical text models remain competitive in the ex-ante patent-level analysis, while the insufficiency and obviousness analyses provide evidence of predictive

information in representations tailored to the relevant ground. Numerical features improve the selected models' performance on the court holdout in all of our prediction tasks. In these three analyses, selected supervised specifications achieve higher accuracy, MCC as well as macro- F_1 scores than the zero-shot GPT benchmark, although their advantage is not uniform across metrics and samples. The encoding comparisons favour preserving separate representations of the relevant texts. However, expanding dependent claims does not consistently outperform their literal wording, and including claims with greater dependency depth does not systematically weaken performance.

These findings also identify limits to the approach. Prediction remains difficult in the general claim-level and novelty analyses, and performance on the court holdout does not consistently carry over to the UKIPO sample. The latter result limits the extent to which models estimated on court disputes can be treated as general measures of patent strength across institutional settings.

The first extension is to investigate whether these predictions contain a trading signal. Forecasting a judicial outcome may have financial value if the forecast contains information that is not already reflected in market prices. A future study could link predictions formed using information available before judgment to subsequent abnormal stock returns, and assess whether a trading strategy based on those predictions remains profitable after transaction costs. I could not conduct this analysis reliably here because the UK sample contains too few relevant litigation events. A larger event sample would make it possible to distinguish predictive accuracy about legal outcomes from the ability to identify mispriced litigation risk.

The second extension is to construct counterfactual predictions for claims that were not challenged for insufficiency. The central question is whether these claims are stronger against insufficiency than claims that were challenged on that ground. Applying the insufficiency model to both groups would allow me to compare their predicted probabilities of surviving a hypothetical insufficiency attack. Comparisons among otherwise similar claims, including claims within the same patent where possible, could help assess whether challengers tend to target claims that the model also identifies as more vulnerable. These would be model-based counterfactual predictions: the outcomes of unmade attacks are unobserved, and the decision to challenge a claim is selective. Accounting for this selection would therefore be essential when interpreting differences between the two groups as evidence about claim strength.

A Definition of Numerical Features

Table A.1: Definition and use of numerical features

Feature	Definition	Used at
claimant_entity	Legal status of the claimant. It equals 1 if the claimant is an individual, a private company, or a non-profit organization such as a university; 2 if the claimant is a publicly listed company; and 3 if the claimant is a public authority, specifically the Comptroller.	Patent and claim
defendant_entity	Legal status of the defendant. It equals 1 if the defendant is an individual, a private company, or a non-profit organization such as a university; 2 if the defendant is a publicly listed company; and 3 if the defendant is a public authority, specifically the Comptroller.	Patent and claim
claimant_headquarter	Location of the claimant’s headquarters. It equals 0 for the United Kingdom; 1 for the European Union or Switzerland, excluding the United Kingdom; and 2 for all other countries.	Patent and claim
defendant_headquarter	Location of the defendant’s headquarters. It equals 0 for the United Kingdom; 1 for the European Union or Switzerland, excluding the United Kingdom; and 2 for all other countries.	Patent and claim
patent_age	Age of the patent, measured in days from the patent’s priority date to the date of the first judgment in the case. Formally, it is calculated as the first judgment date minus the priority date.	Patent and claim
forward_citation	Number of times the patent had been cited by other patents as of the date of the first judgment. Citations occurring after that date are not counted.	Patent and claim
backward_citation	Number of earlier patents cited by the patent.	Patent and claim
claimant_is_owner	Indicator for the relationship between the claimant and the patent owner. It equals 0 if the claimant is the patent owner and 1 otherwise.	Patent and claim
amended	Indicator for whether the claim is an amended version. It equals 0 if the claim was not amended and 1 if it was amended.	Claim only
dependency_depth	Maximum number of dependency links in any path connecting the claim to an independent claim. An independent claim has depth zero. The depth of a dependent claim is one plus the greatest depth among its parent claims, as defined in Definition 3.1.	Claim only

B Derivation of models in section 4

B.1 Regularization and loss function of classification heads

Regularization For a coefficient vector $w = (w_1, \dots, w_d)^\top$, the L_1 (lasso) regularization function takes the form

$$R_{L_1}(w) = \|w\|_1 = \sum_{j=1}^d |w_j|.$$

The L_2 (ridge) regularization function takes the form

$$R_{L_2}(w) = \|w\|_2^2 = \sum_{j=1}^d w_j^2.$$

Elastic-net regularization is a convex combination of the L_1 and L_2 penalties:

$$R_{\text{EN}}(w) = \rho \|w\|_1 + (1 - \rho) \|w\|_2^2, \quad \rho \in [0, 1].$$

Here, $\rho = 1$ corresponds to lasso regularization, while $\rho = 0$ corresponds to ridge regularization.

Loss function for logistic regression. Logistic regression models the probability that observation i belongs to class 1 as

$$p_i = P(y_i = 1 \mid x_i) = \sigma(w^\top x_i + b) = \frac{1}{1 + \exp[-(w^\top x_i + b)]}.$$

The probability assigned to the observed outcome $y_i \in \{0, 1\}$ is

$$P(y_i \mid x_i) = p_i^{y_i} (1 - p_i)^{1-y_i}.$$

Taking the negative logarithm gives the loss for a single observation i :

$$\ell_i(w, b) = -\log P(y_i \mid x_i) = -[y_i \log p_i + (1 - y_i) \log(1 - p_i)].$$

The total weighted and regularized loss is

$$\mathcal{L}_{\text{logit}}(w, b) = -\sum_{i=1}^n a_{y_i} s_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + \lambda R(w).$$

The first term is the weighted binary cross-entropy loss. Minimizing this term is equivalent to maximizing the weighted conditional log-likelihood of the observed outcomes: the model is rewarded for assigning high probability to the observed class and penalized for confident but incorrect predictions. The second term regularizes the coefficient vector to reduce overfitting. I consider L_2 , L_1 , and elastic-net regularization.

At prediction time, an observation is assigned to class 1 if $p_i > 0.5$ and to class 0 otherwise.

Loss function for support vector machines. For a linear SVM, let

$$f_i = w^\top x_i + b$$

denote the decision score for observation i . I transform the binary outcome $y_i \in \{0, 1\}$ into

$$\tilde{y}_i = 2y_i - 1 \in \{-1, +1\}.$$

The product $\tilde{y}_i f_i$ is the functional margin: it is positive when the observation is correctly classified and negative when it is misclassified.

The SVM seeks to classify each observation with a functional margin of at least one:

$$\tilde{y}_i f_i \geq 1.$$

To allow violations of this requirement, introduce a nonnegative slack variable ξ_i such that

$$\tilde{y}_i f_i \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

For given w and b , the smallest feasible slack is $\max\{0, 1 - \tilde{y}_i f_i\}$. This gives the hinge loss for a single observation:

$$\ell_i(w, b) = \max\{0, 1 - \tilde{y}_i f_i\} = \max\{0, 1 - (2y_i - 1)(w^\top x_i + b)\}.$$

The loss is zero when the observation is correctly classified with a functional margin of at least one. It is positive when the observation lies inside the margin or is misclassified, with larger violations receiving larger penalties.

The total weighted and regularized loss is

$$\mathcal{L}_{\text{SVM}}(w, b) = \sum_{i=1}^n a_{y_i} s_i \max\{0, 1 - (2y_i - 1)(w^\top x_i + b)\} + \lambda R(w).$$

The first term is the weighted hinge loss. The second term regularizes the coefficient vector and favors a simpler decision boundary. I consider both L_2 and L_1 regularization for the linear SVM.

At prediction time, an observation is assigned to class 1 if $w^\top x_i + b > 0$ and to class 0 otherwise.

I also consider an SVM with a radial basis function kernel, referred to as the RBF-SVC. It applies the same hinge-loss principle in an implicitly transformed representation space using the kernel

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|_2^2\right).$$

The kernel measures the similarity between observations according to the distance between their representations and allows the SVM to construct nonlinear decision boundaries. A smaller γ makes each observation influence a larger region of the representation space and generally produces a smoother, more global decision boundary. A larger γ makes each observation's influence more local

and permits a more complex decision boundary.

Loss function for Ridge-regularized least-squares classifier. Using the transformed labels $\tilde{y}_i = 2y_i - 1 \in \{-1, +1\}$, the loss for a single observation i is the squared difference between its transformed label and the linear decision score:

$$\ell_i(w, b) = \left(\tilde{y}_i - w^\top x_i - b \right)^2.$$

The total weighted and regularized objective is

$$\min_{w, b} \left\{ \sum_{i=1}^n a_{y_i} s_i \left(\tilde{y}_i - w^\top x_i - b \right)^2 + \lambda R(w) \right\}.$$

Thus, unlike logistic regression and SVMs, the L_2 -regularized least-squares classifier uses squared-error loss. I use L_2 regularization only, with $R(w) = \|w\|_2^2$ and α controlling its strength.

The fitted classifier produces the linear decision score

$$g(x) = w^\top x + b.$$

An observation is assigned to class 1 if $g(x) > 0$ and to class 0 otherwise.

B.2 Classical models

Splitting criterion for Extra Trees. Let S denote the set of training observations reaching the current node, x_{ij} the value of feature j for observation i , and $|A|$ the number of observations in a set A . At each node, the algorithm randomly selects a subset of the d features. For each selected feature j , it then randomly generates a candidate threshold τ . Each candidate split (j, τ) divides the observations into

$$S_L(j, \tau) = \{i \in S : x_{ij} \leq \tau\}$$

and

$$S_R(j, \tau) = \{i \in S : x_{ij} > \tau\}.$$

Thus, observations with values at or below the threshold move to the left child, while those with values above the threshold move to the right child.

For a node A , its Gini impurity is

$$G(A) = 1 - \sum_{c \in \{0, 1\}} p_c(A)^2,$$

where $p_c(A)$ is the share of observations in A belonging to class c . A node has zero impurity when all its observations belong to the same class. Among the randomly generated candidate splits, the

algorithm chooses the pair (j, τ) that maximizes the reduction in impurity,

$$\Delta G(j, \tau) = G(S) - \frac{|S_L|}{|S|} G(S_L) - \frac{|S_R|}{|S|} G(S_R).$$

Derivation of TF-IDF Let the corpus contain n documents and let

$$\mathcal{V} = \{v_1, \dots, v_d\}$$

denote the selected vocabulary of d terms. TF-IDF maps document i into a d -dimensional vector

$$x_i^{\text{TFIDF}} = (\text{tfidf}_{i1}, \dots, \text{tfidf}_{id})^\top,$$

where each dimension corresponds to one term in the vocabulary. For term v_j , let

$$\text{tf}_{ij}$$

denote the number of times that the term appears in document i , and let

$$\text{df}_j = \sum_{i=1}^n \mathbf{1}\{\text{tf}_{ij} > 0\}$$

denote the number of documents containing that term. Its TF-IDF weight is

$$\text{tfidf}_{ij} = \text{tf}_{ij} \log \left(\frac{n}{\text{df}_j} \right).$$

The term-frequency component assigns greater weight to terms that occur frequently within a document, while the inverse-document-frequency component discounts terms that appear in many documents and therefore provide relatively little information for distinguishing among them.

After vectorization, the TF-IDF representation is concatenated with other features z_i ,

$$\tilde{x}_i = \begin{bmatrix} x_i^{\text{TFIDF}} \\ z_i \end{bmatrix},$$

and the resulting vector is passed to one of the classification heads described in the previous subsection for training and prediction.

Complement Naive Bayes. Each i , regardless of its outcome class, is represented using the common vector

$$x_i = (x_{i1}, \dots, x_{id})^\top,$$

where x_{ij} is the TF-IDF weight of term v_j in document i .

For each class c , the classifier aggregates term weights over documents that do *not* belong to

that class:

$$N_{cj}^{\text{comp}} = \sum_{i: y_i \neq c} x_{ij}.$$

It then estimates the relative frequency of term v_j in the complement of class c as

$$q_{cj} = \frac{\alpha + N_{cj}^{\text{comp}}}{\alpha d + \sum_{k=1}^d N_{ck}^{\text{comp}}},$$

where $\alpha > 0$ is a smoothing parameter. In the binary setting, q_{1j} is therefore estimated using documents from class 0, while q_{0j} is estimated using documents from class 1.

A new document is transformed into the same TF-IDF vector

$$x = (x_1, \dots, x_d)^\top$$

and assigned to the class

$$\hat{y} = \arg \min_c \sum_{j=1}^d x_j \log q_{cj}.$$

Derivation of Word2Vec Let $\mathcal{V}$ denote the learned vocabulary and let r denote the embedding dimension. Each word $w \in \mathcal{V}$ is associated with two vectors,

$$v_w \in \mathbb{R}^r \quad \text{and} \quad u_w \in \mathbb{R}^r,$$

corresponding to its input and output representations, respectively. The distinction arises because a word can be used either to predict another word or as the word being predicted.

After tokenization and removal of words not retained in $\mathcal{V}$, document i is represented by the sequence

$$(w_{i1}, \dots, w_{iT_i}),$$

where T_i is the number of retained token occurrences. For the word in position t , let

$$C_{it} = \{s : 1 \leq s \leq T_i, 0 < |s - t| \leq m\}$$

denote the positions within a context window of at most m words on either side.

I consider both the skip-gram and continuous bag-of-words (CBOW) variants of Word2Vec. Under skip-gram, the central word w_{it} is used to predict each surrounding word w_{is} , $s \in C_{it}$. The probability of observing context word o given central word c is

$$P(o \mid c) = \frac{\exp(u_o^\top v_c)}{\sum_{w \in \mathcal{V}} \exp(u_w^\top v_c)}.$$

Thus, a context word receives a higher probability when its output representation has a larger dot

product with the input representation of the central word. The skip-gram loss is

$$\mathcal{L}^{\text{SG}} = - \sum_{i=1}^n \sum_{t=1}^{T_i} \sum_{s \in C_{it}} \log P(w_{is} \mid w_{it}).$$

CBOW reverses the prediction task. It first averages the input representations of the surrounding words,

$$h_{it} = \frac{1}{|C_{it}|} \sum_{s \in C_{it}} v_{w_{is}},$$

and uses the resulting context representation to predict the central word. The corresponding probability is

$$P(w_{it} \mid C_{it}) = \frac{\exp(u_{w_{it}}^\top h_{it})}{\sum_{w \in \mathcal{V}} \exp(u_w^\top h_{it})},$$

and the CBOW loss is

$$\mathcal{L}^{\text{CBOW}} = - \sum_{i=1}^n \sum_{t=1}^{T_i} \log P(w_{it} \mid C_{it}).$$

In either variant, minimizing the loss increases the similarity between words that frequently occur in the same local contexts. For computational efficiency, my implementation replaces the full softmax calculation with negative sampling.

After training, I retain the input embedding v_w of each word. To obtain a fixed-dimensional representation of document i , I consider two pooling methods. Unweighted mean pooling averages the embeddings of all token occurrences in the document:

$$x_i^{\text{W2V}} = \frac{1}{T_i} \sum_{t=1}^{T_i} v_{w_{it}}.$$

I also consider TF-IDF-weighted mean pooling. Let a_{it} denote the inverse-document-frequency weight of token w_{it} . The document representation is

$$x_i^{\text{W2V}} = \frac{\sum_{t=1}^{T_i} a_{it} v_{w_{it}}}{\sum_{t=1}^{T_i} a_{it}}.$$

After vectorization, the representation is concatenated with other features z_i ,

$$\tilde{x}_i = \begin{bmatrix} x_i^{\text{W2V}} \\ z_i \end{bmatrix},$$

and the resulting vector is passed to one of the classification heads described in the previous subsection for training and prediction.

C UKIPO and Combined Test Results

This appendix reports external-test performance on the UKIPO sample and on the combined court and UKIPO sample. Court-only results are reported in the main text. Where two depth regimes are available, Panel A reports depth ≤ 2 and Panel B reports full depth (≤ 100).

C.1 Ex-ante patent-level prediction

Table C.1: External test performance: UKIPO

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Precision (owner loss)	Recall (owner loss)
TF-IDF, without numerical features	0.551	0.541	0.546	0.093	0.540	0.510	0.625
TF-IDF, with numerical features	0.496	0.459	0.471	-0.064	0.441	0.450	0.675
Word2Vec + linear SVM (L_1), without numerical features	0.521	0.518	0.518	0.036	0.517	0.488	0.525
Word2Vec + linear SVM (L_1), with numerical features	0.519	0.529	0.535	0.070	0.528	0.500	0.625
ChatGPT (GPT-5.6 Sol), zero-shot	0.607	0.506	0.526	0.073	0.450	0.486	0.875

$N = 85$: 45 owner wins/survivals and 40 owner losses after binary coding.

Precision and recall treat owner loss as the positive outcome. Bold indicates the highest value in each column.

The ChatGPT prompt is reproduced in Section E.1.

Table C.2: External test performance: overall

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Precision (owner loss)	Recall (owner loss)
TF-IDF, without numerical features	0.565	0.576	0.577	0.156	0.575	0.561	0.648
TF-IDF, with numerical features	0.516	0.521	0.523	0.047	0.512	0.511	0.662
Word2Vec + linear SVM (L_1), without numerical features	0.563	0.556	0.556	0.112	0.555	0.547	0.577
Word2Vec + linear SVM (L_1), with numerical features	0.582	0.590	0.591	0.183	0.589	0.575	0.648
ChatGPT (GPT-5.6 Sol), zero-shot*	0.633	0.521	0.526	0.075	0.452	0.508	0.887

$N = 144$: 73 owner wins/survivals and 71 owner losses after binary coding.

Precision and recall treat owner loss as the positive outcome. Bold indicates the highest value in each column.

*Includes one manually obtained ChatGPT prediction reported by the user; all other predictions are from the API.

ChatGPT ROC-AUC uses 143 observations; the manual prediction has no recorded probability.

The ChatGPT prompt is reproduced in Section E.1.

C.2 Insufficiency

Table C.3: Insufficiency: external-test performance on UKIPO observations

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 37$)								
INS-A1, text only	0.537	0.622	0.500	0.000	0.383	0.778	0.000	0.000
INS-A1, text + numerical	0.512	0.622	0.500	0.000	0.383	0.778	0.000	0.000
INS-A2, text only	0.528	0.622	0.500	0.000	0.383	0.778	0.000	0.000
INS-A2, text + numerical	0.571	0.622	0.500	0.000	0.383	0.778	0.000	0.000
GPT-5.6 Sol, zero-shot	0.483	0.595	0.478	-0.130	0.373	0.750	0.000	0.000
Panel B: Full depth (≤ 100; $n = 60$)								
INS-A1, text only	0.623	0.450	0.402	-0.273	0.334	0.741	0.111	0.038
INS-A1, text + numerical	0.649	0.583	0.519	0.149	0.403	0.793	1.000	0.038
INS-A2, text only	0.595	0.567	0.500	0.000	0.362	0.789	0.000	0.000
INS-A2, text + numerical	0.613	0.567	0.500	0.000	0.362	0.789	0.000	0.000
GPT-5.6 Sol, zero-shot	0.428	0.550	0.485	-0.114	0.355	0.763	0.000	0.000

INS-A1: expanded claim and description. INS-A2: expanded claim, complete claim set, and description. Supervised models are refitted on all training and validation data at the matching depth, using a 20-epoch schedule and stopping at epoch 7 (INS-A1) or 9 (INS-A2). GPT is zero-shot; its predictions are subsetted by depth. Precision and recall refer to owner loss on the insufficiency ground; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel. The GPT prompt is reproduced in Section E.2.

Table C.4: Insufficiency: external-test performance on combined court and UKIPO observations

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 83$)								
INS-A1, text only	0.629	0.554	0.522	0.047	0.514	0.649	0.458	0.314
INS-A1, text + numerical	0.636	0.590	0.561	0.131	0.556	0.677	0.520	0.371
INS-A2, text only	0.649	0.578	0.550	0.107	0.546	0.655	0.500	0.371
INS-A2, text + numerical	0.658	0.627	0.592	0.206	0.586	0.726	0.591	0.371
GPT-5.6 Sol, zero-shot	0.619	0.590	0.530	0.095	0.474	0.688	0.556	0.143
Panel B: Full depth (≤ 100; $n = 124$)								
INS-A1, text only	0.643	0.532	0.521	0.042	0.520	0.665	0.469	0.418
INS-A1, text + numerical	0.674	0.605	0.582	0.180	0.575	0.693	0.583	0.382
INS-A2, text only	0.641	0.597	0.575	0.163	0.568	0.693	0.568	0.382
INS-A2, text + numerical	0.658	0.613	0.589	0.197	0.581	0.705	0.600	0.382
GPT-5.6 Sol, zero-shot	0.542	0.532	0.489	-0.033	0.423	0.688	0.400	0.109

INS-A1: expanded claim and description. INS-A2: expanded claim, complete claim set, and description. Supervised models are refitted on all training and validation data at the matching depth, using a 20-epoch schedule and stopping at epoch 7 (INS-A1) or 9 (INS-A2). GPT is zero-shot; its predictions are subsetted by depth. Precision and recall refer to owner loss on the insufficiency ground; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel. The GPT prompt is reproduced in Section E.2.

C.3 Obviousness

Table C.5: Obviousness: external-test performance on UKIPO observations

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 232$)									
OBV-A1, text only	0.493	0.810	0.497	-0.031	0.448	0.788	0.694	0.000	0.000
OBV-A1, text + numerical	0.427	0.448	0.401	-0.154	0.382	0.432	0.527	0.124	0.326
OBV-A2, text only	0.527	0.806	0.495	-0.044	0.446	0.782	0.689	0.000	0.000
OBV-A2, text + numerical	0.425	0.418	0.364	-0.211	0.354	0.378	0.475	0.103	0.279
GPT-5.6 Sol, zero-shot	0.594	0.405	0.608	0.195	0.403	0.437	0.435	0.229	0.930
Panel B: Full depth (≤ 100; $n = 379$)									
OBV-A1, text only	0.378	0.847	0.495	-0.037	0.459	0.822	0.699	0.000	0.000
OBV-A1, text + numerical	0.358	0.269	0.474	-0.046	0.267	0.274	0.421	0.137	0.764
OBV-A2, text only	0.431	0.842	0.492	-0.048	0.457	0.814	0.693	0.000	0.000
OBV-A2, text + numerical	0.363	0.266	0.480	-0.036	0.265	0.272	0.437	0.139	0.782
GPT-5.6 Sol, zero-shot	0.613	0.327	0.569	0.119	0.324	0.356	0.434	0.167	0.909

OBV-A1: literal claim, complete claim set, and prior-art material. OBV-A2: expanded claim, complete claim set, and prior-art material. Supervised models are trained at the same depth as the evaluation sample. GPT is zero-shot; its predictions are subsetted by depth. Metrics are computed at the attack level. Patent-weighted accuracy is the mean of within-patent attack accuracies, with patents grouped within proceedings. Precision and recall refer to owner loss on the obviousness attack; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel at the reported precision. The GPT prompt is reproduced in Section E.3.

Table C.6: Obviousness: external-test performance on combined court and UKIPO observations

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 460$)									
OBV-A1, text only	0.472	0.743	0.499	-0.027	0.426	0.738	0.703	0.000	0.000
OBV-A1, text + numerical	0.519	0.530	0.460	-0.072	0.456	0.513	0.614	0.214	0.316
OBV-A2, text only	0.488	0.741	0.497	-0.039	0.426	0.734	0.701	0.000	0.000
OBV-A2, text + numerical	0.519	0.526	0.457	-0.077	0.453	0.499	0.595	0.211	0.316
GPT-5.6 Sol, zero-shot	0.638	0.452	0.599	0.199	0.452	0.465	0.441	0.304	0.897
Panel B: Full depth (≤ 100; $n = 685$)									
OBV-A1, text only	0.438	0.728	0.495	-0.051	0.421	0.722	0.683	0.000	0.000
OBV-A1, text + numerical	0.516	0.425	0.554	0.110	0.425	0.411	0.521	0.292	0.829
OBV-A2, text only	0.459	0.727	0.494	-0.056	0.421	0.719	0.685	0.000	0.000
OBV-A2, text + numerical	0.521	0.410	0.539	0.080	0.410	0.393	0.514	0.284	0.812
GPT-5.6 Sol, zero-shot	0.633	0.439	0.589	0.188	0.439	0.447	0.453	0.309	0.906

OBV-A1: literal claim, complete claim set, and prior-art material. OBV-A2: expanded claim, complete claim set, and prior-art material. Supervised models are trained at the same depth as the evaluation sample. GPT is zero-shot; its predictions are subsetting by depth. Metrics are computed at the attack level. Patent-weighted accuracy is the mean of within-patent attack accuracies, with patents grouped within proceedings. Precision and recall refer to owner loss on the obviousness attack; precision is zero when there are no positive predictions. Bold indicates column maxima within each panel at the reported precision. The GPT prompt is reproduced in Section E.3.

C.4 Insufficiency: separate versus combined encoding

The following tables extend the comparison in Table 8 to the UKIPO and combined samples.

Table C.7: Insufficiency: separate versus combined encoding on UKIPO observations

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 37$)								
INS-A2, separate	0.571	0.622	0.500	0.000	0.383	0.778	0.000	0.000
INS-X1, combined	0.475	0.432	0.348	-0.377	0.302	0.729	0.000	0.000
INS-X2, combined	0.494	0.432	0.348	-0.377	0.302	0.729	0.000	0.000
INS-X3, combined	0.540	0.432	0.348	-0.377	0.302	0.729	0.000	0.000
Panel B: Full depth (≤ 100) ($n = 60$)								
INS-A2, separate	0.613	0.567	0.500	0.000	0.362	0.789	0.000	0.000
INS-X1, combined	0.507	0.433	0.382	-0.343	0.302	0.737	0.000	0.000
INS-X2, combined	0.516	0.433	0.382	-0.343	0.302	0.737	0.000	0.000
INS-X3, combined	0.566	0.433	0.382	-0.343	0.302	0.737	0.000	0.000

INS-A2 encodes the expanded claim, complete claim set, and description separately. INS-X1 concatenates the literal claim, complete claim set, and description; INS-X2 concatenates the expanded claim, complete claim set, and description; INS-X3 concatenates the expanded claim and description. Each INS-X specification uses one frozen Qwen-8B encoding of the concatenated text and the same MLP classification-head design as INS-A2. All specifications include numerical features, use equal patent weights ($\beta = 1$), and use a 20-epoch learning-rate schedule with training stopped at epoch 9. Training and evaluation depths match within each panel. The classification threshold is 0.5. The INS-A2 results are reproduced from Table C.3. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

Table C.8: Insufficiency: separate versus combined encoding on combined court and UKIPO observations

Model	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 83$)								
INS-A2, separate	0.658	0.627	0.592	0.206	0.586	0.726	0.591	0.371
INS-X1, combined	0.571	0.470	0.429	-0.160	0.407	0.614	0.286	0.171
INS-X2, combined	0.570	0.458	0.419	-0.181	0.399	0.602	0.273	0.171
INS-X3, combined	0.578	0.470	0.426	-0.175	0.396	0.618	0.263	0.143
Panel B: Full depth (≤ 100) ($n = 124$)								
INS-A2, separate	0.658	0.613	0.589	0.197	0.581	0.705	0.600	0.382
INS-X1, combined	0.587	0.516	0.480	-0.050	0.439	0.641	0.391	0.164
INS-X2, combined	0.582	0.516	0.480	-0.050	0.439	0.641	0.391	0.164
INS-X3, combined	0.599	0.524	0.489	-0.027	0.452	0.648	0.417	0.182

INS-A2 encodes the expanded claim, complete claim set, and description separately. INS-X1 concatenates the literal claim, complete claim set, and description; INS-X2 concatenates the expanded claim, complete claim set, and description; INS-X3 concatenates the expanded claim and description. Each INS-X specification uses one frozen Qwen-8B encoding of the concatenated text and the same MLP classification-head design as INS-A2. All specifications include numerical features, use equal patent weights ($\beta = 1$), and use a 20-epoch learning-rate schedule with training stopped at epoch 9. Training and evaluation depths match within each panel. The classification threshold is 0.5. The INS-A2 results are reproduced from Table C.4. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

C.5 Obviousness: separate versus combined encoding

Table C.9: Obviousness: separate versus combined encoding on UKIPO observations

Model	Training depth	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 232$)										
OBV-A1, separate	2	0.427	0.448	0.401	-0.154	0.382	0.432	0.527	0.124	0.326
OBV-A2, separate	2	0.425	0.418	0.364	-0.211	0.354	0.378	0.475	0.103	0.279
OBV-X1, combined	2	0.629	0.685	0.448	-0.113	0.443	0.699	0.612	0.083	0.070
OBV-X2, combined	2	0.631	0.694	0.471	-0.061	0.469	0.709	0.642	0.132	0.116
Panel B: Full depth (≤ 100) ($n = 379$)										
OBV-A1, separate	100	0.358	0.269	0.474	-0.046	0.267	0.274	0.421	0.137	0.764
OBV-A2, separate	100	0.363	0.266	0.480	-0.036	0.265	0.272	0.437	0.139	0.782
OBV-X1, combined	100	0.591	0.712	0.500	-0.001	0.497	0.732	0.636	0.145	0.200
OBV-X2, combined	100	0.596	0.712	0.507	0.013	0.503	0.734	0.637	0.154	0.218

OBV-X1 combines the literal claim, complete claim set, and prior-art material; OBV-X2 uses the expanded claim in place of the literal claim. All four specifications include numerical features. The separate-encoding models are trained at the evaluation depth and use patent-size weighting with $\beta = 1/16$. All models in Panel A are trained and evaluated at depth ≤ 2 ; all models in Panel B are trained and evaluated at full depth (≤ 100). Both combined-encoding models use uniform loss weights and stop at epoch 2. Training and evaluation depths therefore match within each panel, although the separate- and combined-encoding specifications differ in loss weighting. The OBV-A1 and OBV-A2 results are reproduced from Table C.5. Metrics are computed at the attack level. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

Table C.10: Obviousness: separate versus combined encoding on combined court and UKIPO observations

Model	Training depth	ROC-AUC	Accuracy	Balanced accuracy	MCC	Macro- F_1	Claim-weighted accuracy	Patent-weighted accuracy	Precision (owner loss)	Recall (owner loss)
Panel A: Depth ≤ 2 ($n = 460$)										
OBV-A1, separate	2	0.519	0.530	0.460	-0.072	0.456	0.513	0.614	0.214	0.316
OBV-A2, separate	2	0.519	0.526	0.457	-0.077	0.453	0.499	0.595	0.211	0.316
OBV-X1, combined	2	0.385	0.550	0.411	-0.176	0.412	0.561	0.526	0.125	0.128
OBV-X2, combined	2	0.385	0.554	0.420	-0.159	0.421	0.567	0.541	0.139	0.145
Panel B: Full depth (≤ 100) ($n = 685$)										
OBV-A1, separate	100	0.516	0.425	0.554	0.110	0.425	0.411	0.521	0.292	0.829
OBV-A2, separate	100	0.521	0.410	0.539	0.080	0.410	0.393	0.514	0.284	0.812
OBV-X1, combined	100	0.358	0.547	0.427	-0.144	0.428	0.558	0.499	0.162	0.171
OBV-X2, combined	100	0.358	0.546	0.428	-0.142	0.429	0.558	0.498	0.165	0.177

OBV-X1 combines the literal claim, complete claim set, and prior-art material; OBV-X2 uses the expanded claim in place of the literal claim. All four specifications include numerical features. The separate-encoding models are trained at the evaluation depth and use patent-size weighting with $\beta = 1/16$. All models in Panel A are trained and evaluated at depth ≤ 2 ; all models in Panel B are trained and evaluated at full depth (≤ 100). Both combined-encoding models use uniform loss weights and stop at epoch 2. Training and evaluation depths therefore match within each panel, although the separate- and combined-encoding specifications differ in loss weighting. The OBV-A1 and OBV-A2 results are reproduced from Table C.6. Metrics are computed at the attack level. Precision and recall refer to owner loss. Bold indicates column maxima within each panel at the reported precision.

D LLM Prompts

Sections D.1–D.5 reproduce the four extraction prompts and the extra prompt. Sections E.1–E.3 reproduce the zero-shot prediction prompts. Source wording, examples, and output instructions are retained; document formatting is adapted for the appendix.

D.1 First prompt: patent-level extraction

IP JUDGMENT EXTRACTION PROMPT — PATENT ONLY — RIGHT-LEVEL VALIDITY DATASET

Right-level UK patent-validity dataset — revised with a mandatory Google Patents visual citation example

You are extracting structured data from UK patent judgments at the patent-right level.

STRICT RULE: NEVER USE "" IN TEXT FIELDS

STRICT RULE: NEVER USE "" IN TEXT FIELDS

STRICT RULE: NEVER USE "" IN TEXT FIELDS

This prompt produces only the validity dataset. Do not produce an infringement dataset or infringement rows.

This is a right-level dataset, not a claim-level dataset.

* Each row corresponds to one disputed patent or patent application, aggregated at patent level.

* Do not create separate rows merely for individual claims.

* Claim-level reasoning must be summarized inside fields 18, 25, 27, and 29.

ABSOLUTE OUTPUT RESTRICTIONS

* Output exactly one fenced code block labelled csv.

* Put all CSV rows inside that csv code block.

* Do not include a header.

* Do not include notes, explanations, citations, footnotes, source references, or any text before or after the csv code block.

* Every row must contain exactly 32 fields.

* Wrap every field in double quotes.

* Escape all internal double quotes using double double-quotes.

* Lists must look like "["A", "B"]" when placed inside a CSV field.

* Dictionaries must use double quotes for all internal keys and strings, escaped correctly for CSV.

* No field may contain a URL unless that field explicitly asks for a URL.

* If the case must be skipped, output exactly one CSV row. Put "skip" in every field except field 30, which must contain the judgment URL.

* Do not hallucinate.

* If a value is not available from the judgment or required public research, leave it as "".

* Date format must be YYYY-MM-DD.

* If only month and year are available, use YYYY-MM.

* If only year is available, use YYYY.

* If a date-list field has no available information, output "[]", not "".

- * Field 30 must contain the raw judgment URL only.
- * Every CSV record must occupy exactly one physical line in the output.
- * Never insert an actual newline or carriage-return character inside any CSV field.
- * Whenever this prompt specifies \n, output the two literal characters backslash and n, not an actual line break.
- * Field 26 must separate passages using the literal text \n—\n.
- * In field 29, place the required patent-level outcome first, followed by the literal text \n and then the remainder of the summary.

SCOPE FILTER — ONLY THREE CASE

- * be careful do not make double quote ""
- * link of patent must not be empty. same as all field of citations. if GB versions not available, find WO version

Extract a row only if the case belongs to one of these three types:

1. An infringement action in which the court actually considers or decides patent validity. Extract only the validity result. Do not extract infringement.
2. A revocation action or counterclaim in which patent validity is considered or decided.
3. A patent-application case in which the court considers patentability or another validity-type issue, including novelty, inventive step/obviousness, insufficiency, added matter, excluded subject matter/non-patentability, industrial applicability, priority entitlement, or another ground bearing on whether the application may proceed to grant.

An express concession that the whole patent is invalid counts as a validity outcome under Type 1 or Type 2 even if the court does not conduct a full merits analysis of individual validity grounds. Include the patent at right level. A mere withdrawal, abandonment, non-maintenance, or agreed revocation without an express concession of invalidity does not count by itself.

For Type 3 patent-application cases, the Comptroller/UK IPO is mandatory but not necessarily the only defendant/respondent or adverse party:

- * Field 6 must include "Comptroller-General of Patents, Designs and Trade Marks" and must also include every separately represented third-party respondent/opponent identified in the case.
- * Field 8 must include "Comptroller-General of Patents, Designs and Trade Marks" and must also include every third-party opponent that directly opposes or challenges the patent application in this row.
- * Field 10 must contain the Comptroller object and separate verified objects for all qualifying third-party opponents in field 8, subject to the historical parent/group collapse rules.
- * Do not omit a qualifying third-party opponent merely because the Comptroller is already present.
- * Do not add a third party merely because it appears in the caption; it must directly oppose or challenge the application.
- * Field 28 = 0 when the applicant is the claimant/appellant.

Skip:

- * a pure infringement or non-infringement case with no live validity issue considered, decided, or expressly conceded invalid;
- * registered-design cases;
- * damages, inquiry as to damages, costs-only, consequential-order, enforcement-only, or other post-merits judgments;
- * a merely interlocutory judgment that does not finally or summarily decide a live validity, revocation, refusal, or patentability issue;

- * permission-to-appeal decisions following a non-merits interlocutory ruling;
- * any strike-out, dismissal, stay, jurisdiction, service, forum, standing, party, pleading-amendment, patent-amendment-procedure, costs, or case-management ruling that does not directly and summarily decide patent validity, revocation, refusal, or patentability;
- * a row where the revocation claim, counterclaim, or only live invalidity argument is withdrawn, abandoned, not maintained, or no longer pursued and the court consequently does not decide it;
- * any case outside Types 1–3 above.

SUMMARY JUDGMENT / STRIKE-OUT

Apply this section before extracting any row. First identify (a) who made the strike-out or summary-disposition application, (b) who won that application, (c) whether the application directly concerned patent validity, revocation, refusal, or patentability, and (d) whether a later hearing or judgment in the same case decided validity.

DIRECT-VALIDITY REQUIREMENT

- * The strike-out or summary-disposition application must directly seek a final or summary determination of a live patent-validity, revocation, refusal, or patentability issue.
- * A threshold merits assessment may count even though it is shorter than a full trial, but the court must assess the validity challenge itself rather than merely decide a collateral procedural question.
- * Always skip a strike-out or other application concerning only jurisdiction, service, forum conveniens, a stay, standing, joinder or identity of parties, duplication or utility of claims, permission to amend pleadings, permission or procedure for patent amendment without deciding validity, costs, timetabling, disclosure, evidence, or other case management.

MANDATORY STRIKE-OUT DECISION MATRIX

- * Challenger applies and challenger wins: **INCLUDE**. Treat the result as an adverse-side right-level win; field 13 = 0 and field 25 = 1. This applies when the ruling summarily disposes of the patent or the live validity issue against the owner.
- * Challenger applies and patent owner wins: **INCLUDE** only if no later hearing or judgment in the same case decides the same patent's validity. Treat the result as an owner-side right-level win; field 13 = 0 and field 25 = 0. If a later validity hearing or judgment exists, **SKIP** this preliminary row and use the later validity decision instead.
- * Patent owner applies and patent owner wins: **INCLUDE** only if no later hearing or judgment in the same case decides the same patent's validity. Treat the result as an owner-side right-level win; field 13 = 0 and field 25 = 0. If a later validity hearing or judgment exists, **SKIP** this preliminary row and use the later validity decision instead.
- * Patent owner applies and challenger wins: **SKIP**. A refusal to strike out the challenger's validity case merely allows that case to continue and is not itself a patent-validity outcome.
- * If the identity of the applicant, the winner, the direct validity issue, or the existence of a later same-case validity decision cannot be established reliably, do not guess. **SKIP**.

WITHDRAWN OR ABANDONED REVOCATION ARGUMENTS

- * If a party withdraws, abandons, does not maintain, or no longer pursues a revocation claim, revocation counterclaim, or validity argument and the court consequently does not assess or decide it, do not treat the withdrawal as a validity outcome.
- * If no other live validity, revocation, refusal, or patentability issue is finally or summarily decided for that patent, **SKIP** the entire right-level row.
- * If only one ground is withdrawn but one or more other grounds are actually decided, keep the row, leave the withdrawn ground blank in field 27, and code only the grounds the court decides.
- * A formal statement that a patent is valid, maintained, or not revoked does not override this rule when it

merely records the consequence of an unopposed withdrawal or amendment and the court does not assess any live validity ground.

OUTPUT FOR INCLUDED PRE-TRIAL RULINGS

- * Output one right-level row per disputed patent or patent application.
- * Fill all right-level fields that can be established and make field 29 state that the result is a summary or strike-out outcome rather than a full merits judgment.
- * Do not invent claim-level rows or validity grounds that the court did not assess.

ROW UNIT RULE

Each row must correspond to:

- * one disputed patent or patent application;
- * one owner/applicant/enforcement side; and
- * one adverse validity challenger side.

Do not split rows merely because:

- * different claims of the same patent have different outcomes;
- * the same side contains multiple co-owners, co-applicants, exclusive licensees, enforcement-side parties, or joint validity challengers; or
- * the same patent is attacked on multiple validity grounds.

Split rows only when the legal/economic unit changes: a different patent/application, a different owner/applicant/enforcement side, or a different adverse validity challenger side.

If different claims of the same patent have different outcomes, aggregate at patent level. For field 25, count only claims or claim sets that were actually attacked and substantively decided. Unattacked claims, unargued claims, withdrawn or abandoned attacks, and unresolved or undecided claims do not count as surviving claims for field 25. Apply the mandatory attacked-claim and amendment rules under field 25 below.

WHOLE-PATENT INVALIDITY CONCESSION — RIGHT-LEVEL RULE

- * If the judgment, final order, or reliable litigation record expressly states that the patent owner concedes that the whole patent is invalid, include one right-level row for that patent.
- * A whole-patent invalidity concession is a valid adverse validity outcome even if no full merits analysis or ground-by-ground decision follows.
- * Field 25 must be 1 because the patent owner loses the whole patent.
- * Identify the exact historical patent/version covered by the concession.
- * Field 18 must state "patent as a whole; whole patent conceded invalid" and list the independent claims of the conceded version if they can be verified.
- * Field 27 must record the identified grounds. If no ground is identified, set "Others" = 1 and leave the named grounds blank.
- * Field 29 must state that invalidity was conceded, identify when the concession occurred, and distinguish it from a judicial merits finding.
- * An express whole-patent invalidity concession recorded before the substantive validity trial still counts.
- * Mere withdrawal of a revocation application, abandonment of a patent, non-maintenance of claims, or agreed revocation without an express admission/concession that the patent is invalid does not count and must not be coded as a right-level validity loss.

PATENT-VERSION RULE — MANDATORY

The extracted patent/application must be the version legally operative and actually addressed by the judgment on the judgment date.

- * Identify whether the judgment concerns the application claims, granted claims, a previously amended specification, or proposed amended claims.
- * Use the correct patent/application number and publication/grant suffix applicable at that date, for example A, A1, B, or B1 where relevant.
- * If the judgment does not identify the version clearly, research the official UKIPO/EPO record and contemporaneous patent documents.
- * Do not silently use the current Google Patents version when it differs from the version before the court.
- * Field 15 must name the correct version.
- * Field 18 must identify the actual claim version attacked or decided.
- * Field 29 must explain any old/granted/amended/proposed-amended distinction and identify which version was operative.
- * If the court substantively assesses both the granted and amended versions, record both in fields 18 and 29 but retain one right-level row.
- * If the correct historical version cannot be verified, do not guess; state the uncertainty briefly in field 29.

REFERENCE DATES

- * `FIRST_HEARING_DATE` = the earliest hearing date in field 2 for the judgment being extracted.
- * `FIRST_JUDGMENT_DATE` = the earliest judgment date in field 1 for the judgment being extracted.
- * `ENTITY_REFERENCE_DATE` for fields 9 and 10 = `FIRST_JUDGMENT_DATE`.
- * The four forward-citation counts use only `FIRST_HEARING_DATE` and `FIRST_JUDGMENT_DATE` as specified in fields 19, 20, 31, and 32.

FIELDS

1. "date_of_judgment"

List of judgment date(s), in YYYY-MM-DD format.

Example: ["2024-07-15"].

2. "date_of_hearing"

List of hearing date(s), in YYYY-MM-DD format.

If unavailable, output [].

Do not use ranges. Expand ranges into separate dates where possible.

3. "date_filing_the_complaint"

List of claim-form, revocation-application, counterclaim, or appeal filing date(s).

If unavailable, output [].

If only month and year are available, use YYYY-MM.

If only year is available, use YYYY.

4. "judges"

Names of all judges. Return a list of strings.

Do not include prefixes such as Sir, Mr, Mrs, The Hon, or Lord Justice.

If the judgment uses only a judicial title, identify the actual name if reliable public research permits.

5. "claimants"

Names of all claimants/appellants in the judgment caption or case title. Return a list of strings.

6. "defendants"

Names of all defendants/respondents in the judgment caption or case title. Return a list of strings.

For every Type 3 patent-application case, the list must include "Comptroller-General of Patents, Designs and Trade Marks". Also include every other defendant/respondent that separately opposes the application. The Comptroller is mandatory, not exclusive.

FIELDS 7–10: LEGAL PARTIES AND ECONOMIC ENTITY REPRESENTATIVES

Fields 5 and 6 are caption-level fields.

Fields 7 and 8 are row-level legal-party fields.

Fields 9 and 10 are row-level economic-entity fields measured at ENTITY_REFERENCE_DATE.

Fields 7–10 must not mechanically follow claimant/defendant status.

* Field 7 identifies the owner/applicant/enforcement legal side for the patent/application in field 15.

* Field 8 identifies the adverse validity challenger legal side.

* Fields 9 and 10 may collapse legal subsidiaries or related parties in fields 7 and 8 to their common parent/group representative under the mandatory parent rules below.

* Therefore, the standardized parent name in field 9 or 10 need not be identical to the legal-party name in field 7 or 8.

7. "right_owner_or_applicant_side"

List only the legal parties that own, apply for, exclusively license, or directly enforce the patent/application in field 15.

Rules:

* This is a patent-right owner/applicant-side field, not a claimant/defendant field.

* Include only parties connected to the specific patent/application in this row.

* Do not include parties merely because they appear in the caption.

* Do not include parties active only on another patent, another issue, damages, costs, or a formal/nominal role.

* If aligned co-owners, co-applicants, exclusive licensees, or enforcement-side parties are treated together, list them together.

* If the owner/applicant side is a defendant, still list it here.

* If ownership or applicant status cannot be determined, output [].

8. "other_side_or_validity_challenger_side"

List only the legal parties that attack, oppose, seek revocation of, or directly challenge the validity/patentability of the patent/application in field 15.

Rules:

* Include only parties connected to the specific patent/application in this row.

* Do not include parties merely because they appear in the caption.

* If several parties jointly challenge the same right and the court treats them together, list them together.

* For every Type 3 patent-application case, include "Comptroller-General of Patents, Designs and Trade Marks" and every third-party opponent that directly opposes or challenges the application in this row.

* The Comptroller is mandatory but does not replace or absorb a separate private or public opponent.

* If the adverse side cannot be determined, output [].

9. "right_owner_or_applicant_party_type_ticker_and_headquarter"

Return a dictionary indexed consecutively from "1". Each object must contain exactly these four fields and

no others:

```
{"1":{"name":"","type":1,"ticker_at_time":[],"headquarter":0}}
```

10. "other_side_or_validity_challenger_party_type_ticker_and_headquarter"

Use the same JSON structure and the same mandatory verification rules as field 9.

For every Type 3 patent-application case, field 10 must contain at least:

```
{"1":{"name":"Comptroller-General of Patents, Designs and Trade Marks","type":3,"ticker_at_time":[],"headquarter":0}}
```

If field 8 contains one or more third-party opponents, add consecutively numbered objects for them. Verify and, where required, collapse each opponent under the historical entity/parent/group rules below. The Comptroller remains a separate mandatory object and must never be collapsed with another opponent.

MANDATORY FIELD 9 / FIELD 10 VERIFICATION RULES

Verify every entry as of ENTITY_REFERENCE_DATE. Do not use current ownership, listing status, ticker, or headquarters when it differs from the historical position on that date.

JSON format:

- * Keys are "1", "2", and so on.
- * Every object has exactly: "name", "type", "ticker_at_time", and "headquarter".
- * "type" and "headquarter" must be numbers, never strings.
- * "ticker_at_time" must always be a list, for example ["NYSE:ABC"]. Use [] when there was no verified ticker.
- * If field 7 or field 8 is [], return {} for the corresponding field 9 or field 10.

Type coding:

- * 0 = individual/person.
- * 1 = private or non-listed company/entity.
- * 2 = listed company. This includes a legal party that is a subsidiary of a listed parent at ENTITY_REFERENCE_DATE; collapse it to the listed parent.
- * 3 = government, UKIPO, Comptroller, or public authority.

Headquarter coding:

- * 0 = United Kingdom.
- * 1 = European Union excluding the United Kingdom, with Switzerland included in code 1.
- * 2 = outside the United Kingdom and the code-1 region.
- * For an individual/person, always use headquarter = 0 unless the judgment, the input, or reliable public research explicitly establishes a different residence or domicile.
- * For the Comptroller or UKIPO, use headquarter = 0.

Parent/group representative rule:

- * If a legal party was a subsidiary of a listed parent at ENTITY_REFERENCE_DATE, collapse it to the listed parent and use the parent's legal name, historical ticker, and headquarters.
- * If multiple legal entities on the same side belonged to the same corporate group or ultimate parent at ENTITY_REFERENCE_DATE, collapse them into one entry.
- * A UK subsidiary of Pfizer is represented by Pfizer Inc., not by the UK subsidiary.
- * A German subsidiary of Clariant is represented by Clariant AG, not by the German subsidiary.
- * If the ultimate parent/group was private, collapse to the private parent/group when the relationship is clear.

* If both parent and subsidiary appear on the same side, normally retain only the parent representative after collapse.

* Do not collapse entities merely because their names are similar. Require reliable evidence of common control or parentage at ENTITY_REFERENCE_DATE.

* If the relevant parent cannot be verified, retain the legal entity separately and use the safest verifiable type/headquarter coding. Mention "insufficient evidence to collapse" briefly in field 29.

Historical ticker rule:

* The ticker must be the ticker that existed at ENTITY_REFERENCE_DATE.

* Use an exchange prefix, for example NYSE:, NASDAQ:, LSE:, EPA:, AMS:, FWB:, SIX:, SWX:, TSE:, TYO:, KRX:, BSE:, NSE:, BIT:, or EBR:.

* If a company had not yet completed its IPO at ENTITY_REFERENCE_DATE, use type = 1 and ticker_at_time = [].

* If a company had already been acquired or delisted before ENTITY_REFERENCE_DATE, use the acquiring/group parent at that date when the parent rule applies.

* Do not use a present-day ticker for an earlier date.

Name rule:

* Standardize the legal name without unnecessary rewriting.

* When collapsing, use the parent's legal name.

* Preserve the correct suffix, such as Inc., Ltd, plc, AG, GmbH, S.p.A., N.V., Corporation, or Limited.

11. "claimant_lawyers"

Names of all lawyers representing the claimants/appellants. Return a list of strings.

If unavailable, output [].

12. "defendant_lawyers"

Names of all lawyers representing the defendants/respondents. Return a list of strings.

If unavailable, output [].

13. "stage"

Return:

* 0 = a summary judgment, strike-out judgment, or other dispositive pre-trial ruling that is INCLUDED under the mandatory SUMMARY JUDGMENT / STRIKE-OUT decision matrix above.

* 1 = merits judgment after trial or a substantive final hearing.

Skip merely interlocutory or post-merits judgments under the scope rules above.

14. "right_type"

Always return 0.

0 = patent or patent application.

Registered designs and all other rights are out of scope.

15. "right_name"

Name, title, and full publication/grant/application number of the patent or patent application.

* Use the exact historically correct version actually addressed by the judgment.

* Include prefixes and suffixes such as GB, EP, UK, WO, A, A1, B, or B1 where applicable.

* If the judgment is unclear, verify the contemporaneous version using official UKIPO/EPO records and patent documents.

* Do not substitute a later granted or later amended version for the version before the court.

16. "right_short_description"

A concise factual description of the invention or patent application.

Use the judgment's wording where possible.

Do not speculate.

17. "google_patent_url"

Raw Google Patents URL for the patent or patent application.

You must research this online.

If no Google Patents page exists, use the best official UKIPO/EPO patent-record URL.

If no reliable URL is found, leave "".

18. "patent_unit"

Begin with "patent as a whole" or "patent application as a whole".

Then identify:

* the claims actually attacked, defended, amended, proposed for amendment, or decided;

* whether each relevant claim set is the application, old/granted, previously amended, or proposed-amended version; and

* whether revocation/refusal applies to the whole right even though only representative claims were analysed.

Examples:

* "patent as a whole; validity attacked on granted claims 1, 3, and 5"

* "patent as a whole; granted claim 1 invalid; proposed amended claim 1 allowed"

* "patent application as a whole; application claims 1–20 refused for obviousness"

* "patent as a whole; whole patent conceded invalid; independent granted claims 1, 10, and 15"

* "patent as a whole; attacked claims not individually identified"

Do not infer that unargued claims were individually invalid merely because the whole patent was revoked. An express whole-patent invalidity concession is recorded as a right-level loss under the special rule above; listing its verified independent claims in field 18 does not create separate rows in this right-level dataset.

FORWARD-CITATION RULES FOR FIELDS 19, 20, 31, AND 32

You must research Google Patents and use only the family-level "Families Citing this family" table.

General rules for all four counts:

* Count citing families, not individual publication rows.

* Each citing family counts once per field.

* Do not use the separate publication-level "Cited by" table for these four fields.

* Do not subtract self-family or self-citation entries shown by Google Patents.

* Apply the relevant cutoff date inclusively: a date equal to the cutoff qualifies.

* If there are no qualifying citing families, return 0.

* If FIRST_HEARING_DATE is unavailable, leave fields 19 and 31 empty rather than inventing a date.

19. "forward_citations_first_hearing_priority_date"

Count families in "Families Citing this family" whose priority date is on or before FIRST_HEARING_DATE.

20. "forward_citations_first_judgment_priority_date"

Count families in "Families Citing this family" whose priority date is on or before FIRST_JUDGMENT_DATE.

21. "backward_citations_1_family_to_family"

Count only the family-to-family portion of the backward citations shown by Google Patents: the rows under "Family Cites Families" or below the "Family To Family Citations" divider.

- * Count the rows in this family-to-family portion only; each displayed cited family counts once.
- * Exclude every ordinary patent-citation row in the first table, meaning the rows above the "Family To Family Citations" divider or before the separate "Family Cites Families" heading.
- * Do not use the forward-looking "Families Citing this family" section. The direction of the words matters: "Family Cites Families" is backward; "Families Citing this family" is forward.
- * Field 21 is therefore field 22 with the first ordinary patent-citation table removed.
- * If the family-to-family portion is empty, return 0 even when the first ordinary patent-citation table contains rows.
- * Do not impose a first-hearing or first-judgment cutoff.
- * If there are no such citations, return 0.

22. "backward_citations_2_all_rows_in_patent_citations"

FIELD 22 IS ALWAYS THE NUMBER IN PARENTHESES IMMEDIATELY BESIDE "Patent Citations": "Patent Citations (N)" means field 22 = N. Extract the parenthesized N, not any number beside another heading.

- * Mandatory verification formula: field 22 = number of ordinary rows in the first patent-citation table + field 21.
- * The parenthesized N beside "Patent Citations" is the value to extract for field 22. Verify it by counting all backward rows: the ordinary first-table rows plus the family-to-family rows.
- * Never use the forward-looking "Cited By (N)" total for field 22. "Cited By" is not a source for either backward field.
- * Count the rows in each backward table exactly once within that table. Do not subtract a row merely because a related citation appears elsewhere.
- * Do not impose a first-hearing or first-judgment cutoff.
- * Field 22 must be greater than or equal to field 21. The difference field 22 - field 21 must equal the number of ordinary rows in the first patent-citation table.
- * If both the first ordinary table and the family-to-family portion are empty, return 0.

Mandatory examples:

- * If the first table has 3 ordinary rows, the family-to-family portion has 0 rows, and "Cited By (9)" is shown, then field 21 = 0 and field 22 = 3. The number 9 is irrelevant.

MANDATORY VISUAL EXAMPLE — FIELDS 21–22

READ THE DIRECTIONS LITERALLY: FIELD 22 is always the parenthesized number immediately beside "Patent Citations" — "Patent Citations (N)" means field 22 = N. Backward 1 excludes the first ordinary patent-citation table; backward 2 includes the whole backward-citation block. Always verify field 22 >= field 21 and never use "Cited By (N)".

Patent Citations (3)

FIELD 22: EXTRACT THIS NUMBER

Publication number	Priority date	Publication date	Assignee	Title
GB2279414A *	1993-06-23	1995-01-04	Frank John Hopson Percival	Machine fluid collector
CA2226014A1 *	1997-12-23	1999-06-23	Nicholas Paul Bada	Oil spill containment system
GB2366343A *	2000-09-05	2002-03-06	Urban Environmental Services L	Oil collection device
Family To Family Citations				

* Cited by examiner, † Cited by third party

Cited By (9)

Publication number	Priority date	Publication date	Assignee	Title
Family To Family Citations				
GB2476093A *	2009-12-11	2011-06-15	Manvers Engineering Ltd	Oil Collecting Device Made Of Hydrophobic And Oleophilic Materials.
GB2487936B *	2011-02-09	2015-08-26	Kevin Laker	Swarf tray
DE102011052506A1 *	2011-08-09	2013-02-14	Christian Schröder	catcher
US20160178124A1 *	2014-12-23	2016-06-23	HalenHardy, LLC	Oil and Grease Recovery Device
GB2568865A	2017-10-15	2019-06-05	Manvers Engineering Ltd	Composite absorbent materials and devices incorporating such materials
GB201803602D0 *	2018-03-06	2018-04-18	Manvers Engineering Ltd	Fluid collector
US11124960B1	2019-03-22	2021-09-21	Halenhardy Llc	Drain protector
GB202012048D0	2020-08-03	2020-09-16	Manvers Engineering Ltd	Oil and fuel collector
CN115111516B *	2022-07-20	2024-05-31	中南大学	A rail lubrication system based on guardrail structure

GB2428032B — the first patent-citation table has 3 ordinary backward rows; the family-to-family portion is empty; "Cited By (9)" is forward and irrelevant. Therefore field 21 = 0 and field 22 = 3.

* If the page shows "Patent Citations (61)", the first ordinary table has 1 row, and the family-to-family portion has 60 rows, then field 21 = 60 and field 22 = 61. The parenthesized 61 is field 22; any "Cited By" total is irrelevant.

23. "grant_date"

Date the patent was granted, in YYYY-MM-DD format.

For a pending or refused application that never reached grant, leave "".

24. "priority_date"

Priority date of the patent/application, in YYYY-MM-DD format.

You must research this online.

If unavailable, leave "".

25. "right_level_winner_owner_vs_other"

Code the outcome from the perspective of the patent owner or applicant:

* 0 = owner/applicant wins at right level: every attacked claim in the operative claim set survives every live validity ground asserted and decided against it; or, after amendment, every attacked claim in the final replacement amended claim set that is substantively assessed survives every live ground asserted and decided against it. A fully successful replacement amended claim set is coded 0 even if old/granted claims were deleted, abandoned, conceded invalid, or held invalid.

* 0.5 = mixed or partial outcome among attacked claims: at least one attacked claim is affirmatively upheld

and survives every live validity ground directed to that claim, but the result does not qualify for 0 because at least one other attacked claim or attacked claim set fails or remains unresolved. A claim that wins one ground but loses another, or has a material live attack left undecided, does not count as surviving.

* 1 = adverse side wins at right level: all attacked claims that are substantively decided fail, and no attacked claim is affirmatively upheld through all live grounds directed to it; this remains 1 even if other claims remain in the patent solely because they were not attacked, were not pursued, or were left undecided. Also code 1 for unconditional revocation, refusal/rejection of the application, or an express concession that the whole patent is invalid, unless a final replacement amended claim set is substantively assessed and fully survives.

MANDATORY ATTACKED-CLAIM OUTCOME RULES:

* Determine the outcome claim-by-claim first, then aggregate only across attacked claims.

* Count only claims or claim sets actually attacked on validity/patentability. A claim that was never attacked cannot rescue the owner from 1 and cannot create 0.5.

* An attacked claim counts as a survivor only if the tribunal affirmatively resolves every live validity ground that it actually decides against that claim in the owner/applicant's favour. If any decided ground succeeds against the claim, that claim is lost. If a material attack on that claim is left unresolved or undecided, do not count that claim as a survivor for field 25.

* Do not infer survival from silence. Unattacked, unargued, withdrawn, abandoned, not-maintained, or undecided claims/grounds do not create an owner win for field 25.

* A bare "revoked unless amended" order does not automatically mean 0.5. If all attacked claims decided in the judgment lose and no replacement amended claim set is substantively assessed and maintained in that judgment, code 1.

* AMENDMENT OVERRIDE: if the old/granted claims fail or are deleted, but a replacement amended claim set is then substantively assessed and every attacked amended claim survives every live ground decided against it, code 0, not 0.5. If the attacked amended set is mixed, code 0.5; if all attacked amended claims fail, code 1.

26. "product_or_process_judicial_quotes"

This field must contain raw judicial text, not a human-written summary.

Extract every materially relevant passage in which the judge describes:

* the patented invention, product, process, manufacturing method, material, component, apparatus, device, or technical field;

* a commercial embodiment; or

* a prior-art product, process, or embodiment relevant to the validity dispute.

Rules:

* Quote literally from the judgment.

* Do not paraphrase, summarize, simplify, or add words.

* Prefer fuller technical passages over generic labels.

* Put all relevant passages in the same field.

* Separate passages using the literal text \n—\n. Do not insert actual line breaks.

* Do not add ellipses unless the ellipsis appears in the judgment.

* Escape internal double quotes correctly for CSV.

* If different claim versions are in issue, include the judicial passages that identify or compare those versions. Put the human-written explanation of the version history in field 29, not here.

* Leave empty only if the judgment genuinely contains no relevant product/process/invention/prior-art description.

27. "issue_dictionary"

Return this dictionary exactly:

```
{  
  "Lack of novelty": "",  
  "Lack of inventive step / obviousness": "",  
  "Insufficiency": "",  
  "Added matter": "",  
  "Excluded subject matter / non-patentability": "",  
  "Priority not entitled": "",  
  "Lack of industrial applicability": "",  
  "Others": ""  
}
```

For each ground:

* 0 = patent owner/applicant wins on that ground at right level.

* 0.5 = mixed or partial outcome on that ground at right level.

* 1 = patent owner/applicant loses on that ground at right level.

* "" = the ground was not live or not decided in the present judgment.

If a ground is withdrawn, abandoned, not maintained, or not pursued and the court does not decide it, leave that ground as "". If no other ground or validity issue is decided, SKIP the entire row under the withdrawal rule above.

Code every live ground separately. Do not invent grounds.

The dictionary must be substantively consistent with field 25 but need not mechanically equal it.

For a whole-patent invalidity concession, code every expressly identified validity ground. If no specific ground is identified, set "Others" = 1 and leave all named grounds as "".

28. "claimant_is_owner_side"

Return:

* 0 if the claimant/appellant side is the patent owner or applicant side.

* 1 if the claimant/appellant side is not the patent owner or applicant side.

For revocation claims brought by a challenger, this will usually be 1.

For patent-application appeals brought by the applicant, this must be 0.

If unclear, leave "".

29. "summary"

Keep the summary under 500 words.

The first segment, before the literal text \n, must state the patent-level validity outcome clearly, such as:

* "whole patent maintained";

* "whole patent revoked";

* "whole patent conceded invalid";

* "patent partially revoked";

* "patent maintained only in amended or narrowed form";

- * "revoked unless amended";
- * "patent application allowed"; or
- * "patent application refused".

The summary must also:

- * identify the case as Type 1, 2, or 3 under the scope filter;
- * identify whether the judgment follows trial/substantive final hearing or is a dispositive pre-trial ruling;
- * name the patent/application and state the correct historical version considered by the judgment;
- * identify the owner/applicant side and adverse validity challenger side;
- * explain why the case is within scope;
- * if invalidity was conceded, state expressly that the concession counts as the right-level validity outcome, identify its timing, and distinguish it from a judicial merits finding;
- * identify the claims and claim versions attacked or decided without treating unargued claims as individually invalid;
- * distinguish application, old/granted, previously amended, and proposed-amended claims where relevant;
- * explain the result on every live validity ground;
- * explain precisely why field 25 is 0.5 if the outcome is partial or mixed;
- * explain why fields 7 and 8 contain the correct row-level legal parties when their alignment differs from claimant/defendant status;
- * briefly state any Fields 9–10 parent/group collapse that is not obvious from fields 7–8;
- * state "insufficient evidence to collapse" if a possible parent relationship could not be verified at ENTITY__REFERENCE__DATE;
- * identify relevant prior art and the claims against which it was compared;
- * distinguish same-case judgments from background/comparator judgments where useful;
- * explain multiple complaint dates if field 3 contains more than one date; and
- * contain no URLs.

30. "link"

Raw URL of the judgment only.

31. "forward_citations_first_hearing_publication_date"

Count families in the Google Patents "Families Citing this family" table whose publication date is on or before FIRST_HEARING_DATE.

Apply all general forward-citation rules stated above field 19.

32. "forward_citations_first_judgment_publication_date"

Count families in the Google Patents "Families Citing this family" table whose publication date is on or before FIRST_JUDGMENT_DATE.

Apply all general forward-citation rules stated above field 19.

FINAL VALIDATION BEFORE OUTPUT

Before returning the csv code block, verify all of the following:

1. The case belongs to exactly one of Types 1–3.
2. The output contains validity only and no infringement row.
3. Type 3 includes the Comptroller/UKIPO in fields 6, 8, and 10 and also retains every qualifying third-party opponent in those fields.

4. An express whole-patent invalidity concession is included with field 25 = 1, while withdrawal, abandonment, non-maintenance, or agreed revocation without an express invalidity concession is not misclassified.
- 4A. Every summary judgment or strike-out row passes the mandatory applicant/winner matrix, directly concerns validity, and has been checked for a later same-case validity hearing or judgment.
- 4B. If revocation or the only live validity argument was withdrawn or abandoned and the court did not decide it, the row is skipped; a withdrawn individual ground is left blank when other grounds were decided.
- 4C. Field 25 follows the mandatory attacked-claim rule: no attacked claim affirmatively survives all of its live grounds = 1; at least one attacked claim affirmatively survives all of its live grounds, but the attacked set is not a complete win = 0.5; all attacked claims in the operative set win = 0; and a fully successful substantively assessed replacement amended claim set = 0 even if old/granted claims were deleted or invalid.
5. The patent/application version is correct as of the judgment date and is the version actually addressed by the judgment.
6. Fields 9 and 10 use historical entity, parent, ticker, and headquarter information as of FIRST__JUDGMENT__DATE.
7. Fields 9 and 10 use numeric type/headquarter values and list-valued ticker_at_time.
8. Forward fields 19, 20, 31, and 32 are family-to-family counts and use the correct date field and cutoff date.
9. Field 21 counts only the backward family-to-family portion and excludes every ordinary row in the first patent-citation table; it never uses the forward "Families Citing this family" section.
10. Field 22 is always the parenthesized N immediately beside "Patent Citations". Verify that N equals the number of ordinary first-table rows plus field 21; never use "Cited By (N)".
11. Field 22 is greater than or equal to field 21, and field 22 minus field 21 equals the number of rows in the first ordinary patent-citation table. If either check fails, recount before output.
12. Every row has exactly 32 double-quoted fields.
13. Every CSV record occupies exactly one physical line, with no actual newline or carriage-return character inside any field.
14. The answer contains exactly one csv code block and no text outside it.

D.2 Second prompt: claim-level validity attacks

Validity-Attack Extraction Prompt

Claim-level patent validity rows | eight attack categories | max-parent dependency_depth | prior-art unit IDs

Updated version: dependency_depth is computed by the max-parent rule. Independent claims are 0. A dependent claim has dependency_depth = 1 + the maximum dependency_depth of its direct parent claims. claim_text_expanded must be a standalone technical definition, must preserve alternative dependency paths as alternatives rather than cumulative limitations, and must contain no patent-claim cross-reference. An express concession must be classified under a specific attack category whenever the judgment identifies one; attack category 8 is only the residual category for expressly conceded invalidity with no identified ground in categories 1-7. For novelty and obviousness, each prior-art document/component is output on its own CSV row, while the final field prior_art_unit_id groups consecutive component rows that the judgment treats together as one prior-art unit. Mandatory discrepancy warnings are now a controlling exception to the eight-block-only output rule: every input claim with zero output rows must be named visibly, and every mismatch between issue_dictionary and the judgment must also be reported. Claim-set completeness is checked only within the patent or patents actually submitted in the current request: warn for missing or extra input claims within each submitted patent, but never warn about claims from other patents in the same judgment that were not submitted.

1. Inputs

You will receive three inputs:

One or more claims text files, covering either one patent from the judgment or multiple patents from the judgment.

A BAILII, Casemine, archive, or other judgment text file or judgment URL.

One or more CSV input rows.

Each CSV input row contains, in order:

claims_text_file_link

judgment_url

google_patents_url

target_claim_label

index

issue_dictionary

Example input row:

file:///Users/nghiapham/Documents/claim/text/EP0958447B1_claims.txt

<https://www.bailii.org/ew/cases/EWPC/2006/49.html>

<https://patents.google.com/patent/EP0958447B1/en>

claim 1

10000

```
'{"Lack of novelty":0,"Lack of inventive step / obviousness":0,"Insufficiency":"","Added matter":"","Excluded subject matter / non-patentability":"","Priority not entitled":"","Lack of industrial applicability":"","Others":"","Conceded invalidity":""}'
```

The issue_dictionary is to double check. If there is a number rather than null (“”) it means it had this ground when you did this the first time. It is not ground truth, you might have made a mistake the first time but make sure to check it. Its possible you hallucinated things that didn’t exist, it is also possible you forgot a ground of attack.

2. Task

Your task is to extract validity-attack data for the target claim identified in target_claim_label. If multiple CSV input rows are provided, process each target claim independently and preserve the first five fields of each row exactly.

3. Eight possible validity attacks

Section name | Attack type

novelty/ só 1 | Novelty

obviousness/ só 2 | Obviousness / inventive step

added matter/ só 3 | Added matter

lack of industrial applicability/ só 4 | Lack of industrial applicability

excluded subject matter/ só 5 | Excluded subject matter

extension by amendment/ só 6 | Extension by amendment

insufficiency/ só 7 | Insufficiency

concede invalid/ só 8 | Conceded invalidity - unspecified ground only

4. Output format

Output exactly eight sections, in the exact order shown above.

Under each section name, output exactly one CSV code block.

If a validity attack is actually addressed for the target claim, put the relevant CSV row or rows inside that attack's CSV code block.

If a validity attack is not actually addressed for the target claim, still output that attack section and still output a CSV code block, but leave the CSV code block completely empty.

Do not output JSON.

Do not include a CSV header.

Do not include explanation, comments, bullet points, or extra text outside the eight attack section names and the eight CSV code blocks, except for the mandatory discrepancy WARNING lines required by Section 4A.

4A. Mandatory discrepancy warnings - explicit exception to the eight-block rule

Before final output, determine the submitted-patent scope, build an expected-claim inventory for each submitted patent, and build an internal claim-coverage ledger for every CSV input row. For each input target claim, identify every validity ground actually addressed by the judgment and count all output rows across the eight CSV blocks. Check `issue_dictionary` against the judgment, but treat the judgment as ground truth.

Submitted-patent scope: the current request may contain all target claims for only one patent from a multi-patent judgment, or it may contain all target claims for multiple or all patents in that judgment. Treat only the patent or patents represented by the claims text file(s), Google Patents URL(s), and CSV input rows deliberately supplied in the current request as submitted. Do not assume that every patent mentioned in the judgment has been submitted merely because the judgment discusses it.

Perform the completeness check separately for each submitted patent. If the current request submits only Patent A, check missing and extra claims only for Patent A. Do not issue any missing-claim, missing-row, or missing-patent warning for Patent B, Patent C, or any other patent from the same judgment that was not submitted in the current request. If multiple patents are submitted, repeat the check independently for each one.

For each submitted patent, build `expected_claim_inventory` from every claim or separately treated claim version of that patent whose validity the judgment actually addresses on at least one of the eight grounds. Distinct granted, amended, proposed-amended, corrected, or other claim versions count as distinct targets only when the judgment treats them separately. Do not include claims of an unsubmitted patent in this inventory.

Compare the CSV input rows for each submitted patent against that patent's `expected_claim_inventory`. Match by patent identity and the complete `target_claim_label`, including any version qualifier such as granted, amended, proposed amended, corrected, claim-set label, or dependency qualifier. Do not match merely by bare claim number when the judgment distinguishes versions.

If an expected claim or separately treated claim version of a submitted patent has no CSV input row, output exactly one visible warning in this format: `WARNING - MISSING INPUT CLAIM: patent="[patent identifier]"; expected_target_claim_label="[expected target claim label]";` the judgment addresses this claim of a patent submitted in the current request on at least one of the eight validity grounds, but no CSV input row was supplied; no CSV row can be produced for this missing input claim.

If a CSV input row does not correspond to any claim or separately treated claim version in `expected_claim_inventory` for its submitted patent, output exactly one visible warning in this format: `WARNING - EXTRA INPUT CLAIM: patent="[patent identifier]"; target_claim_label="[target_claim_label]"; index="[index]";` this input claim is not an expected validity target for this submitted patent in the judgment; review or remove this CSV input row.

A duplicate CSV input row for the same submitted patent and the same complete `target_claim_label` is also extra unless the judgment genuinely treats the rows as distinct claim versions and the labels distinguish those versions. Multiple output rows generated from one legitimate input claim because of different prior-art units or component documents do not make the input claim extra.

The missing/extra completeness check concerns the CSV input claim set, not the complete numbered claim set printed in the patent. A claim that merely exists in the claims text file is not expected unless the judgment actually addresses its validity on at least one of the eight grounds. Conversely, a claim of a submitted patent that the judgment does address on such a ground must be reported as missing if its CSV input row is absent.

If an input target claim produces zero CSV rows because the judgment does not attack that claim on any of the eight validity grounds, output exactly one visible warning for that input row in this format: WARNING - ZERO OUTPUT ROW: target_claim_label="[target_claim_label]"; index="[index]"; the judgment does not attack this target claim on any of the eight validity grounds; no CSV row is output for this input claim.

The ZERO OUTPUT ROW warning is mandatory even when issue_dictionary is entirely empty. An input target claim that produces no CSV row must never silently disappear from the final answer.

If the judgment attacks an input target claim on a ground whose issue_dictionary field is empty, output one visible warning in this format: WARNING - MISSED GROUND: target_claim_label="[target_claim_label]"; index="[index]"; issue_dictionary leaves "[attack category]" empty, but the judgment attacks this target claim on that ground; the corresponding CSV row(s) are included.

If issue_dictionary lists a ground for an input target claim but the judgment does not attack that target claim on that ground, output one visible warning in this format: WARNING - HALLUCINATED GROUND: target_claim_label="[target_claim_label]"; index="[index]"; issue_dictionary lists "[attack category]", but the judgment does not attack this target claim on that ground; no CSV row is output for that ground.

Apply the missing-input-claim, extra-input-claim, zero-output-row, missed-ground, and hallucinated-ground warning tests as specified above. Output one WARNING line for each applicable discrepancy. Place all WARNING lines in the final answer immediately before novelty/ só 1. These WARNING lines are mandatory and are the only permitted text outside the eight section names and eight CSV code blocks. This exception overrides every instruction in this prompt that says "nothing else", "no extra text", or "only the eight sections". Never hide a required warning in analysis, reasoning, commentary, a CSV block, or any content that may be collapsed.

If none of these discrepancies exists, output no WARNING line and follow the normal eight-section-only format.

5. Row granularity

For each validity attack that is actually addressed for the target claim, output the CSV row or rows required by the rules below.

For novelty and obviousness, first identify each distinct prior-art unit considered by the judgment. A prior-art unit may consist of one document, multiple documents treated together as one combined reference/disclosure/attack, or Common general knowledge. Output one CSV row per component document, not one row per prior-art unit. A one-document unit therefore produces one row; a multi-document unit produces one row for each component, with all of those component rows grouped by the same prior_art_unit_id. Exception: if the patent owner expressly concedes that the target claim lacks novelty or is obvious but the judgment identifies no prior-art unit for that concession, output exactly one row for the target claim under that specific attack and leave prior_art_name_less_than_20_words, prior_art_link_status, prior_art_full_text_link, and prior_art_unit_id empty.

For obviousness/inventiveness, Common general knowledge may be output as its own single-row unit if the judgment actually addresses it as part of the obviousness analysis for the target claim. Give that row its own prior_art_unit_id.

Do not combine Common general knowledge and a named prior-art document in the same CSV row. If the judgment treats them together as one combined obviousness unit, output separate component rows and assign the same prior_art_unit_id; otherwise assign different IDs.

Do not collapse genuinely distinct prior-art units together. If the judgment treats two or more documents together as one prior-art reference, one combined disclosure, or one combined attack, output each component document on its own CSV row and give every component row of that unit the same prior_art_unit_id.

Keep the component rows for one unit consecutive.

Treat multiple documents as the same prior-art unit when they are copies, translations, versions, parts, appendices, or evidentiary records of the same underlying prior-art disclosure and the judgment analyzes them as one unit. In that case, output separate component rows with the same `prior_art_unit_id`. Do not assign the same ID merely because documents concern the same technology, belong to the same patent family, have the same author, or are cited near each other if the judgment analyzes them as distinct prior art.

For a multi-document prior-art unit, `prior_art_name_less_than_20_words` must identify only the specific component document represented by that row. Do not concatenate multiple component names with " + ", semicolons, pipes, or any other delimiter. Keep each component name under 20 words.

For novelty and obviousness, assign `prior_art_unit_id` as a zero-based integer string in judgment order, starting at "0" separately for each target claim and each attack section. All rows belonging to the same prior-art unit must have the same ID and must be consecutive. Different prior-art units must have different IDs. Example: if documents A and B are treated together as one prior-art unit and document C is a separate unit, output A with `prior_art_unit_id` = "0", B immediately after it with `prior_art_unit_id` = "0", and C with `prior_art_unit_id` = "1".

Do not group different validity attacks together. `prior_art_unit_id` is local to the target claim and attack section and does not link rows across novelty and obviousness.

For added matter, lack of industrial applicability, excluded subject matter, extension by amendment, insufficiency, and residual concede invalid, output one CSV row for the attack if that attack is actually addressed for the target claim, unless the judgment clearly treats multiple distinct attacks of the same type separately for the same target claim.

Apply ground-specific classification before concede invalid. If the judgment states that the patent owner concedes, accepts, or no longer contests invalidity for lack of novelty, obviousness, added matter, lack of industrial applicability, excluded subject matter, extension by amendment, or insufficiency, output the row only in the corresponding specific section (1-7), with `patent_owner_loses` = "1". The fact that the result arose by concession does not create an additional concede invalid row.

Use concede invalid (section 8) only as a residual category: the judgment must expressly concede, accept, or no longer contest the invalidity of the target claim, and must not identify any corresponding ground in sections 1-7 for that concession. Do not infer a concession from amendment, withdrawal, silence, non-appearance, or the general result of the case. Do not infer a specific ground from an unspecified concession.

For every residual concede invalid row, `patent_owner_loses` must be "1" and `prior_art_name_less_than_20_words`, `prior_art_link_status`, `prior_art_full_text_link`, and `prior_art_unit_id` must all be empty. Classify separately claim by claim: a ground-specific concession for one target claim does not determine the classification of another target claim.

6. Output row schema

Each output row must begin with the exact same five fields from the input CSV row, in the same order. After those five original fields, append the following extra fields in this exact order:

`claim_text_literal`

`claim_text_expanded`

`dependency_depth`

`patent_owner_loses`

`prior_art_name_less_than_20_words`

`amended_claim`

`prior_art_link_status`

`prior_art_full_text_link`

prior_art_unit_id

7. Field definitions

Field | Instruction

`claim_text_literal` | Output the literal text of the target claim. Prefer the wording as reproduced in the judgment. If the judgment does not reproduce the full claim text, use the claims text file. Do not paraphrase. For `amended_claim = "1"`, the controlling text is the amended, proposed-amended, corrected, or otherwise modified claim version reproduced, approved, or analyzed in the judgment, including any schedule, annex, or order. Use the claims text file only as a fallback for wording that the judgment does not reproduce and only after confirming that the relevant wording was not amended, deleted, renumbered, disclaimed, or replaced. Never substitute superseded wording from the old claim set.

`claim_text_expanded` | For `amended_claim = "1"`, first reconstruct the controlling amended claim set from the judgment, including all renumbering, changed dependency references, inserted limitations, deletions, substitutions, provisos, and disclaimers. Build `claim_text_expanded` from that amended claim set and use the judgment's amended wording wherever it is available. Do not trace a direct parent or ancestor by the numbering or dependency graph of a superseded claims text file. The claims text file is only a fallback for the exact wording of a portion that the judgment does not reproduce and that is demonstrably unchanged. Never mix wording or dependency nodes from different claim versions in one expansion. If `dependency_depth = "0"`, this must be exactly identical to `claim_text_literal`. If `dependency_depth` is greater than "0", construct a standalone technical definition that preserves every substantive technical limitation of the target claim and every inherited limitation required by each applicable dependency path. First identify the target claim's direct parents from its dependency wording. Recursively expand each direct parent along its own path until an independent claim is reached; do not stop at the immediate parent. Remove or minimally rewrite only patent-claim cross-references and dependency scaffolding such as "Claim 1:", "according to claim 4", "claims 2-4", or "the solution of claim 5". Preserve the substantive wording literally as far as possible and add only the minimum bridging language needed to produce grammatical standalone text. Do not remove the word "claim" when it is part of the substantive technical subject matter, such as an insurance-claim processing system. For alternative direct parents `p1`, `p2`, ..., first construct one complete expansion for each path. The result must be logically equivalent to $[E(p1) + N(c)]$ OR $[E(p2) + N(c)]$ OR ..., where $E(p)$ is the recursively expanded parent text and $N(c)$ contains the substantive limitations newly added by the target claim. Present these branches in one field using "or", "any one of the following alternatives", or equivalent standalone wording. Identical wording shared by every branch may be factored out only when the result remains exactly logically equivalent. Never combine branch-specific limitations cumulatively. Every limitation newly added by the target claim must apply to every alternative. Neutral bridging language such as "no additional limitation as to dosage form" may be introduced only when needed to describe a branch that adds no limitation on that dimension; it must not add new technical substance.

`dependency_depth` | Output the dependency depth using the max-parent rule: independent claim = "0"; dependent claim = 1 + the maximum `dependency_depth` of all direct parents selected by the target claim's dependency wording. Claims inherited only through a direct parent are ancestors and enter recursively; do not add them again as direct parents. Use "1", "2", "3", etc. Do not cap at "2" unless explicitly instructed. See Section 8 for examples. For `amended_claim = "1"`, identify every direct parent and calculate every depth from the controlling amended claim set, never from the old claims file.

`patent_owner_loses` | Output "0" if the patent owner wins on this validity attack for this target claim. Output "1" if the patent owner loses. For novelty and obviousness, decide the outcome separately for each prior-art unit or Common general knowledge unit. If one prior-art unit has multiple component rows, repeat the same `patent_owner_loses` value on every row sharing that `prior_art_unit_id` because those rows represent one combined attack. A ground-specific concession and a residual concede invalid row both require "1" because the patent owner has expressly conceded, accepted, or ceased to contest invalidity of the target claim on the relevant basis. Do not use outcomes from other claims. Do not guess from the general result of the case.

`prior_art_name_less_than_20_words` | For novelty and obviousness, output the name of the specific prior-art document/component represented by this row, or "Common general knowledge" for a Common

general knowledge row. If multiple documents form one prior-art unit, do not combine their names in one field; each component must have its own row and its own concise name under 20 words. The shared `prior_art_unit_id`, not name concatenation, indicates that multiple rows belong to one combined prior-art unit. If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, output an empty CSV field. For all other validity attacks, output an empty CSV field.

`amended_claim` | Output "1" if the target claim is an amended claim. Output "0" if the target claim is not an amended claim.

`prior_art_link_status` | For novelty and obviousness only, assess this field separately for the specific component document represented by the row. Output "2" if `prior_art_full_text_link` provides full text for that component; output "1" if it provides only an abstract, summary, extract, citation record, bibliographic record, citation in another document, or another identifiable non-full-text description; output "0" if no identifiable source link is available for that component. For a multi-document prior-art unit, do not calculate one aggregate link status for the whole unit; each component row has its own status. For Common general knowledge, output "0". If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, output an empty CSV field rather than "0". For all other validity attacks, output an empty CSV field.

`prior_art_full_text_link` | For novelty and obviousness, output the single best available link for the specific prior-art document/component represented by this row. Prefer full text. For patent prior art, prefer a Google Patents full-text link if the patent number is clearly identified. For non-patent prior art, use the best available reliable link if full text is unavailable, including an abstract, summary, bibliographic record, citation record, or another supplied document that clearly identifies the same specific reference. Never place links for two or more prior-art components in one CSV field, whether separated by semicolons, " | ", commas, "+", or any other delimiter. If multiple documents form one prior-art unit, output separate component rows and group them with the same `prior_art_unit_id`. For Common general knowledge and all non-novelty/non-obviousness attacks, output an empty CSV field. If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, also output an empty CSV field.

`prior_art_unit_id` | For novelty and obviousness only, use this final field to identify which component rows belong to the same prior-art unit. For each target claim and each attack section separately, number distinct prior-art units in judgment order using zero-based integer strings: "0", "1", "2", and so on. All component rows of one multi-document unit must share the same ID and be consecutive. A one-document unit has one row with its own ID. Common general knowledge, when output as its own unit, also receives its own ID. Do not reuse an ID for a different unit within the same target claim and attack section. Numbering may restart at "0" for a different target claim or for the other attack section. If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit, output an empty CSV field. For added matter, lack of industrial applicability, excluded subject matter, extension by amendment, insufficiency, and residual concede invalid, output an empty CSV field.

8. Dependency-depth rules

`dependency_depth = "0"`: target claim is independent; `claim_text_expanded` must be identical to `claim_text_literal`.

Mandatory source-of-truth rule for amended claims: when `amended_claim = "1"`, treat the amended or proposed-amended claim set stated in the judgment, schedule, annex, or order as the controlling and closed dependency graph. Every claim number in a dependency reference means the claim with that number in that amended set. NEVER trace the target, a direct parent, or an ancestor through the numbering or dependency wording of the old or granted claims file.

For an amended claim, use the judgment's wording for the target claim and for every amended parent or ancestor wherever that wording is provided. The claims text file may supply only exact wording omitted from the judgment and confirmed to be unchanged. If the judgment changes claim 1 but leaves a dependent claim's own wording unchanged, expand that dependent claim through the amended claim 1, not the old claim 1.

Apply every amendment that changes substantive scope, even if it appears outside the numbered claim text: include inserted disclaimers, provisos, substitutions, deletions, and limitations stated in the judgment, schedule, annex, or order. Do not omit such text merely because it is absent from the claims text file.

Renumbering/version check: before expanding an amended target, make an internal mapping of each amended claim number to its direct parent or parents. If the judgment omits, deletes, consolidates, or renumbers old claims, do not carry the old numbering forward and do not assume that amended claim N corresponds to old claim N.

Mandatory trace check for amended claims with `dependency_depth` of "2" or greater: write an internal path ledger for every alternative path, listing the target, each direct parent, every ancestor, the claim-set version/source used for that node, and the calculated depth. Verify that every path ends at an independent claim in the same amended claim set and that no node was silently taken from a superseded claim set. If the amended dependency path cannot be reconstructed unambiguously from the available materials, do not guess or import the old path.

For any dependent claim, first identify its direct parents from the dependency wording in the applicable controlling claim set. A direct parent is a claim expressly selected by the dependency phrase. An earlier claim inherited only through a direct parent is an ancestor, not an additional direct parent. Calculate recursively from that same controlling claim set: $\text{dependency_depth}(\text{independent claim}) = 0$; $\text{dependency_depth}(\text{dependent claim}) = 1 + \max(\text{dependency_depth}(\text{direct parent})$ for every direct parent selected by the target claim).

If a claim refers to "any preceding claim", "any one of the preceding claims", "any of claims 1 to N", or similar wording, treat every claim in that referenced range as a direct-parent alternative for the target claim. Ancestors of those alternatives enter only through recursive expansion.

`dependency_depth = "1"`: every expressly listed parent claim is independent, e.g. claim 3 according to claim 1 or claim 2 where both claim 1 and claim 2 are independent.

`dependency_depth = "2"`: at least one expressly listed parent has `dependency_depth` "1" and no expressly listed parent has a greater depth, e.g. claim 3 according to claim 1 or claim 2 where claim 1 is independent and claim 2 depends on claim 1.

Use "3", "4", etc. for deeper dependency chains. Do not cap at "2" unless explicitly instructed.

Do not use the shortest path where alternative direct parents have different depths. Use the maximum direct-parent depth because `claim_text_expanded` must account for every dependency alternative selected by the target claim.

Do not create extra rows merely because a dependent claim has alternative dependency paths, unless the judgment actually treats those alternatives differently for validity. Preserve the "or" dependency language in one expanded text where appropriate.

When expanding a dependent target with alternative direct parents, construct each complete path separately before combining the paths with "or". Write `claim_text_expanded` as one standalone technical definition using wording such as "any one of the following alternatives: ...; ...; or ...". Do not label branches as patent claims and do not leave patent-claim cross-references in the expanded text.

If preserving literal dependency wording would leave a patent-claim cross-reference inside `claim_text_expanded`, rewrite only the dependency-reference wording into standalone language. Preserve every substantive technical limitation. The first priority is logical equivalence: no missing inherited limitation, no new technical limitation, and no misleading cumulative combination of alternative branches.

Alternative-path check: if the direct parents are p1, p2, ..., verify that the expanded text means $[E(p1) + N(c)]$ OR $[E(p2) + N(c)]$ OR ..., not $E(p1) + E(p2) + \dots + N(c)$.

Worked example: claim 1 is independent; claim 2 depends on claim 1; claims 3 and 4 each depend on claim 2; and claim 5 depends on any one of claims 2-4 while adding a concentration limitation. The direct parents of claim 5 are claims 2, 3, and 4. Its longest path is 5 -> 3 or 4 -> 2 -> 1, so `dependency_depth = "3"`.

For the worked example, the three alternatives are: the composition inherited through claim 2 with no additional dosage-form limitation; the same composition inherited through claim 3 and restricted to a cream; or the same composition inherited through claim 4 and restricted to a gel. The concentration limitation added by claim 5 applies to all three alternatives.

A correct `claim_text_expanded` for the worked example is: "A pharmaceutical composition containing calcipotriol monohydrate: (1 α , 3 β , 5Z, 7E, 22E, 24S)-24-cyclopropyl-9,10-secochola-5,7,10(19),22-tetraene-1,3,24-triol, monohydrate, and having an active-component content of 1-100 μ g/g of the composition, where either (i) no further limitation is imposed on the dosage form; (ii) the composition is a cream; or (iii) the composition is a gel."

9. Prior-art link rules

Do not leave `prior_art_full_text_link` empty merely because the available source is only an abstract, summary, extract, citation record, bibliographic record, or citation in another document.

The link must identify the same specific prior-art reference being compared.

Do not output an unrelated modern page, generic search result, or source that does not clearly identify the prior-art reference.

If a prior-art component is identified but no identifiable link is available for that component, output an empty `prior_art_full_text_link` and set that row's `prior_art_link_status` to "0". If a ground-specific concession of lack of novelty or obviousness identifies no prior-art unit at all, leave `prior_art_name_less_than_20_words`, `prior_art_link_status`, `prior_art_full_text_link`, and `prior_art_unit_id` empty.

When multiple documents form one prior-art unit, output one consecutive CSV row per component document. Put only that component's best available link in `prior_art_full_text_link`. Never put multiple component links side by side in one field. Give all component rows of the same unit the same `prior_art_unit_id`.

For a multi-document prior-art unit, determine `prior_art_link_status` separately on each component row: "2" if full text is available for that component, "1" if only an identifiable non-full-text source is available for that component, and "0" if no identifiable source link is available for that component. Do not downgrade or upgrade one component because of link availability for another component.

10. Important rules

Always output all eight attack sections and all eight CSV code blocks.

Do not omit an attack section merely because the attack is not addressed.

If an attack is not addressed for the target claim, its CSV code block must be empty.

Do not infer an attack merely because it is mentioned elsewhere in the judgment.

The validity attack must be applied to the target claim.

A concession must first be assigned to any specific validity ground expressly identified by the judgment. For example, a claim conceded to be obvious belongs only in obviousness/ *só* 2, with `patent_owner_loses` = "1"; do not duplicate it in concede invalid/ *só* 8. Apply the same rule to every specific ground in sections 1-7.

A concede invalid row requires an express concession or acceptance of invalidity as to the target claim and is permitted only when no specific ground in sections 1-7 is identified for that concession. Do not infer it from a concession concerning another claim, from a proposed amendment, or from the patent-level outcome.

For residual concede invalid, `patent_owner_loses` must be "1". The prior-art name, link-status, link, and `prior_art_unit_id` fields must be empty.

Preserve the first five input fields exactly as CSV fields.

Quote every CSV field with double quotes.

Escape internal double quotes by doubling them.

Do not put line breaks inside any CSV field. Each CSV row must be one single physical line.

For an empty CSV field, output "".

The `prior_art_link_status` field must appear immediately before `prior_art_full_text_link`, and `prior_art_unit_id` must appear immediately after `prior_art_full_text_link` as the final field of every output row.

For novelty and obviousness rows tied to an identified prior-art unit, `prior_art_link_status` must be "2", "1", or "0" for the specific component row and `prior_art_unit_id` must be a zero-based integer string identifying

its unit. For a ground-specific concession of lack of novelty or obviousness with no identified prior-art unit, `prior_art_name_less_than_20_words`, `prior_art_link_status`, `prior_art_full_text_link`, and `prior_art_unit_id` must all be empty. For all other validity attacks, `prior_art_link_status` and `prior_art_unit_id` must be empty CSV fields.

Do not output null, None, N/A, unknown, or any other placeholder.

Be careful not to output malformed CSV such as quadruple quotes.

The final answer must contain exactly eight section names and exactly eight CSV code blocks. The only permitted additional text is the mandatory discrepancy WARNING lines required by Section 4A.

11. Mandatory internal validation before final output

Create the submitted-patent scope, per-patent expected `claim_inventory`, and claim-coverage ledger required by Section 4A before drafting the final answer. The ledger must contain one entry for every CSV input row, keyed by patent identity, `target_claim_label`, and index.

For each submitted patent, compare expected `claim_inventory` against the CSV input rows for that patent. Emit one MISSING INPUT CLAIM warning for every expected claim or separately treated claim version with no input row, and one EXTRA INPUT CLAIM warning for every supplied row that does not correspond to the expected inventory or is an unjustified duplicate.

Recheck the patent-scope boundary before emitting completeness warnings. Never emit a missing warning for a claim belonging only to a patent from the same judgment that was not submitted in the current request.

For every input row, count its CSV rows across all eight attack blocks. If the count is zero, verify that the final answer contains the mandatory ZERO OUTPUT ROW warning for that input row, even when `issue_dictionary` is entirely empty.

Compare the judgment-determined attack grounds with `issue_dictionary` for every input target claim. Emit a MISSED GROUND warning for every judgment-only ground and a HALLUCINATED GROUND warning for every `issue_dictionary`-only ground.

Do not finalize while any submitted patent's expected claim is missing without a MISSING INPUT CLAIM warning, any extra input claim lacks an EXTRA INPUT CLAIM warning, any input target claim has silently disappeared, or any other required discrepancy warning is absent. Required WARNING lines must be visible immediately before novelty/ só 1 in the final answer, not merely mentioned in analysis or commentary.

This validation checklist is internal only. Do not output the checklist. Output any mandatory discrepancy WARNING lines required by Section 4A, followed by exactly the required eight sections and eight CSV code blocks.

Reconstruct the dependency chain for every target claim before drafting rows. For an unamended claim, use the applicable claims text. For `amended_claim = "1"`, use the controlling amended claim set from the judgment, schedule, annex, or order; use the claims text file only for wording that is omitted and confirmed unchanged.

For every `amended_claim = "1"` row, verify that the target, every direct parent, and every ancestor belong to the same controlling amended claim set. Fail the row if any dependency node or claim number was taken from the old or granted claim set merely because it was available in the claims text file.

For every amended claim with `dependency_depth` of "2" or greater, inspect the complete internal path ledger node by node and branch by branch. Verify the amended numbering, direct-parent choices, maximum depth, independent endpoint, and source/version for every node before accepting `claim_text_expanded`.

For every amended claim, compare `claim_text_literal` and `claim_text_expanded` against the judgment's amendment language and verify that all inserted limitations, disclaimers, provisos, substitutions, deletions, and renumbering effects are reflected. Do not allow old wording to overwrite or dilute amended wording.

If `dependency_depth = "0"`, verify `claim_text_expanded` is exactly identical to `claim_text_literal`.

If `dependency_depth > "0"`, verify `claim_text_expanded` is not identical to `claim_text_literal`.

If `dependency_depth > "0"`, verify `claim_text_expanded` preserves every substantive technical limitation of

the target dependent claim and incorporates every inherited limitation required by each applicable dependency path. Patent-claim cross-references and dependency scaffolding may and must be removed or minimally rewritten.

If `dependency_depth > "0"`, verify `claim_text_expanded` contains no patent-claim cross-reference or dependency scaffolding such as "claim 1", "claims 2-4", or "according to claim 3". Do not reject a row merely because the word "claim" forms part of the substantive technical subject matter.

If the target has alternative dependency paths, verify `claim_text_expanded` preserves the alternative logic using "or", "any one of the following", or equivalent standalone wording, rather than stacking all parent alternatives cumulatively.

For multi-level dependencies, do not stop at the immediate parent claim. Continue until an independent claim is reached.

Verify the numeric `dependency_depth` follows the recursive max-parent rule: independent claims are 0; dependent claims equal 1 plus the maximum `dependency_depth` of every direct parent selected by the dependency wording.

If `dependency_depth` and `claim_text_expanded` are inconsistent, the row is invalid and must be fixed before output.

Verify `amended_claim` is correct for the target claim. If `amended_claim = "1"`, also verify that `claim_text_literal`, `claim_text_expanded`, `dependency_depth`, and every dependency path were derived from the same controlling amended claim set rather than the old claims file.

Verify `patent_owner_loses` is decided separately for each attack and each prior-art unit. If a unit has multiple component rows, verify every row sharing its `prior_art_unit_id` repeats the same `patent_owner_loses` value.

For novelty and obviousness, verify that every specific prior-art document/component has its own CSV row. If multiple documents are treated together as one prior-art unit, verify that their rows are consecutive and share the same `prior_art_unit_id`. Verify that genuinely distinct prior-art units have different IDs.

For every multi-document prior-art unit, verify that no component names or links are concatenated in one field; each row contains only one component name, one component-specific `prior_art_link_status`, and at most one best link. Verify that all component rows share the same zero-based `prior_art_unit_id`, are consecutive, and that IDs increase in judgment order starting at "0" for each target claim and attack section.

For every express concession, verify first whether the judgment identifies a specific ground in sections 1-7. If it does, verify that the row appears only in that specific section, `patent_owner_loses` is "1", and no duplicate row appears in concede invalid/ só 8.

For a ground-specific concession of lack of novelty or obviousness with no identified prior-art unit, verify that exactly one row is output for the target claim in the specific section and that `prior_art_name_less_than_20_words`, `prior_art_link_status`, `prior_art_full_text_link`, and `prior_art_unit_id` are all empty; do not invent a prior-art name, unit ID, or use link status "0".

For every concede invalid row, verify that the concession is express, applies to the target claim, identifies no ground in sections 1-7, and has `patent_owner_loses = "1"`.

For every residual concede invalid row, verify that `prior_art_name_less_than_20_words`, `prior_art_link_status`, `prior_art_full_text_link`, and `prior_art_unit_id` are empty.

Verify no CSV row contains line breaks, every field is double-quoted, internal double quotes are escaped by doubling them, and no malformed quadruple quotes appear.

12. Required final answer skeleton

[zero or more mandatory WARNING lines under Section 4A, only when applicable]

novelty/ só 1

““csv

““

obviousness/ só 2

“‘csv

“‘

added matter/ só 3

“‘csv

“‘

lack of industrial applicability/ só 4

“‘csv

“‘

excluded subject matter/ só 5

“‘csv

“‘

extension by amendment/ só 6

“‘csv

“‘

insufficiency/ só 7

“‘csv

“‘

concede invalid/ só 8

“‘csv

“‘

Final check: make sure claim_expanded is the same for the same claims

Final check: with the exception of novelty and obviousness, each claim only has 1 row for other fields.

Final check: if the judgment copies the exact prior art then it is also type 2

Final check: same prior arts must have the same link. For example link for prior art A used for claim 1 must have the same link as prior art A used for claim 2

Final check: verify the submitted-patent scope, each submitted patent's expected_claim_inventory, and the claim-coverage ledger. Every expected validity target of every submitted patent must have an input row or a visible MISSING INPUT CLAIM warning; every extra input claim must have an EXTRA INPUT CLAIM warning. Do not check or warn about missing claims from patents in the judgment that were not submitted in the current request. Every input target claim must either produce at least one CSV row or a visible ZERO OUTPUT ROW warning. Every ground mismatch between issue_dictionary and the judgment must produce a visible MISSED GROUND or HALLUCINATED GROUND warning. Place these warnings immediately before novelty/ só 1; this requirement overrides every "nothing else" rule in this prompt.

D.3 Third prompt: patent text extraction

Patent extraction

attention: before extracting make sure this is the correct patent in the case

You are given a Google Patents link or a patent PDF. Extract only the English patent text and create exactly two downloadable UTF-8 plain text files.

IMPORTANT SOURCE RULE:

- If a Google Patents link is provided, use the Google Patents HTML text as the primary source.
- Use the PDF only as a fallback when the HTML text is missing, incomplete, or clearly corrupted.
- Use OCR only if both HTML text and PDF text layer are unavailable or unusable.
- Do not OCR chemical structures, formula images, drawings, or tables into garbage text.

File naming rule:

- Infer the patent publication number from the patent, e.g. EP0383425B1.
- Create exactly:

1. EP0383425B1_description.txt
2. EP0383425B1_claims.txt

Do not create JSON, CSV, DOCX, or PDF. Create only .txt files.

1. DESCRIPTION FILE

Put only the English Description section into the description file.

Start at the heading "Description", "Specification", or the equivalent English heading that begins the substantive patent description.

Stop immediately before the heading "Claims".

Do not include:

- front-page bibliographic metadata,
- abstract-only front matter,
- publication number repeated in headers,
- page numbers,
- column line numbers,
- OCR artifacts,
- drawings-only pages,
- table-only pages,
- German/French claim translations,
- "Patentansprüche",
- "Revendications".

2. CLAIMS FILE

Put only the English Claims section into the claims file.

Start at the heading "Claims".

Stop immediately before any non-English claim heading, such as "Patentansprüche" or "Revendications", or before drawings-only/table-only pages.

Preserve claim numbers exactly.

Preserve claim dependencies exactly.

Keep each claim separated by a blank line.

3. TABLES

Do not reproduce tables.

Do not convert tables into plain text.

Do not include table image content.

Keep table references in the description text, e.g. "Table I shows..." or "as shown in Table II".

Keep table captions/titles if they appear as part of the English description text.

For table-only pages or table blocks, insert only:

[TABLES_NOT_INCLUDED]

Do not describe the tables from visual inspection unless the surrounding description text itself describes them.

4. FORMULAS / CHEMICAL STRUCTURES / EQUATIONS

The goal is to produce clean NLP-friendly patent text for embedding models. Do not preserve chemical-structure garbage, layout-based formulas, or OCR formula fragments.

Keep ordinary machine-readable chemical and technical text, including:

- chemical names, material names, compound names, polymer names, and reagent names;
- simple molecular formulas only when they appear inline as ordinary text, e.g. H₂O, CO₂, NaCl, HCl;
- substituent definitions, e.g. R is hydrogen, alkyl or aryl;
- chemical class or range expressions, e.g. C1-C6 alkyl, C1-C4 alcohol, halogen, aryl, alkoxy;
- composition ranges, ratios, concentrations, percentages, units, temperatures, pressures, pH ranges, and reaction conditions;
- simple arithmetic equations or inequalities using +, -, ×, /, =, <, >, ≤, ≥, powers, percentages, ratios, ranges, and units.

Remove chemical formulas or structures that are not useful as ordinary text embeddings, including:

- chemical structure drawings;
- Markush structures;
- reaction schemes;
- structural formulas represented by bonds, rings, branches, or diagram layout;
- isolated chemical fragments such as =C—C=C=, =C—C—H, —CH₂—CH₂—, unless they occur as clean inline text inside a normal sentence;
- multi-line chemical formulas;
- formula images;
- OCR-corrupted chemical formulas;
- ASCII-art or layout-based chemical structures.

Replacement rules for chemical content:

- If a chemical structure, Markush formula, reaction scheme, or structural formula is removed, write:

[CHEMICAL_STRUCTURE_REMOVED]

- If a formula is OCR-corrupted or uncertain, write:

[FORMULA_OCR_REMOVED]

- If several adjacent chemical-structure lines belong to the same formula block, replace the whole block with a single placeholder only.
- Do not convert chemical structure images into LaTeX.
- Do not attempt to reconstruct chemical structures from visual inspection.
- Do not include ASCII-art formula fragments.

- Do not invent missing formula content.

For mathematical formulas:

- Keep simple arithmetic equations, ratios, powers, inequalities, and simple systems of equations if they are clean and machine-readable.
- Keep simple expressions such as $x + y = z$, $a^2 + b^2 = c^2$, $A:B = 1:3$, or $0.1 \leq x \leq 5$.
- Remove complex mathematical notation, including integrals, derivatives, summations/products, vectors, matrices, tensors, and multi-line displayed equations that cannot be represented cleanly as simple text.
- Replace removed complex mathematical formulas with:

[COMPLEX_FORMULA_REMOVED]

5. FIGURES / DRAWINGS

Do not include image files or attempt to reproduce drawings.

Keep figure references in the description text, e.g. "Figure 1 shows..."

Keep figure captions if they appear in the English description.

For drawings-only pages or drawing-only blocks, insert only:

[DRAWINGS_NOT_INCLUDED]

Do not describe the figures from visual inspection unless the description text itself describes them.

6. OCR / SCANNED PDF RULES

The PDF may be image-only or scanned.

Use OCR/vision only if needed.

Preserve the correct reading order, especially for two-column patent pages.

Fix broken line wraps and obvious OCR hyphenation, for example:

"con-\ntact" -> "contact"

"polymerisa-\ntion" -> "polymerisation"

But do not change technical terms, claim scope, numbers, reference numerals, units, or legal wording.

If ordinary text is too blurry or uncertain, mark it with:

[OCR_UNCLEAR: ...]

Do not guess.

Do not use OCR to transcribe tables, chemical structures, complex formulas, or drawings into noisy text.

7. CLEANING RULES

Remove repeated headers, footers, page numbers, publication numbers, and column numbers.

Normalize whitespace.

Keep paragraph order.

Keep reference numerals in parentheses, e.g. (20,22;50,51).

Do not summarize.

Do not paraphrase.

Do not modernize spelling.

Do not translate.

8. OUTPUT RULES

Create the two .txt files as downloadable files.

Do not paste the full extracted contents into the chat unless asked.

After creating the files, report only:

- the two filenames,
- the download links,
- the file path name in the form "text/filename",
- whether Google Patents HTML text was used,
- whether PDF text layer was used,
- whether OCR was needed,
- whether any [OCR_UNCLEAR] markers were inserted,
- whether any tables/formulas/figures were found and how they were handled.

Give the filenames in an easy way for me to copy-paste.

Here is a guidance

After giving me the links for download, give me the file path names in the form:

text/filename

you need 2 fields, both in csv

first field for description

second field for claim file

Final check. the name of the file should match the right version.

D.4 Fourth prompt: prior-art extraction

Prior Art Extraction - Single-Row Input / Eight-Field Output

Purpose: Process one prior-art dataset row, resolve the actual prior-art document from the judgment URL, create the required text files, and return one eight-field CSV row.

If cant find patent then say skip in all 4 fields that you have to fill in. But try your best.

INPUT FORMAT

You are given exactly one CSV row containing exactly four fields, in this order:

1. index
2. judgment_url
3. prior_art_name_less_than_20_words
4. prior_art_full_text_link

index,judgment_url,prior_art_name_less_than_20_words,prior_art_full_text_link

Example:

17,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2008/3070,Oshlack,https://patents.google.com/patent/EP0253104A1/en

The judgment_url is the judgment in which the prior art is discussed. The prior_art_full_text_link is only a candidate source and may be correct, incorrect, incomplete, a different patent-family member, a later or earlier patent publication, a non-patent document, or a placeholder source.

Do not expect the full judgment text to be pasted into the prompt. Retrieve and read the judgment directly from judgment_url. The judgment is the primary authority for identifying the prior-art document actually

considered by the court.

IMMUTABLE INPUT FIELDS RULE

Fields 1-4 are immutable. The output row **MUST** begin with the exact original input row string, copied verbatim, character-for-character. Do not parse and reserialize the first four fields in a way that changes their textual representation. Construct the final row by taking the original four-field input row exactly as supplied and appending only: `,field5,field6,field7,field8`.

Do not correct, normalize, canonicalize, trim, rewrite, replace, translate, re-quote, unquote, change capitalization, alter URL paths, add or remove trailing slashes, change language suffixes, or otherwise modify any character in fields 1-4, even when a field is inaccurate.

Any corrected or resolved prior-art URL belongs **ONLY** in field 8 (`resolved_prior_art_link`). If field 4 points to the wrong patent publication or uses a non-/en Google Patents URL, field 4 must still remain exactly unchanged; only field 8 may contain the corrected canonical URL.

1. TASK OVERVIEW

For the single supplied CSV row:

1. Read the judgment at `judgment_url`.
2. Identify the exact prior-art reference represented by `prior_art_name_less_than_20_words`.
3. Determine whether the prior art is a patent document or a non-patent document.
4. Resolve the correct full-text source.
5. Extract the required text.
6. Create the required downloadable UTF-8 `.txt` file or files.
7. Output exactly one CSV code block containing exactly one line and exactly eight fields.

The first four output fields are IMMUTABLE. They must be copied from the input row verbatim, character-for-character, with no normalization or reserialization. Append exactly four new fields:

1. `description_file_path`
2. `claims_file_path`
3. `patent_full_file_path`
4. `resolved_prior_art_link`

`index,judgment_url,prior_art_name_less_than_20_words,prior_art_full_text_link,description_file_path,claims_file_path,patent_full_file_path,resolved_prior_art_link`

Do not output a CSV header.

2. IDENTIFY THE PRIOR ART FROM THE JUDGMENT

Read the judgment before relying on the candidate prior-art link. Find the passages discussing the prior art named in `prior_art_name_less_than_20_words`.

Determine as precisely as possible:

the prior-art name or shorthand used by the court;

whether it is a patent or non-patent reference;

the document actually relied upon by the court;

publication number, application number, or patent number where applicable;

author, title, inventor, applicant, proprietor, or other identifying information;

jurisdiction and patent-family relationship where applicable;

publication or version information;

the date of judgment.

The judgment is the primary authority for prior-art identity. The candidate `prior_art_full_text_link` is only a starting point and must not be assumed to be correct.

3. PATENT PRIOR ART

If the prior art is a patent document, resolve the exact patent publication relied upon by the judgment.

The candidate Google Patents link may point to:

the wrong patent-family member;

the wrong jurisdiction;

a granted patent instead of the published application relied upon in the judgment;

an application instead of a granted patent discussed by the judgment;

a later publication, correction, republication, grant, or amended version;

a document that did not yet exist on the judgment date;

a document with a similar title, inventor, applicant, or priority number but a different publication identity.

3.1 Resolution hierarchy

Express publication number

If the judgment expressly gives a publication number and kind code, use that exact publication. Example: if the judgment identifies EP1234567A1, use EP1234567A1. Do not replace it with a later B1 merely because that version is now easier to access.

Publication number without kind code

If the judgment identifies a patent or application number but omits the publication kind code, use bibliographic history and the judgment context to determine the relevant published version.

Expressly discussed version

If the judgment expressly discusses the granted patent, use the granted version if it existed by the judgment date. If it discusses the published application, use that publication instead of a later grant.

Judgment-date restriction

Never resolve to a publication, grant, correction, amendment, or republication that first became public after the judgment date.

Latest relevant pre-judgment version

If the judgment does not expressly specify the publication version, select the latest relevant public version that existed by the judgment date, belongs to the correct patent family, corresponds to the document actually discussed by the judgment, and does not materially change the identity of the relied-upon prior art.

Jurisdiction

Prefer the jurisdiction expressly used in the judgment. Do not replace an EP publication with WO, GB with US, WO with a national-stage patent, or one national publication with another merely because another family member has cleaner text.

Family-equivalent fallback

A family-equivalent English patent may be used as a text source only when the resolved publication has no usable English text. The final resolved link and filenames must still identify the publication supported by the judgment.

Ambiguity

Do not invent a publication identity. If exact resolution is genuinely uncertain, choose the best-supported version only when the evidence strongly favors it.

4. PATENT SOURCE PRIORITY

For patent prior art, use sources in this order:

1. Google Patents HTML for the resolved publication.
2. PDF text layer for the resolved publication.
3. Official patent-office HTML or text.
4. English family-equivalent publication, only as a justified fallback.
5. OCR or visual transcription only as a last resort.

For patent prior art, resolved_prior_art_link (field 8) MUST be the canonical Google Patents English URL for the resolved publication, and MUST end in /en:

https://patents.google.com/patent/PUBLICATION_NUMBER/en

Patent field-8 rule: always report exactly https://patents.google.com/patent/PUBLICATION_NUMBER/en for the resolved patent publication. Do not use /zh, /de, /fr, another language suffix, a bare Google Patents URL without /en, a PDF URL, or an alternative patent-office URL in field 8 when a canonical Google Patents page exists.

This rule applies to field 8 only. Field 4 remains immutable and must never be changed to match field 8.

5. PATENT OUTPUT FILES

If the prior art is a patent, create exactly three downloadable UTF-8 plain-text files:

INDEX-PUBLICATION_NUMBER_description.txt

INDEX-PUBLICATION_NUMBER_claims.txt

INDEX-PUBLICATION_NUMBER_patent_full.txt

Example for input index 17:

17-EP0253104A1_description.txt

17-EP0253104A1_claims.txt

17-EP0253104A1_patent_full.txt

The dataset paths placed in the output CSV must be:

text_prior_art/17-EP0253104A1_description.txt

text_prior_art/17-EP0253104A1_claims.txt

text_prior_art/17-EP0253104A1_patent_full.txt

The index prefix is mandatory. Use the resolved publication number, not automatically the publication number appearing in the candidate link.

6. PATENT DESCRIPTION FILE

The Description file must contain only the complete cleaned English Description. Start at Description, Specification, or the equivalent heading beginning the substantive patent description. Stop immediately before Claims.

Do not include:

front-page bibliographic metadata or abstract-only front matter;

search reports or examination reports;

headers, footers, page numbers, repeated patent titles, repeated publication numbers, or column line numbers;

non-English duplicate sections;

judgment text or prior-art URLs.

Do not summarize or paraphrase.

7. PATENT CLAIMS FILE

The Claims file must contain only the complete cleaned English Claims section. Preserve claim numbers, dependencies, ordering, and substantive wording.

Do not silently substitute:

later amended claims;

later granted claims;

later corrected claims;

claims from another family member;

for the claims belonging to the resolved publication. Do not summarize or paraphrase.

8. PATENT FULL FILE

The patent-full file must contain the complete Description followed by the complete Claims, using this structure:

Description

[complete cleaned English Description]

Claims

[complete cleaned English Claims]

It must contain the same substantive patent text as the two individual files. Do not add metadata, summaries, judgment text, source reports, URLs, or commentary.

9. NON-PATENT PRIOR ART

If the judgment shows that the prior art is not a patent, do not create Description or Claims files.

Instead:

1. Resolve the actual prior-art document.
2. Locate the best available full-text source.
3. Create exactly one downloadable UTF-8 plain-text full-text file.
4. Leave `description_file_path` and `claims_file_path` empty.
5. Populate `patent_full_file_path` with the non-patent full-text file path.
6. Populate `resolved_prior_art_link` with the resolved source URL.

The non-patent full-text file should be named:

`INDEX-SHORT_IDENTIFIER_full.txt`

Examples:

`150-Roy_full.txt`

`151-Kabachnik_1978_full.txt`

`152-Frontline_spray_full.txt`

Use a short filename-safe identifier derived from the prior-art name. Replace spaces with underscores, remove filename-invalid characters, and do not make the identifier unnecessarily long.

The dataset path must be:

`text_prior_art/INDEX-SHORT_IDENTIFIER_full.txt`

For non-patent prior art, the last four output fields therefore have this pattern:

`„text_prior_art/INDEX-SHORT_IDENTIFIER_full.txt,RESOLVED_SOURCE_URL`

Example complete row:

150,https://caselaw.nationalarchives.gov.uk/ewhc/pat/2009/213,Roy,https://pubmed.ncbi.nlm.nih.gov/8742940/,,,text_prior_art/150-Roy_full.txt,https://pubmed.ncbi.nlm.nih.gov/8742940/

10. NON-PATENT FULL-TEXT EXTRACTION

For non-patent prior art, extract the substantive text of the document itself. Preserve the original substantive wording. Do not summarize or paraphrase.

Remove:

website navigation, menus, advertisements, and unrelated page material;

repeated headers and footers;

citation-manager UI, download controls, and cookie notices;

obvious HTML extraction artifacts.

Where reasonably available, preserve the title, authors, section headings, and substantive body text. Do not include the judgment merely because the candidate link points to the judgment. If the candidate link is only a placeholder and the actual document can be identified from the judgment, resolve and use the actual document.

11. TABLES

Do not reproduce tables as noisy plain text. For a continuous table block, insert:

[TABLES_NOT_INCLUDED]

Use one placeholder per continuous table block. Keep ordinary textual references to tables.

12. CHEMICAL STRUCTURES AND FORMULAS

Keep ordinary machine-readable technical text, including chemical names, compound names, material names, simple molecular formulas, concentrations, percentages, units, ranges, and simple equations.

For chemical structures, Markush structures, reaction schemes, or diagrammatic formulas that cannot be represented reliably as ordinary text, use:

[CHEMICAL_STRUCTURE_REMOVED]

For OCR-corrupted formulas, use:

[FORMULA_OCR_REMOVED]

For complex mathematical formulas that cannot be cleanly represented, use:

[COMPLEX_FORMULA_REMOVED]

Do not invent missing formula content.

13. FIGURES AND DRAWINGS

Do not reproduce drawings or figure images. Keep textual references to figures. For drawing-only material, use:

[DRAWINGS_NOT_INCLUDED]

Do not infer technical details from drawings.

14. OCR

Use OCR only as a last resort. Fix obvious OCR line-wrap and hyphenation artifacts without changing substantive wording. If ordinary text cannot be read reliably, use:

[OCR_UNCLEAR: ...]

Do not guess.

15. CLEANING RULES

Normalize whitespace and obvious extraction artifacts. Preserve paragraph order, technical terminology, legal wording, numbers, units, claim wording, spelling conventions, and useful document structure.

Do not:

summarize;

paraphrase;

translate;

modernize;

rewrite;

add analysis;

change terminology to match the judgment.

16. VALIDATION

Immutable fields validation

Before output, compare the literal prefix through field 4 against the supplied input row and verify that every character is identical. The output prefix through field 4 must be exactly the original input row. Only fields 5-8 may be newly generated.

For patent prior art

The resolved patent is the prior art discussed in the judgment.

The publication belongs to the correct patent family.

The relevant version existed on or before the judgment date.

No improper later grant or correction was substituted.

The jurisdiction matches the judgment unless a justified text fallback was necessary.

resolved_prior_art_link identifies the selected publication and, for patent prior art, is exactly the canonical Google Patents English URL ending in /en.

All three filenames contain the correct index and resolved publication number.

Description and Claims belong to the same resolved version.

The patent-full file contains the same substantive Description and Claims as the two separate files.

For non-patent prior art

The resolved document corresponds to the prior art discussed in the judgment.

The full-text source represents the actual document rather than merely a secondary citation when the primary document is available.

The filename begins with the supplied index.

Exactly one full-text file is created.

description_file_path and claims_file_path are empty. For non-patent prior art, resolved_prior_art_link is the resolved non-patent source URL; the /en requirement does not apply.

17. FINAL RESPONSE

Provide the downloadable file links first.

For patent prior art, provide exactly three downloadable files. For non-patent prior art, provide exactly one downloadable full-text file.

After the download link or links, output exactly one CSV code block. The CSV code block must:

NON-NEGOTIABLE OUTPUT CONSTRUCTION: use the exact original input row as the literal prefix of the output row, then append fields 5-8. Never regenerate fields 1-4 from parsed values.

use the csv language;
 contain exactly one line;
 contain exactly eight fields;
 contain no header;
 contain no commentary;
 contain no numbering;
 begin with the exact original four-field input row verbatim, character-for-character; do not reconstruct, normalize, or reserialize fields 1-4;
 append `description_file_path`, `claims_file_path`, `patent_full_file_path`, and `resolved_prior_art_link` as fields 5-8; for patent prior art, field 8 must be the canonical Google Patents URL ending in `/en`.

Patent example

17,<https://caselaw.nationalarchives.gov.uk/ewhc/pat/2008/3070>,Oshlack,<https://patents.google.com/patent/EP0253104A1/en>,`text_prior_art/17-EP0253104A1_description.txt`,`text_prior_art/17-EP0253104A1_claims.txt`,`text_prior_art/17-EP0253104A1_patent_full.txt`,<https://patents.google.com/patent/EP0253104A1/en>

Non-patent example

150,<https://caselaw.nationalarchives.gov.uk/ewhc/pat/2009/213>,Roy,<https://pubmed.ncbi.nlm.nih.gov/8742940/>,`text_prior_art/150-Roy_full.txt`,<https://pubmed.ncbi.nlm.nih.gov/8742940/>

The final CSV code block must be the last thing in the response. Do not output any additional code block after it.

D.5 Extra prompt

IP JUDGMENT EXTRACTION PROMPT — STEP 2 — CLAIM-LEVEL VALIDITY ROWS

STRICT RULE: NEVER USE "" IN TEXT FIELDS

You are converting patent-level validity rows produced by Step 1 into claim-level validity rows for UK patent judgments.

This prompt produces only the validity dataset. Do not produce infringement rows.

CORE PURPOSE

Step 1 has already extracted and verified the reusable case-level and patent-level information, including dates, judges, parties, lawyers, historical entity classifications, parents, tickers, headquarters, patent metadata, citation counts, and source URLs.

Do not research, reconstruct, recode, standardize, or otherwise change those reusable fields in Step 2.

Your tasks in Step 2 are only to:

1. identify every patent claim or separately assessed claim version that was actually attacked, decided, expressly conceded invalid, or otherwise given a clean claim-level validity outcome;
2. create one output row for each qualifying claim unit;
3. copy every reusable field exactly from the corresponding Step-1 patent-level row;
4. complete the claim-specific fields 18, 25, 27, and 29; and
5. keep every overall claim-level outcome in field 25 strictly binary: "0" or "1".

This prompt does not create a `claim_index`. Any stable `claim_index` needed for later matching must be assigned outside this prompt by a persistent registry or deterministic data-processing script. Never invent a

claim_index.

INPUTS

You will receive:

1. A judgment text file or judgment URL.
2. The claims text file for each patent or patent application covered by the supplied Step-1 row or rows.
3. One or more Step-1 patent-level CSV rows.

Each Step-1 patent-level CSV row has exactly 32 fields in this order:

1. date_of_judgment
2. date_of_hearing
3. date_filing_the_complaint
4. judges
5. claimants
6. defendants
7. right_owner_or_applicant_side
8. other_side_or_validity_challenger_side
9. right_owner_or_applicant_party_type_ticker_and_headquarter
10. other_side_or_validity_challenger_party_type_ticker_and_headquarter
11. claimant_lawyers
12. defendant_lawyers
13. stage
14. right_type
15. right_name
16. right_short_description
17. google_patent_url
18. patent_unit
19. forward_citations_first_hearing_priority_date
20. forward_citations_first_judgment_priority_date
21. backward_citations_1_family_to_family
22. backward_citations_2_all_rows_in_patent_citations
23. grant_date
24. priority_date
25. right_level_winner_owner_vs_other
26. product_or_process_judicial_quotes
27. issue_dictionary
28. claimant_is_owner_side
29. summary
30. link
31. forward_citations_first_hearing_publication_date
32. forward_citations_first_judgment_publication_date

COPY-EXACTLY RULE

For every claim-level row derived from a Step-1 patent-level row, copy these fields exactly and character-for-character from that Step-1 row:

Fields 1–17, fields 19–24, field 26, field 28, and fields 30–32.

Do not change spelling, capitalization, ordering, lists, dictionaries, dates, names, entity coding, quotation content, citation counts, URLs, or CSV escaping in those copied fields.

Only fields 18, 25, 27, and 29 may be newly completed or rewritten in Step 2.

In the Step-2 output, these four positions change to claim-level meanings:

- * field 18 = claim_unit;
- * field 25 = claim_level_winner_owner_vs_other;
- * field 27 = the nine-key claim-level issue_dictionary defined below; and
- * field 29 = the claim-level summary defined below.

If several claim-level rows come from the same Step-1 patent-level row, repeat the copied fields exactly in every such row.

ABSOLUTE OUTPUT RESTRICTIONS

- * Output exactly one fenced code block labelled csv.
- * Put every output row for every patent, claim, and claim version inside the same csv code block.
- * Do not include a header.
- * Do not include notes, explanations, citations, footnotes, source references, or any text before or after the csv code block.
- * Every output row must contain exactly 32 fields.
- * Wrap every field in double quotes.
- * Escape every internal double quote by doubling it.
- * Do not produce malformed CSV such as quadruple quotes.
- * Lists and dictionaries copied from Step 1 must retain their existing valid CSV escaping.
- * No CSV field may contain an actual carriage return or newline.
- * Each CSV row must occupy exactly one physical line.
- * If field 29 needs several statements, put them on the same physical line separated by spaces.
- * Preserve literal backslash-n sequences already contained in copied fields; do not convert them into actual line breaks.
- * Do not output JSON outside a CSV field.
- * Do not hallucinate.
- * If no qualifying claim-level row exists for a supplied patent-level row, do not create a claim row for it.
- * If no qualifying claim-level row exists for any supplied patent-level row, output one empty csv code block.

CLAIM-LEVEL ROW UNIT — MANDATORY

Each output row must correspond to:

- * one patent or patent application;
- * exactly one identifiable claim number;
- * exactly one identifiable claim version or other outcome-homogeneous claim unit;
- * the owner/applicant/enforcement side already recorded in the Step-1 row; and

* the adverse validity-challenger side already recorded in the Step-1 row.

Create a row only for a claim or claim version that meets one of the following conditions:

* at least one validity or patentability ground was raised against that claim or claim version, AND the judgment adjudicates its outcome on that ground;

* its invalidity was expressly conceded or expressly no longer contested, subject to the concession rules below; or

* it is a corresponding independent claim in another patent to which the judgment applies its validity determination for a representative patent. Assign the same outcome to that corresponding claim.

Do not create a row for a claim merely because it:

* appears in the patent;

* is mentioned as dependency or technical background;

* is relevant only to infringement;

* is referred to only as an unargued claim;

* is withdrawn, abandoned, not maintained, or not separately defended without a clean validity outcome; or

* belongs to a patent that was revoked when the judgment does not identify or necessarily decide that claim at claim level.

If claims 1–3 are separately identifiable but decided together, output three rows: one for claim 1, one for claim 2, and one for claim 3.

Never write a grouped unit such as "claims 1–3" when the individual claim numbers can be identified.

If the judgment genuinely refers only to "the asserted claims", "the relevant claims", or equivalent language and individual claim numbers cannot be identified after checking the judgment and claims text, output one fallback row with field 18 = "attacked claims not individually identified" and explain the limitation in field 29.

Do not create duplicate rows for the same claim unit merely because several validity grounds were argued. All grounds for the same outcome-homogeneous claim unit belong in field 27 of the same row.

OLD / GRANTED / AMENDED CLAIM VERSIONS

Treat each separately assessed claim version as a distinct claim unit.

* An old/granted claim and an amended claim are different claim units.

* Two separately assessed proposed amendments to the same numbered claim are different claim units.

* Conditional and unconditional amended versions are different claim units when both are substantively assessed.

* If the judgment considers only the amended claim and gives no separate outcome to the old/granted claim, output only the amended claim row.

* If the old/granted claim is invalid and a separately assessed amended version survives, output two rows: field 25 = "1" for the old/granted row and field 25 = "0" for the amended row.

* Do not use one mixed row to combine old/granted and amended outcomes.

Use precise field-18 labels, for example:

* "application claim 1"

* "old/granted claim 1"

* "amended claim 1"

* "conditional amended claim 1"

* "conditional amended claim 12 — revised amendment 1"

* "conditional amended claim 12 — revised amendment 2"

* "old/granted claim 1; whole patent conceded invalid"

WHOLE-PATENT INVALIDITY CONCESSION — SPECIAL RULE

If the judgment, final order, or reliable litigation record expressly states that the patent owner concedes that the whole patent is invalid:

- * identify the exact patent version covered by the concession;
- * identify every independent claim in that exact version from the claims text or contemporaneous official patent document;
- * output one row for each independent claim;
- * set field 25 = "1" for every such row;
- * determine whether the concession expressly identifies one or more ordinary validity grounds;
- * if the concession identifies an ordinary ground, code that ground = 1 and leave "Conceded invalidity" = "";
- * if the concession identifies several ordinary grounds, code each identified ground = 1 and leave "Conceded invalidity" = "";
- * if the concession identifies no ordinary validity ground, set "Conceded invalidity" = 1; and
- * explain in field 29 that the row was created under the express whole-patent invalidity-concession rule.

Do not add dependent claims solely because the whole patent was conceded invalid. Add a dependent claim only if it also has a separately identifiable attack, decision, or individual invalidity concession.

A withdrawal, abandonment, non-maintenance, agreed revocation, or decision not to defend a patent does not trigger this rule unless the record expressly concedes or accepts invalidity.

If the whole-patent concession is clear but the independent claim numbers cannot be verified after best efforts, output one fallback row with field 18 = "independent claims not individually identified; whole patent conceded invalid" and field 25 = "1". Apply the same ground-specific concession rule: code an identified ordinary ground = 1 and leave "Conceded invalidity" = ""; use "Conceded invalidity" = 1 only when no ordinary ground of concession is identified. Explain the failed verification in field 29. Never guess claim numbers.

BINARY OUTCOME RULE FOR FIELD 25 — MANDATORY

Field 25, `claim_level_winner_owner_vs_other`, must always be either "0" or "1".

Never output "0.5" in field 25.

Use:

- * "0" = the patent owner/applicant wins for the precise claim unit in field 18. That claim unit remains valid or patentable as presented.
- * "1" = the adverse side wins for the precise claim unit in field 18. That claim unit is invalid, revoked, refused, rejected, expressly conceded invalid, or otherwise does not remain in valid form.

If the owner loses on any ground that legally invalidates the claim unit, field 25 must be "1", even if the owner wins on other grounds.

If the owner wins on every live ground and the claim unit survives, field 25 must be "0".

Do not use "0.5" to describe partial survival, narrowing, severance, different alternatives, different versions, or different claim constructions.

OUTCOME-SPLITTING RULE — MANDATORY

Every row must be outcome-homogeneous.

If the court divides what would otherwise appear to be one numbered claim or claim version into two separately identifiable claim units and gives them different overall outcomes, create separate rows.

For example, if the judgment treats one separable part, alternative, category, construction, or version of

claim 1 as valid and another as invalid:

- * create one row for the valid claim unit with field 25 = "0"; and

- * create one row for the invalid claim unit with field 25 = "1".

Field 18 must distinguish the two units precisely, using the judgment's terminology where possible. Field 29 must explain the distinction and why the rows were split.

Do not keep one row and assign field 25 = "0.5".

Split only when the judgment itself supplies a legally meaningful and separately identifiable division. Do not invent artificial claim fragments merely to force a binary outcome.

If a genuinely mixed result remains within a single validity ground after every legally meaningful claim unit has been separated, field 27 may use 0.5 for that ground. Field 25 must nevertheless remain "0" or "1" according to the final legal effect on the claim unit.

FIELD 18 — CLAIM_UNIT

Identify exactly one claim number and one claim version or other outcome-homogeneous unit.

Field 18 must:

- * name the precise claim number when identifiable;

- * identify whether it is application, old/granted, previously amended, conditional amended, unconditional amended, or another separately assessed version;

- * distinguish court-defined claim parts, alternatives, constructions, or categories when the outcome-splitting rule applies; and

- * remain concise but sufficiently specific to prevent two different rows from appearing to represent the same claim unit.

FIELD 25 — CLAIM_LEVEL_WINNER_OWNER_VS_OTHER

Output only "0" or "1" under the binary outcome rule above.

FIELD 27 — ISSUE_DICTIONARY

Return this dictionary with exactly these nine keys and no others:

```
{
  "Lack of novelty": "",
  "Lack of inventive step / obviousness": "",
  "Insufficiency": "",
  "Added matter": "",
  "Excluded subject matter / non-patentability": "",
  "Priority not entitled": "",
  "Lack of industrial applicability": "",
  "Others": "",
  "Conceded invalidity": ""
}
```

For the precise claim unit in field 18, code each ordinary validity ground as follows:

- * 0 = the patent owner/applicant wins on that ground.

- * 0.5 = the result on that ground is genuinely mixed or partial within the same outcome-homogeneous claim unit and cannot be represented more accurately by applying the mandatory outcome-splitting rule.

- * 1 = the patent owner/applicant loses on that ground.

* "" = the ground was not live or was not decided for that claim unit.

The value 0.5 is permitted only inside field 27. It is never permitted in field 25.

CONCEDED INVALIDITY — FALLBACK CODING ONLY

"Conceded invalidity" is a fallback dataset category for an express invalidity concession that does not identify any ordinary validity ground.

* First determine whether the patent owner expressly concedes, accepts, admits, or expressly no longer contests invalidity on a specific ordinary ground.

* If the concession is expressly based on lack of novelty, set "Lack of novelty" = 1 and leave "Conceded invalidity" = "".

* If the concession is expressly based on lack of inventive step or obviousness, set "Lack of inventive step / obviousness" = 1 and leave "Conceded invalidity" = "".

* Apply the same rule to insufficiency, added matter, excluded subject matter/non-patentability, priority not entitled, lack of industrial applicability, or an identifiable ground falling within "Others".

* If several ordinary grounds are expressly conceded, code every conceded ordinary ground = 1 and leave "Conceded invalidity" = "".

* Do not double-code the same concession under both an ordinary ground and "Conceded invalidity".

* Set "Conceded invalidity" = 1 only when the patent owner expressly concedes, accepts, admits, or expressly no longer contests the invalidity of the precise claim unit without identifying any ordinary validity ground for that concession.

* Whenever "Conceded invalidity" = 1, field 25 must be "1".

* "Conceded invalidity" may contain only 1 or "".

* Never set "Conceded invalidity" to 0 or 0.5.

* If no express concession exists, leave it as "".

* If an opponent alleges that a concession occurred but the court finds that no concession or admission existed, leave "Conceded invalidity" as ""; do not code 0.

* Do not infer conceded invalidity from amendment, withdrawal, abandonment, silence, non-appearance, non-maintenance, deletion, failure to defend, or the patent-level result alone.

* If an express concession identifies no ordinary validity ground, set "Conceded invalidity" = 1 and leave the ordinary grounds according to whether they were separately live or decided. Do not automatically set "Others" = 1 merely because the concession lacks a named ground.

* A ground separately litigated or decided is still coded on its own merits. The no-double-coding rule applies to the basis of the concession, not to an independent judicial decision on another ground.

Field 27 must be substantively consistent with field 25:

* If any ordinary ground is coded 1, field 25 must be 1.

* If "Conceded invalidity" is coded 1, field 25 must be 1.

* Field 25 may be 1 even when no ordinary ground is coded 1, for example when conceded invalidity is the only adverse basis.

* If an ordinary ground is coded 1 because it was expressly conceded, "Conceded invalidity" must be "".

* Different ordinary grounds may have different results in the same row. For example, novelty may be 0 and obviousness may be 1; field 25 is then 1.

FIELD 29 — CLAIM-LEVEL SUMMARY

Rewrite field 29 for each claim-level row. Keep it under 500 words and on one physical line.

The first sentence must state:

- * the overall binary outcome for the precise claim unit; and
- * the patent/application-level consequence, where available.

Examples:

- * "old/granted claim 1 valid; whole patent maintained."
- * "old/granted claim 1 invalid; whole patent revoked."
- * "old/granted claim 1 invalid; amended claim 1 considered separately."
- * "amended claim 1 valid; patent maintained in amended form."
- * "independent old/granted claim 1 conceded invalid; whole patent conceded invalid."
- * "application claim 1 refused; patent application refused."

The summary must also:

- * identify the precise claim number, claim version, and any court-defined split unit in field 18;
- * explain every live validity ground and its result for that unit;
- * explain any express concession of invalidity, identify its ground when one was specified, and state when the ground-neutral "Conceded invalidity" fallback was used;
- * explain why two or more rows were created when the outcome-splitting rule applies;
- * distinguish old/granted and amended versions;
- * explain separately whether the whole patent/application was maintained, revoked, amended, narrowed, refused, restored, or upheld;
- * avoid treating unargued claims as individually invalid;
- * contain no URLs; and
- * contain no actual line breaks.

Do not repeat entity-research explanations, ticker research, citation methodology, or other metadata work already completed in Step 1 unless it is strictly necessary to explain the claim-level result.

FINAL VALIDATION BEFORE OUTPUT

Before returning the single csv code block, verify all of the following:

1. Every output row derives from a supplied Step-1 patent-level row.
2. Fields 1–17, 19–24, 26, 28, and 30–32 are copied exactly from the corresponding Step-1 row.
3. Only fields 18, 25, 27, and 29 have been newly completed or rewritten.
4. Every separately identifiable attacked, decided, or expressly conceded claim version has its own row.
5. Claims mentioned only as background, dependency, infringement, abandonment, withdrawal, non-maintenance, or unargued claims have not been added without a clean claim-level validity outcome.
6. Old/granted and amended versions are separate whenever they receive separate consideration or outcomes.
7. Every row is outcome-homogeneous.
8. If one numbered claim or version received different outcomes for separately identifiable units, those units were split into separate rows.
9. Field 25 contains only "0" or "1" in every row.
10. Field 25 never contains "0.5" or any other value.
11. Field 27 contains exactly the nine required keys.
12. The value 0.5 appears only inside field 27 and only when a ground is genuinely mixed after all meaningful row splitting.

13. "Conceded invalidity" contains only 1 or "".
14. Whenever "Conceded invalidity" = 1, field 25 = "1" and the concession identifies no ordinary validity ground.
15. If the concession identifies an ordinary validity ground, that ground is coded 1 and "Conceded invalidity" is "".
16. The same concession is never double-coded under both an ordinary ground and "Conceded invalidity".
17. A rejected allegation of concession is not coded as conceded invalidity or as a conceded ordinary ground.
18. No claim_index was invented.
19. Every row contains exactly 32 double-quoted CSV fields.
20. Every row occupies exactly one physical line and contains no actual carriage return or newline.
21. All rows appear inside one csv code block with no header.
22. The final answer contains exactly one csv code block and no text outside it.

E ChatGPT zero-short prompt

E.1 Patent

You are performing a zero-shot prediction task about patent-level outcomes in UK patent validity or patentability proceedings.

Your task is to predict whether all claims at issue would fail on one or more applicable grounds of invalidity or non-patentability.

You are supplied with the full patent claim set and the patent description.

A claim fails if at least one applicable ground succeeds against it. Consider the following seven grounds:

1. Lack of novelty (anticipation).
2. Lack of inventive step / obviousness.
3. Insufficiency.
4. Lack of industrial applicability.
5. Excluded subject matter / non-patentability.
6. Added matter.
7. Impermissible extension of protection by post-grant amendment.

The “skilled person” is the hypothetical person or team with ordinary skill and knowledge in the relevant technical field. “Common general knowledge” means the technical knowledge that such a skilled person would ordinarily be expected to know at the relevant time.

Use the supplied patent texts together with technical knowledge that you judge would be common general knowledge of the skilled person. Do not search for or rely on any

case-specific judicial determination of what constitutes common general knowledge. Do not use any remembered information about this patent’s litigation history, judicial findings, or actual outcome, including information encountered during training. Do not use external tools or search.

If a claim depends on another claim, interpret it together with the limitations of the claim or claims on which it depends.

Prediction target:

- 0 = SURVIVES: at least one claim at issue survives or is allowed, including partial success.
- 1 = INVALIDATED: all claims at issue fail, including refusal of all claims at issue in an application.

This is an ex-ante prediction task. You are NOT given the judgment, court or IPO reasoning, expert evidence, pleadings, or eventual outcome.

Briefly explain the main reasons for your prediction. End your response with **FINAL_JSON**: followed by a JSON object containing exactly these two fields:

- “prediction”: an integer, either 0 or 1.
- “probability_invalidated”: your estimated probability that all claims at issue fail, expressed as a number from 0 to 1.

Use prediction = 1 if and only if **probability_invalidated** ≥ 0.5 .

FULL CLAIM SET

{patent_claims}

PATENT DESCRIPTION

{patent_description}

E.2 Insufficiency

You are performing a zero-shot prediction task about UK patent insufficiency.

Your task is to predict whether the **TARGET CLAIM** would be found invalid for insufficiency.

You are supplied with the target claim, the full patent claim set, and the patent description.

Insufficiency means that the patent disclosure, together with the skilled person’s common general knowledge, does not enable the claimed invention to be performed across substantially the full scope of the target claim without undue burden.

The “skilled person” is the hypothetical person or team with ordinary skill and knowledge in the relevant technical field. “Common general knowledge” means the technical knowledge that such a skilled person would ordinarily be expected to know at the relevant time.

Use the supplied patent texts together with technical knowledge that you judge would be common general knowledge of the skilled person. Do not search for or rely on any case-specific judicial determination of what constitutes common general knowledge. Do not use any remembered information about this patent’s litigation history, judicial findings, or actual outcome, including information encountered during training. Do not use external tools or search.

The supplied TARGET CLAIM is the claim to be assessed. Use the full patent claim set only as context to help interpret the target claim and its dependencies. The target claim may not appear verbatim in that claim set. Do not treat its absence as evidence of insufficiency, and do not replace it with a different claim from the claim set. Where their wording differs, assess the target claim as supplied.

If the target claim depends on another claim, interpret it together with the limitations of the claim or claims on which it depends.

Prediction target:

- 0 = NOT INSUFFICIENT: the target claim survives an insufficiency challenge.
- 1 = INSUFFICIENT: the target claim fails for insufficiency.

This is an ex-ante prediction task. You are NOT given the judgment, court or IPO reasoning, expert evidence, pleadings, or eventual outcome.

Briefly explain the main reasons for your prediction. End your response with **FINAL_JSON**: followed by a JSON object containing exactly these two fields:

- “prediction”: an integer, either 0 or 1.
- “probability_insufficient”: your estimated probability that the target claim fails for insufficiency, expressed as a number from 0 to 1.

Use $\text{prediction} = 1$ if and only if $\text{probability_insufficient} \geq 0.5$.

TARGET CLAIM

{target_claim}

FULL CLAIM SET

{patent_claims}

PATENT DESCRIPTION

{patent_description}

E.3 Obviousness

You are performing a zero-shot prediction task about UK patent obviousness.

Your task is to predict whether the TARGET CLAIM would be found invalid for obviousness on the basis of the supplied prior-art document or documents.

You are supplied with the target claim, the full patent claim set, and one to three prior-art documents for a single obviousness attack.

Obviousness means that the skilled person, using their common general knowledge and without hindsight, would have arrived at the claimed invention from the supplied prior art without exercising inventive ingenuity.

The “skilled person” is the hypothetical person or team with ordinary skill and knowledge in the relevant technical field. “Common general knowledge” means the technical knowledge that such a skilled person would ordinarily be expected to know at the relevant time.

Use the supplied claim and prior-art texts together with technical knowledge that you judge would be common general knowledge of the skilled person. Do not search for or rely on any case-specific judicial determination of what constitutes common general knowledge. Do not use any remembered information about this patent’s litigation history, judicial findings, or actual outcome, including information encountered during training. Do not use external tools or search.

The supplied TARGET CLAIM is the claim to be assessed. Use the full patent claim set only as context to help interpret the target claim and its dependencies. The target claim may not appear verbatim in that claim set. Do not treat its absence as evidence of obviousness, and do not replace it with a different claim from the claim set. Where their wording differs, assess the target claim as supplied.

If the target claim depends on another claim, interpret it together with the limitations of the claim or claims on which it depends.

Return one prediction for the supplied attack as a whole. Where the attack relies on combining documents, assess whether the skilled person would have had reason, without hindsight, to consider their teachings together. Finding individual claim features across different documents is not, by itself, sufficient to establish obviousness.

Prediction target:

- 0 = NOT OBVIOUS: the target claim survives this obviousness attack.
- 1 = OBVIOUS: the target claim fails for obviousness on the basis of this attack.

This is an ex-ante prediction task. You are NOT given the judgment, court or IPO reasoning, expert evidence, pleadings, or eventual outcome.

Briefly explain the main reasons for your prediction. End your response with **FINAL_JSON**: followed by a JSON object containing exactly these two fields:

- “prediction”: an integer, either 0 or 1.
- “probability_obvious”: your estimated probability that the target claim fails for obviousness on the basis of this attack, expressed as a number from 0 to 1.

Use prediction = 1 if and only if `probability_obvious` $\geq$ 0.5.

TARGET CLAIM

{target_claim}

FULL CLAIM SET

{patent_claims}

PRIOR-ART DOCUMENTS FOR THIS ATTACK

{numbered_prior_art_documents}

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